Lie Groups

— *EYNTKA* Series —

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Part I

Lie Groups and Their Lie Algebras

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Matrix Lie Groups and the Classical Groups

In the middle of the nineteenth century Galois had shown that the solvability of a polynomial equation is controlled by a finite group of symmetries: the permutations of the roots preserving every algebraic relation among them. Sophus Lie set out to build the analogue for differential equations, where the natural symmetries form not a finite set but a continuous family, the rotations of the plane, the translations along a flow, the scalings carrying one solution of an equation to another. The object governing such a family is a group that is at the same time a smooth manifold, so that one may differentiate along it, and the interplay between the group law and the differentiable structure is the source of the theory's power. Lie's own goal, a theory of integration by continuous symmetry parallel to Galois's theory of solvability, proved harder to carry through; but the objects he introduced organize a large part of modern mathematics, from the harmonic analysis of functions on a symmetric space, through the gauge fields of physics and the automorphic forms at the center of number theory, to the classification of the algebraic groups.

This book approaches Lie theory from its analytic and differential-geometric side: a Lie group is a manifold carrying a compatible group law, and the aim is to understand such groups, their infinitesimal counterparts (the Lie algebras), and their representations. The parallel algebraic development, in which the same groups are studied as varieties over an arbitrary field, is the subject of the companion volume [3]; the two meet at the theory of root systems in Part II.

The present chapter takes the most concrete route in. Before a Lie group is defined in the generality of an abstract manifold, the task of the next chapter, the classical groups are met as what they are at the surface: sets of matrices cut out by explicit polynomial or analytic constraints. The general linear group, and inside it the orthogonal, unitary, and symplectic groups, are exhibited directly as smooth manifolds, and the two objects around which the whole theory turns, the matrix exponential and the Lie algebra, are produced by differentiating matrix-valued functions rather than by invoking any abstract apparatus. The reward for this concreteness is that when the general theory arrives in the next chapter it finds a fully worked family of examples already in place. The chapter closes with the group SU(2) and its two-to-one relation with the rotation group of three-space, the example carried through the rest of the book.

1.1 The General Linear Group as a Manifold

The starting object of the whole subject is the set of all $n \times n$ matrices over a field, and the invertible ones among them. Throughout this chapter $\mathbb{F}$ denotes either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$, and $M_n(\mathbb{F})$ denotes the set of $n \times n$ matrices with entries in $\mathbb{F}$. Addition and scalar multiplication make $M_n(\mathbb{F})$ an $\mathbb{F}$-vector space of dimension n^2, and recording a matrix by its n^2 entries identifies it with $\mathbb{F}^{n^2}$; matrix multiplication makes it moreover an associative algebra with unit the identity matrix I. The identification with $\mathbb{F}^{n^2}$ equips $M_n(\mathbb{F})$ with a topology and a smooth structure, those of a finite-dimensional vector space, and every construction below is made against this fixed background.

What singles out the invertible matrices is that they, and only they, admit a two-sided multiplicative inverse; they form a group. The point of this section is that this group is at the same time a smooth manifold, and that the manifold structure is not imposed by hand but inherited for free from the ambient space of all matrices. This is the first place where the two structures a Lie group must carry, that of a group and that of a manifold, sit together, and it is the ambient stage on which every other group in the chapter appears.

Definition 1.1.1: General Linear Group

Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. The *general linear group* of rank n over $\mathbb{F}$ is the group of invertible matrices

$$\mathrm{GL}_n(\mathbb{F}) = \{A \in M_n(\mathbb{F}) \mid \det A \neq 0\}$$

under matrix multiplication, with identity element the identity matrix I and the inverse of A its matrix inverse A^{-1}.

That the condition $\det A \neq 0$ characterizes invertibility is the content of the cofactor formula: a matrix has a two-sided inverse if and only if its determinant is a unit of $\mathbb{F}$, and every nonzero element of a field is a unit. The determinant is a polynomial in the n^2 entries of A, so it is a continuous, indeed smooth, function $\det\colon M_n(\mathbb{F}) \to \mathbb{F}$. The set $\mathbb{F} \setminus \{0\}$ is open in $\mathbb{F}$, and $\mathrm{GL}_n(\mathbb{F})$ is its preimage under this continuous map. An open subset of a smooth manifold is itself a smooth manifold of the same dimension, an *open submanifold*, with charts obtained by restricting the ambient charts (see [2]). The general linear group therefore acquires a manifold structure at no cost.

Proposition 1.1.2: The General Linear Group is an Open Submanifold

The group $\mathrm{GL}_n(\mathbb{R})$ is an open subset of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, hence a smooth manifold of dimension n^2. The group $\mathrm{GL}_n(\mathbb{C})$ is an open subset of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$, hence a smooth manifold of real dimension $2n^2$. In each case the group operations

$$m\colon \mathrm{GL}_n(\mathbb{F}) \times \mathrm{GL}_n(\mathbb{F}) \to \mathrm{GL}_n(\mathbb{F}), \quad (A, B) \mapsto AB, \qquad \iota\colon \mathrm{GL}_n(\mathbb{F}) \to \mathrm{GL}_n(\mathbb{F}), \quad A \mapsto A^{-1}$$

are smooth.

***Proof*:**

Openness and the resulting manifold structure were established above: $\mathrm{GL}_n(\mathbb{F}) = \det^{-1}(\mathbb{F} \setminus \{0\})$ is the preimage of an open set under the continuous map det, hence open, and an open subset of the vector space $M_n(\mathbb{F})$ is a smooth manifold of the same dimension. Over $\mathbb{R}$ this dimension is n^2; over $\mathbb{C}$, identifying each complex entry with its real and imaginary parts identifies $M_n(\mathbb{C})$ with $\mathbb{R}^{2n^2}$, so the real dimension is $2n^2$.

For the multiplication, the (i, j) entry of AB is $\sum_{k=1}^{n} A_{ik}B_{kj}$, a polynomial in the entries of A and of B; a map into $\mathbb{F}^{n^2}$ whose every component is a polynomial in the coordinates is smooth, so m is smooth. For the inversion, Cramer's rule gives

$$A^{-1} = \frac{1}{\det A}\,\mathrm{adj}(A)$$

where the *adjugate* $\mathrm{adj}(A)$ has as its entries the $(n-1) \times (n-1)$ signed minors of A, themselves polynomials in the entries of A. On $\mathrm{GL}_n(\mathbb{F})$ the denominator $\det A$ never vanishes, so each entry of A^{-1} is a ratio of a polynomial by a nowhere-zero polynomial, hence a smooth function of the entries of A. Over $\mathbb{C}$ the same formulas exhibit the two maps as smooth in the real and imaginary parts of the entries. Thus m and ι are smooth, completing the proof.

A manifold that is also a group whose multiplication and inversion are smooth is precisely what the next chapter will call a *Lie group*, and proposition 1.1.2 is the statement that $\mathrm{GL}_n(\mathbb{F})$ is one. Every group met in this chapter is a subgroup of some $\mathrm{GL}_n(\mathbb{F})$, and the general linear group is the ambient object inside which the classical groups will be cut out; for that reason its own geometry is worth settling first. The most basic geometric invariant, the number of connected pieces, already distinguishes the real from the complex case.

Proposition 1.1.3: Connected Components of the General Linear Group

The group $\mathrm{GL}_n(\mathbb{C})$ is path-connected. The group $\mathrm{GL}_n(\mathbb{R})$ has exactly two connected components, namely

$$\mathrm{GL}_n^+(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid \det A > 0\}, \qquad \mathrm{GL}_n^-(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid \det A < 0\},$$

each of which is open and path-connected. The component $\mathrm{GL}_n^+(\mathbb{R})$ is a normal subgroup of index two.

***Proof*:**
The argument rests on one fact from linear algebra: every matrix of determinant 1 is a product of *transvections*, the matrices $I + \lambda E_{ij}$ with $i \neq j$ and $\lambda \in \mathbb{F}$, where E_{ij} is the matrix with a single 1 in position (i, j) and zeros elsewhere. This is the statement that Gaussian elimination reduces a determinant-one matrix to the identity using only the row operation that adds a multiple of one row to another. Given any $A \in \mathrm{GL}_n(\mathbb{F})$, set $D = \mathrm{diag}(\det A, 1, \ldots, 1)$; then $D^{-1}A$ has determinant 1, so

$$A = D \cdot S, \qquad D = \mathrm{diag}(\det A, 1, \ldots, 1), \quad S \in \mathrm{SL}_n(\mathbb{F}) \tag{1.1}$$

with $S = D^{-1}A$ a product of transvections.

Step 1: each factor is joined to the identity by a path. A single transvection $I + \lambda E_{ij}$ is joined to I by $t \mapsto I + t\lambda E_{ij}$ for $t \in [0, 1]$, a path lying in $\mathrm{GL}_n(\mathbb{F})$ because each of these matrices has determinant 1. A product of transvections is therefore joined to I by the pointwise product of the corresponding paths, which again lies in $\mathrm{GL}_n(\mathbb{F})$; so every $S \in \mathrm{SL}_n(\mathbb{F})$ is path-connected to I within $\mathrm{SL}_n(\mathbb{F})$. The diagonal factor $D = \mathrm{diag}(d, 1, \ldots, 1)$ with $d = \det A$ is handled by moving d: for any path $d\colon [0, 1] \to \mathbb{F} \setminus \{0\}$ from $d(0) = d$ to $d(1) = 1$, the matrices $\mathrm{diag}(d(t), 1, \ldots, 1)$ form a path in $\mathrm{GL}_n(\mathbb{F})$ from D to I.

Step 2: the complex case. When $\mathbb{F} = \mathbb{C}$ the punctured plane $\mathbb{C} \setminus \{0\}$ is path-connected, so a path from $d = \det A$ to 1 as required in Step 1 always exists. Concatenating the path from $A = DS$ to S (moving D to I) with the path from S to I shows that every $A \in \mathrm{GL}_n(\mathbb{C})$ is path-connected to I. Hence $\mathrm{GL}_n(\mathbb{C})$ is path-connected.

Step 3: the real case splits according to the sign of the determinant. When $\mathbb{F} = \mathbb{R}$ the set $\mathbb{R} \setminus \{0\}$ is the disjoint union of the two open rays $\mathbb{R}_{>0}$ and $\mathbb{R}_{<0}$, and $\det\colon \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R} \setminus \{0\}$ is continuous, so the two preimages $\mathrm{GL}_n^+(\mathbb{R})$ and $\mathrm{GL}_n^-(\mathbb{R})$ are disjoint, nonempty, open, and cover $\mathrm{GL}_n(\mathbb{R})$. No connected set meets both, since det would then take a value of each sign and, being continuous, the value 0 in between, which is impossible on $\mathrm{GL}_n(\mathbb{R})$. It remains to see that each piece is path-connected. If $\det A > 0$, then in (1.1) the scalar $d = \det A$ lies in $\mathbb{R}_{>0}$, which is path-connected, so the path of Step 1 from D to I may be taken with $d(t) > 0$ throughout; the concatenated path from A to I then satisfies $\det(\mathrm{diag}(d(t), 1, \ldots, 1)S(t)) = d(t) > 0$ at every time, so it stays in $\mathrm{GL}_n^+(\mathbb{R})$. Thus $\mathrm{GL}_n^+(\mathbb{R})$ is path-connected. For $\mathrm{GL}_n^-(\mathbb{R})$, fix the reflection $J = \mathrm{diag}(-1, 1, \ldots, 1)$, which has determinant -1. Given A with $\det A < 0$, the matrix $J^{-1}A$ has determinant > 0, hence is joined to I by a path γ in $\mathrm{GL}_n^+(\mathbb{R})$; then $t \mapsto J\gamma(t)$ joins J to A and has determinant $-\det\gamma(t) < 0$ throughout, so it lies in $\mathrm{GL}_n^-(\mathbb{R})$. Hence $\mathrm{GL}_n^-(\mathbb{R})$ is path-connected as well.

Finally, $\mathrm{GL}_n^+(\mathbb{R})$ is the kernel of the composite homomorphism $\mathrm{GL}_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R} \setminus \{0\} \xrightarrow{\mathrm{sgn}} \{\pm 1\}$, hence a normal subgroup; its two cosets are $\mathrm{GL}_n^+(\mathbb{R})$ and $\mathrm{GL}_n^-(\mathbb{R}) = J \cdot \mathrm{GL}_n^+(\mathbb{R})$, so it has index two. This exhibits the two path-connected pieces as the two components, completing the proof.

The disconnectedness of $\mathrm{GL}_n(\mathbb{R})$ is the matrix expression of orientation: the sign of the determinant records whether a linear automorphism of $\mathbb{R}^n$ preserves or reverses orientation, and no continuous deformation through invertible maps can turn one into the other without passing through a singular matrix. Over $\mathbb{C}$ this obstruction dissolves, because a single complex scalar can be routed around the origin; the punctured line $\mathbb{C} \setminus \{0\}$ is connected where $\mathbb{R} \setminus \{0\}$ is not. The same split will reappear for the orthogonal group in the next section, where $\mathrm{SO}(n)$ plays the role of the identity component.

Example 1.1.4: The Groups GL_1 and $\mathrm{GL}_2(\mathbb{R})$

For $n = 1$ a matrix is a scalar and invertibility means nonvanishing, so $\mathrm{GL}_1(\mathbb{F}) = \mathbb{F}^\times$, the multiplicative group of the field. Here proposition 1.1.3 recovers a familiar picture: $\mathrm{GL}_1(\mathbb{R}) = \mathbb{R}^\times$ is the disjoint union of the two half-lines $\mathbb{R}_{>0}$ and $\mathbb{R}_{<0}$, while $\mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$ is the connected punctured plane.

For $n = 2$ the group $\mathrm{GL}_2(\mathbb{R})$ is a four-dimensional manifold, the open subset of $M_2(\mathbb{R}) \cong \mathbb{R}^4$ where $ad - bc \neq 0$ for the matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Its identity component $\mathrm{GL}_2^+(\mathbb{R})$ contains every rotation $\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, of determinant 1, and every positive scaling. A reflection such as $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, of determinant -1, lies in the other component $\mathrm{GL}_2^-(\mathbb{R})$ and cannot be connected to the identity without its determinant passing through zero.

Two features of $\mathrm{GL}_n(\mathbb{F})$ separate it from the groups studied in the rest of the chapter and are worth recording now. It is *noncompact*: it contains the matrices $\mathrm{diag}(t, 1, \ldots, 1)$ for all $t \neq 0$, an unbounded family, so $\mathrm{GL}_n(\mathbb{F})$ is an unbounded subset of $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$ and cannot be compact. It is also *dense* in $M_n(\mathbb{F})$: any matrix can be perturbed to an invertible one, since the determinant of $A - tI$ is a nonzero polynomial in t and so vanishes for only finitely many t. The classical groups introduced next trade this openness for the opposite geometry: cut out by equations rather than inequalities, several of them are compact, and it is compactness that will make their representation theory in Part III so tractable.

Exercise 1.1.1

1. Show that $\mathrm{GL}_n(\mathbb{F})$ is dense in $M_n(\mathbb{F})$: given any $A \in M_n(\mathbb{F})$, prove that $A - tI$ is invertible for all but finitely many $t \in \mathbb{F}$, and deduce that every matrix is a limit of invertible ones.
2. Prove that $\mathrm{GL}_n(\mathbb{F})$ is not compact, in two ways: by exhibiting an unbounded sequence in it, and by showing it is a nonempty proper open subset of the connected space $M_n(\mathbb{F})$ and hence not closed.
3. Determine the center of $\mathrm{GL}_n(\mathbb{F})$. Show that a matrix commuting with every element of $\mathrm{GL}_n(\mathbb{F})$ (equivalently, with every transvection $I + E_{ij}$) is a scalar multiple λI of the identity with $\lambda \in \mathbb{F}^\times$.
4. Prove that $\det\colon \mathrm{GL}_n(\mathbb{F}) \to \mathbb{F}^\times$ is a surjective group homomorphism with kernel $\mathrm{SL}_n(\mathbb{F})$, and conclude that $\mathrm{SL}_n(\mathbb{F})$ is a normal subgroup with $\mathrm{GL}_n(\mathbb{F})/\mathrm{SL}_n(\mathbb{F}) \cong \mathbb{F}^\times$.
5. Using proposition 1.1.3, show that $A \mapsto \mathrm{sgn}(\det A)$ induces an isomorphism $\mathrm{GL}_n(\mathbb{R})/\mathrm{GL}_n^+(\mathbb{R}) \cong \{\pm 1\}$, and that $\mathrm{GL}_n^+(\mathbb{R})$ is exactly the set of products of transvections and positive scalings.

1.2 The Classical Groups

The general linear group is cut out of the space of all matrices by an *inequality*, $\det A \neq 0$, which is why it is open. The groups of present interest are cut out instead by *equations*: they are the subgroups of GL_n preserving some piece of structure, a determinant normalization, a symmetric or Hermitian inner product, or a skew form. Preserving structure is a closed condition, so these subgroups are closed rather than open, and several of them are compact. Each is presented here as the solution set of a system of polynomial (or, in the complex case, real-analytic) equations, and the uniform tool that converts such a solution set into a smooth manifold is the regular value theorem.

Recall the statement in the form used throughout: if $F\colon M \to N$ is a smooth map of manifolds and $c \in N$ is a *regular value*, meaning that the differential DF_p is surjective at every $p \in F^{-1}(c)$, then $F^{-1}(c)$ is an embedded submanifold of M of dimension $\dim M - \dim N$, with tangent space $T_pF^{-1}(c) = \ker DF_p$ at each of its points (see [2]). For each classical group the recipe is the same: write the group as $F^{-1}(c)$ for an explicit F into a vector space of "constraint values", check surjectivity of DF along the group by a single substitution, and read off the dimension. The closed-subgroup theorem of chapter 1 will later subsume all of these computations under one statement, that every closed subgroup of a Lie group is a Lie subgroup; the point of doing them by hand now is to make the manifold structure, and above all the dimension, of each principal example explicit and concrete.

Definition 1.2.1: Special Linear Group

The *special linear group* is the kernel of the determinant,

$$\mathrm{SL}_n(\mathbb{F}) = \{A \in \mathrm{GL}_n(\mathbb{F}) \mid \det A = 1\}$$

the group of matrices of determinant 1.

The determinant is the simplest structure a matrix can preserve, namely volume with orientation, and the special linear group is the first application of the regular value recipe.

Proposition 1.2.2: The Special Linear Group is a Submanifold

$\mathrm{SL}_n(\mathbb{R})$ is a closed embedded submanifold of $\mathrm{GL}_n(\mathbb{R})$ of dimension n^2-1, and $\mathrm{SL}_n(\mathbb{C})$ is a closed embedded submanifold of $\mathrm{GL}_n(\mathbb{C})$ of real dimension $2(n^2-1)$ (complex dimension n^2-1).

***Proof*:**

The determinant restricts to a smooth map $\det\colon \mathrm{GL}_n(\mathbb{F}) \to \mathbb{F}^\times$, and $\mathrm{SL}_n(\mathbb{F}) = \det^{-1}(1)$. The derivative of the determinant at A is computed from the expansion

$$\det(A+tX) = \det(A)\det(I+tA^{-1}X) = \det(A)\big(1 + t\operatorname{tr}(A^{-1}X) + O(t^2)\big)$$

valid for A invertible, where the middle equality factors out A and the last uses $\det(I+tB) = 1 + t\operatorname{tr}(B) + O(t^2)$. Differentiating at $t=0$ gives Jacobi's formula

$$D\det_A(X) = \det(A)\operatorname{tr}(A^{-1}X)$$

As X ranges over $M_n(\mathbb{F})$ the quantity $\operatorname{tr}(A^{-1}X)$ ranges over all of $\mathbb{F}$, for instance $X = AE_{11}$ gives $\operatorname{tr}(E_{11}) = 1$; hence $D\det_A$ is surjective onto $\mathbb{F}$ at every $A \in \mathrm{GL}_n(\mathbb{F})$, so every value, in particular 1, is regular. Therefore $\mathrm{SL}_n(\mathbb{F}) = \det^{-1}(1)$ is an embedded submanifold, closed because $\{1\}$ is closed, of dimension $\dim \mathrm{GL}_n(\mathbb{F}) - \dim \mathbb{F}$. Over $\mathbb{R}$ this is n^2-1; over $\mathbb{C}$, where det is a map of real manifolds onto $\mathbb{C} \cong \mathbb{R}^2$, the real codimension is 2, giving real dimension $2n^2-2$, completing the proof.

The next three groups preserve an inner product. Over $\mathbb{R}$ the relevant structure is a symmetric bilinear form; over $\mathbb{C}$, a Hermitian form; and the groups preserving them are the orthogonal and unitary groups. These are the compact classical groups, and the mechanism producing their compactness, that an isometry sends an orthonormal basis to an orthonormal one and so has bounded entries, is visible already in the proofs.

Definition 1.2.3: Orthogonal and Special Orthogonal Groups

The *orthogonal group* is the group of real matrices preserving the standard inner product on $\mathbb{R}^n$,

$$\mathrm{O}(n) = \left\{ A \in \mathrm{GL}_n(\mathbb{R}) \;\middle|\; A^\mathsf{T} A = I \right\}$$

equivalently the matrices whose columns form an orthonormal basis. The *special orthogonal group* is the subgroup of determinant 1,

$$\mathrm{SO}(n) = \mathrm{O}(n) \cap \mathrm{SL}_n(\mathbb{R}) = \{A \in \mathrm{O}(n) \mid \det A = 1\}$$

the group of rotations of $\mathbb{R}^n$.

The relation $A^\mathsf{T} A = I$ forces $(\det A)^2 = \det(A^\mathsf{T})\det(A) = 1$, so an orthogonal matrix has determinant ± 1 and the condition $\det A = 1$ picks out one half. The orthogonal group is the prototype for the regular value computation, carried out here in full detail.

Proposition 1.2.4: The Orthogonal Group is a Compact Submanifold of Dimension $\binom{n}{2}$

$\mathrm{O}(n)$ is a compact embedded submanifold of $M_n(\mathbb{R})$ of dimension $\binom{n}{2} = \frac{n(n-1)}{2}$. It is disconnected: the determinant takes the values ± 1 on it, and the two clopen pieces $\det^{-1}(1) = \mathrm{SO}(n)$ and $\det^{-1}(-1)$ are each a compact submanifold of the same dimension.

***Proof*:**
Let $\mathrm{Sym}_n(\mathbb{R}) = \left\{ S \in M_n(\mathbb{R}) \;\middle|\; S^\mathsf{T} = S \right\}$ be the space of symmetric matrices, a real vector space of dimension $\frac{n(n+1)}{2}$, and define

$$F\colon M_n(\mathbb{R}) \to \mathrm{Sym}_n(\mathbb{R}), \qquad F(A) = A^\mathsf{T} A$$

which lands in $\mathrm{Sym}_n(\mathbb{R})$ because $(A^\mathsf{T} A)^\mathsf{T} = A^\mathsf{T} A$, and is smooth because its entries are polynomials. By construction $\mathrm{O}(n) = F^{-1}(I)$.

Step 1: I is a regular value. Fix $A \in \mathrm{O}(n)$. The differential of F at A is

$$DF_A(X) = X^\mathsf{T} A + A^\mathsf{T} X$$

as one sees by expanding $F(A + tX) = A^\mathsf{T} A + t(X^\mathsf{T} A + A^\mathsf{T} X) + t^2 X^\mathsf{T} X$ and differentiating at $t = 0$. To see that DF_A is onto $\mathrm{Sym}_n(\mathbb{R})$, substitute $X = AZ$ and use $A^\mathsf{T} A = I$:

$$DF_A(AZ) = (AZ)^\mathsf{T} A + A^\mathsf{T}(AZ) = Z^\mathsf{T} A^\mathsf{T} A + A^\mathsf{T} A Z = Z^\mathsf{T} + Z$$

Given any symmetric S, taking $Z = \frac{1}{2} S$ yields $Z^\mathsf{T} + Z = S$; since $X \mapsto Z = A^{-1} X$ is a bijection of $M_n(\mathbb{R})$, the map DF_A is surjective. Hence I is a regular value.

Step 2: the manifold structure and dimension. By the regular value theorem $\mathrm{O}(n) = F^{-1}(I)$ is an embedded submanifold of $M_n(\mathbb{R})$ of dimension

$$\dim M_n(\mathbb{R}) - \dim \mathrm{Sym}_n(\mathbb{R}) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2} = \binom{n}{2}$$

Step 3: compactness. The set $\mathrm{O}(n)$ is closed in $M_n(\mathbb{R})$, being $F^{-1}(I)$ for continuous F. It is bounded: if $A^\intercal A = I$ then the diagonal entries of $A^\intercal A$ give $\sum_i a_{ij}^2 = 1$ for each column j, so the Frobenius norm satisfies $\|A\|^2 = \sum_{i,j} a_{ij}^2 = n$. A closed bounded subset of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$ is compact by the Heine–Borel theorem.

Step 4: the two pieces. The determinant is continuous on $\mathrm{O}(n)$ and takes only the values ± 1, both attained (I has determinant 1 and $\mathrm{diag}(-1, 1, \ldots, 1)$ has determinant -1), so $\det^{-1}(1) = \mathrm{SO}(n)$ and $\det^{-1}(-1)$ are disjoint, nonempty, and both open and closed in $\mathrm{O}(n)$. Each is a submanifold of dimension $\binom{n}{2}$, being open in $\mathrm{O}(n)$, and each is compact, being closed in the compact space $\mathrm{O}(n)$. That $\mathrm{SO}(n)$ is connected, so that these two pieces are exactly the connected components, is proposition 1.2.10 below, completing the proof.

The complex analogue replaces the transpose by the conjugate transpose and the symmetric form by a Hermitian one. Write $A^* = \overline{A}^\intercal$ for the conjugate transpose.

Definition 1.2.5: Unitary and Special Unitary Groups

The *unitary group* is the group of complex matrices preserving the standard Hermitian inner product on $\mathbb{C}^n$,

$$\mathrm{U}(n) = \{A \in \mathrm{GL}_n(\mathbb{C}) \mid A^*A = I\}$$

the matrices whose columns form a Hermitian-orthonormal basis. The *special unitary group* is the subgroup of determinant 1,

$$\mathrm{SU}(n) = \mathrm{U}(n) \cap \mathrm{SL}_n(\mathbb{C}) = \{A \in \mathrm{U}(n) \mid \det A = 1\}$$

The relation $A^*A = I$ gives $|\det A|^2 = \overline{\det A} \det A = \det(A^*) \det(A) = 1$, so a unitary matrix has determinant on the unit circle, and $\mathrm{SU}(n)$ is the fibre of the determinant over $1 \in S^1$.

Proposition 1.2.6: The Unitary and Special Unitary Groups are Compact Submanifolds

$\mathrm{U}(n)$ is a compact embedded submanifold of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$ of real dimension n^2. Its subgroup $\mathrm{SU}(n)$ is a compact embedded submanifold of real dimension $n^2 - 1$.

***Proof*:**
Let $\mathrm{Herm}_n = \{H \in M_n(\mathbb{C}) \mid H^* = H\}$ be the space of Hermitian matrices. This is a *real* vector space (it is not closed under multiplication by i) of real dimension n^2: a Hermitian matrix has n real diagonal entries and $\binom{n}{2}$ free complex entries above the diagonal, contributing $n + 2\binom{n}{2} = n + n(n-1) = n^2$ real parameters. Define

$$F \colon M_n(\mathbb{C}) \to \mathrm{Herm}_n, \qquad F(A) = A^*A$$

a smooth map (its entries are real-polynomial in the real and imaginary parts of the entries of

A) with $\mathrm{U}(n) = F^{-1}(I)$.

Step 1: I is a regular value. For $A \in \mathrm{U}(n)$ the differential is $DF_A(X) = X^*A + A^*X$, and the substitution $X = AZ$ with $A^*A = I$ gives

$$DF_A(AZ) = (AZ)^*A + A^*(AZ) = Z^*A^*A + A^*AZ = Z^* + Z$$

Given Hermitian H, taking $Z = \frac{1}{2}H$ gives $Z^* + Z = H$, so DF_A is surjective onto Herm_n; hence I is a regular value and $\mathrm{U}(n)$ is an embedded submanifold of real dimension $2n^2 - n^2 = n^2$.

Step 2: compactness. As with the orthogonal group, $\mathrm{U}(n) = F^{-1}(I)$ is closed, and the columns of a unitary matrix are unit vectors, so $\|A\|^2 = n$ and $\mathrm{U}(n)$ is bounded; it is therefore compact by Heine–Borel.

Step 3: the special unitary group. By the regular value theorem the tangent space at $A \in \mathrm{U}(n)$ is $T_A\,\mathrm{U}(n) = \ker DF_A = \{AZ \mid Z^* + Z = 0\}$, that is $A \cdot \mathfrak{u}(n)$ where $\mathfrak{u}(n)$ is the space of skew-Hermitian matrices. Consider the smooth map $\det\colon \mathrm{U}(n) \to S^1$. Its derivative at A, along a tangent vector $X = AZ$ with Z skew-Hermitian, is by Jacobi's formula

$$D\underset{A}{\det}(AZ) = \det(A)\,\mathrm{tr}(A^{-1}AZ) = \det(A)\,\mathrm{tr}(Z)$$

A skew-Hermitian matrix has purely imaginary diagonal, so $\mathrm{tr}(Z) \in i\mathbb{R}$, and $\mathrm{tr}(Z)$ takes every value in $i\mathbb{R}$ (for example $Z = \mathrm{diag}(it, 0, \dots, 0)$). Thus $D\det_A$ maps $T_A\,\mathrm{U}(n)$ onto $\det(A) \cdot i\mathbb{R}$, which is exactly the tangent line to S^1 at the point $\det(A)$. Hence $\det\colon \mathrm{U}(n) \to S^1$ is a submersion, 1 is a regular value, and $\mathrm{SU}(n) = \det^{-1}(1)$ is an embedded submanifold of $\mathrm{U}(n)$ of dimension $n^2 - 1$. It is closed in the compact group $\mathrm{U}(n)$, hence compact, as we sought to show.

The last family preserves a skew-symmetric form. Its notation is a persistent source of confusion, so the two groups sharing the name "symplectic" are fixed side by side, together with a convention.

Definition 1.2.7: Symplectic Groups

Let $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right) \in M_{2n}(\mathbb{F})$, the matrix of the standard skew form $\omega(x, y) = x^{\mathsf{T}} J y$ on $\mathbb{F}^{2n}$. The *symplectic group* is the group preserving ω,

$$\mathrm{Sp}(2n, \mathbb{F}) = \left\{ A \in \mathrm{GL}_{2n}(\mathbb{F}) \;\middle|\; A^{\mathsf{T}} J A = J \right\}$$

The *compact symplectic group* is the group of quaternionic-unitary matrices,

$$\mathrm{Sp}(n) = \{A \in M_n(\mathbb{H}) \mid A^*A = I\}$$

where $\mathbb{H}$ is the quaternions and $A^* = \overline{A}^{\mathsf{T}}$ is the quaternionic conjugate transpose.

1.2.8 Convention: Two Meanings of "Symplectic Group" The symbol $\mathrm{Sp}(2n, \mathbb{F})$ denotes the (noncompact) group of $2n \times 2n$ matrices over $\mathbb{F}$ preserving the standard skew form; the symbol $\mathrm{Sp}(n)$ denotes the (compact) group of $n \times n$ quaternionic matrices preserving the standard quaternionic-Hermitian form. Both have dimension $n(2n+1)$, but they are different groups: $\mathrm{Sp}(2n, \mathbb{R})$ is noncompact and $\mathrm{Sp}(n)$ is compact. The compact group is a real form of the complex group $\mathrm{Sp}(2n, \mathbb{C})$; concretely, under the standard identification $\mathbb{H}^n \cong \mathbb{C}^{2n}$ one has $\mathrm{Sp}(n) = \mathrm{U}(2n) \cap \mathrm{Sp}(2n, \mathbb{C})$, the verification of which is a routine matter of unwinding the quaternionic structure, taken up where the compact group is needed in Part III. Throughout this book $\mathrm{Sp}(2n, \mathbb{F})$ always carries an explicit field and even size $2n$; the bare symbol $\mathrm{Sp}(n)$ always means the compact group.

The noncompact group is another instance of the regular value recipe, with the skew-symmetric matrices as the space of constraint values.

Proposition 1.2.9: The Symplectic Group is a Submanifold of Dimension $n(2n+1)$

$\mathrm{Sp}(2n, \mathbb{F})$ is a closed embedded submanifold of $\mathrm{GL}_{2n}(\mathbb{F})$ of dimension $n(2n+1)$ over $\mathbb{R}$ (and complex dimension $n(2n+1)$ over $\mathbb{C}$). The compact group $\mathrm{Sp}(n)$ is a compact subgroup of $\mathrm{U}(2n)$, hence compact, of dimension $n(2n+1)$.

***Proof*:**
Let $\mathrm{Skew}_{2n}(\mathbb{F}) = \{S \in M_{2n}(\mathbb{F}) \mid S^\mathsf{T} = -S\}$, of dimension $\binom{2n}{2} = n(2n-1)$, and define $F\colon M_{2n}(\mathbb{F}) \to \mathrm{Skew}_{2n}(\mathbb{F})$ by $F(A) = A^\mathsf{T} J A$; this is skew-symmetric because $J^\mathsf{T} = -J$, and $\mathrm{Sp}(2n, \mathbb{F}) = F^{-1}(J)$. The differential at A is $DF_A(X) = X^\mathsf{T} J A + A^\mathsf{T} J X$, and for $A \in \mathrm{Sp}(2n, \mathbb{F})$ the substitution $X = AZ$ with $A^\mathsf{T} J A = J$ gives

$$DF_A(AZ) = Z^\mathsf{T} A^\mathsf{T} J A + A^\mathsf{T} J A Z = Z^\mathsf{T} J + J Z$$

Given skew-symmetric S, set $Z = -\frac{1}{2} J S$. Using $J^\mathsf{T} = -J$, $J^2 = -I$, and $S^\mathsf{T} = -S$, the transpose is $Z^\mathsf{T} = -\frac{1}{2} S^\mathsf{T} J^\mathsf{T} = -\frac{1}{2}(-S)(-J) = -\frac{1}{2} S J$, so

$$Z^\mathsf{T} J + J Z = -\tfrac{1}{2} S J^2 - \tfrac{1}{2} J^2 S = \tfrac{1}{2} S + \tfrac{1}{2} S = S$$

Hence DF_A is surjective onto $\mathrm{Skew}_{2n}(\mathbb{F})$, the value J is regular, and $\mathrm{Sp}(2n, \mathbb{F}) = F^{-1}(J)$ is a closed embedded submanifold of dimension

$$(2n)^2 - n(2n-1) = 4n^2 - 2n^2 + n = 2n^2 + n = n(2n+1)$$

For the compact group, the identification $\mathbb{H}^n \cong \mathbb{C}^{2n}$ of box 1.2.8 realizes $\mathrm{Sp}(n)$ as $\mathrm{U}(2n) \cap \mathrm{Sp}(2n, \mathbb{C})$, a closed subset of the compact group $\mathrm{U}(2n)$, hence compact; its dimension $n(2n+1)$ is computed from its Lie algebra in chapter 1. This completes the proof.

It remains to settle the connectivity left open for $\mathrm{SO}(n)$, $\mathrm{U}(n)$, and $\mathrm{SU}(n)$. Each is proved by the same idea: diagonalize, or bring to real normal form, and then contract the eigenvalues to 1 along a path that stays inside the group.

Proposition 1.2.10: Connectivity of the Compact Classical Groups

The groups $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and $\mathrm{SO}(n)$ are path-connected. Consequently $\mathrm{O}(n)$ has exactly two connected components, $\mathrm{SO}(n)$ and $\det^{-1}(-1)$.

***Proof*:**
Step 1: the unitary group. A unitary matrix is normal, so by the spectral theorem it is unitarily diagonalizable: $A = PDP^{-1}$ with $P \in \mathrm{U}(n)$ and $D = \mathrm{diag}(e^{i\theta_1}, \ldots, e^{i\theta_n})$ for real $\theta_1, \ldots, \theta_n$, the eigenvalues lying on the unit circle because A is unitary. The path $D(s) = \mathrm{diag}(e^{is\theta_1}, \ldots, e^{is\theta_n})$ for $s \in [0, 1]$ runs inside $\mathrm{U}(n)$ from $D(1) = D$ to $D(0) = I$, and conjugating, $s \mapsto PD(s)P^{-1}$ is a path in $\mathrm{U}(n)$ from A to I. Hence $\mathrm{U}(n)$ is path-connected.

Step 2: the special unitary group. For $A \in \mathrm{SU}(n)$ diagonalize as above; now $\det A = \prod_j e^{i\theta_j} = 1$, so $\sum_j \theta_j \in 2\pi\mathbb{Z}$, and subtracting a suitable integer multiple of 2π from one of the angles (which does not change the corresponding eigenvalue) arranges $\sum_j \theta_j = 0$ exactly. Then the same path $D(s)$ satisfies $\det D(s) = e^{is\sum_j \theta_j} = 1$ for all s, so it lies in $\mathrm{SU}(n)$, and $s \mapsto PD(s)P^{-1}$ joins A to I within $\mathrm{SU}(n)$. Hence $\mathrm{SU}(n)$ is path-connected.

Step 3: the special orthogonal group. A matrix $A \in \mathrm{SO}(n)$ is real orthogonal, hence normal, and the real spectral theorem for orthogonal matrices provides $Q \in \mathrm{SO}(n)$ with $Q^{-1}AQ$ block-diagonal, the blocks being 2×2 rotations $R(\theta_k) = \left(\begin{smallmatrix} \cos\theta_k & -\sin\theta_k \\ \sin\theta_k & \cos\theta_k \end{smallmatrix}\right)$ together with diagonal entries ± 1; the real eigenvalues ± 1 occur because they are the only real numbers on the unit circle, and the number of -1 entries is even because $\det A = 1$, so those entries pair up into rotation blocks $R(\pi) = \mathrm{diag}(-1, -1)$. Thus $Q^{-1}AQ$ is a direct sum of rotation blocks $R(\theta_1), \ldots, R(\theta_m)$ and a $+1$ identity block. Replacing each angle θ_k by $s\theta_k$ gives a path of block matrices in $\mathrm{SO}(n)$ from $Q^{-1}AQ$ (at $s = 1$) to I (at $s = 0$), and conjugating by Q produces a path in $\mathrm{SO}(n)$ from A to I. Hence $\mathrm{SO}(n)$ is path-connected.

Since $\mathrm{SO}(n) = \det^{-1}(1)$ is path-connected and $\det^{-1}(-1) = \mathrm{diag}(-1, 1, \ldots, 1) \cdot \mathrm{SO}(n)$ is its homeomorphic image under left multiplication, both clopen pieces of $\mathrm{O}(n)$ from proposition 1.2.4 are connected, so they are precisely the two components of $\mathrm{O}(n)$, completing the proof.

The dimensions and topological types of the classical groups are collected in fig. 1.1 for reference; the entries for $\mathrm{O}(n)$ and $\mathrm{U}(n)$ recur constantly in Part III, where the representation theory of the compact groups is developed. Every entry has been established above, with one exception: the connectedness of the two symplectic groups is recorded as a standard fact, to be proved from the polar decomposition of Part IV, where $\mathrm{Sp}(2n, \mathbb{R})$ and $\mathrm{Sp}(n)$ are analyzed alongside the other real forms.

Group	Defining relation	Field	$\dim_{\mathbb{R}}$	Topology
$\mathrm{GL}_n(\mathbb{R})$	$\det \neq 0$	$\mathbb{R}$	n^2	open, 2 components
$\mathrm{GL}_n(\mathbb{C})$	$\det \neq 0$	$\mathbb{C}$	$2n^2$	open, connected
$\mathrm{SL}_n(\mathbb{R})$	$\det = 1$	$\mathbb{R}$	$n^2 - 1$	closed, connected
$\mathrm{SL}_n(\mathbb{C})$	$\det = 1$	$\mathbb{C}$	$2(n^2 - 1)$	closed, connected
$\mathrm{O}(n)$	$A^{\mathsf{T}}A = I$	$\mathbb{R}$	$\binom{n}{2}$	compact, 2 components
$\mathrm{SO}(n)$	$A^{\mathsf{T}}A = I,\ \det = 1$	$\mathbb{R}$	$\binom{n}{2}$	compact, connected
$\mathrm{U}(n)$	$A^*A = I$	$\mathbb{C}$	n^2	compact, connected
$\mathrm{SU}(n)$	$A^*A = I,\ \det = 1$	$\mathbb{C}$	$n^2 - 1$	compact, connected
$\mathrm{Sp}(2n, \mathbb{R})$	$A^{\mathsf{T}}JA = J$	$\mathbb{R}$	$n(2n+1)$	closed, connected
$\mathrm{Sp}(n)$	$A^*A = I$ (over $\mathbb{H}$)	$\mathbb{H}$	$n(2n+1)$	compact, connected

Figure 1.1: The classical groups, their defining relations, real dimensions, and topological types

The smallest orthogonal and unitary groups already exhibit the two shapes that recur throughout: a circle and its higher analogues.

Example 1.2.11: The Circle Groups $\mathrm{SO}(2)$ and $\mathrm{U}(1)$

The group $\mathrm{SO}(2)$ consists of the plane rotations

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in \mathbb{R}$$

and $\theta \mapsto R(\theta)$ is a surjective homomorphism $\mathbb{R} \to \mathrm{SO}(2)$ with kernel $2\pi\mathbb{Z}$, so $\mathrm{SO}(2) \cong \mathbb{R}/2\pi\mathbb{Z} \cong S^1$, a circle, of dimension $\binom{2}{2} = 1$ as predicted. Likewise $\mathrm{U}(1) = \{z \in \mathbb{C} \mid |z| = 1\} = S^1$ is the unit circle in $\mathbb{C}$, of real dimension 1, and $z \mapsto \left(\begin{smallmatrix} \operatorname{Re} z & -\operatorname{Im} z \\ \operatorname{Im} z & \operatorname{Re} z \end{smallmatrix}\right)$ is an isomorphism $\mathrm{U}(1) \xrightarrow{\sim} \mathrm{SO}(2)$ realizing the standard fact that multiplication by a unit complex number is a rotation of the plane. The next dimension up, the group $\mathrm{SU}(2)$, turns out to be a 3-sphere and is the subject of section 1.5.

Exercise 1.2.1

1. Verify directly that $\mathrm{SO}(3)$ has dimension 3 and $\mathrm{SU}(2)$ has dimension 3, and that $\mathrm{O}(1) = \{\pm 1\}$ is zero-dimensional with two components, consistent with $\binom{1}{2} = 0$.
2. Show that $\det\colon \mathrm{U}(n) \to S^1$ is a surjective group homomorphism with kernel $\mathrm{SU}(n)$, and deduce that $\mathrm{U}(n)/\mathrm{SU}(n) \cong S^1$.
3. Prove that the center of $\mathrm{SU}(n)$ is the group of scalar matrices ζI with $\zeta^n = 1$, so that $Z(\mathrm{SU}(n)) \cong \mu_n$, the group of nth roots of unity.
4. Show that $A \in \mathrm{Sp}(2n, \mathbb{F})$ if and only if A preserves the skew form $\omega(x, y) = x^{\mathsf{T}}Jy$, that is $\omega(Ax, Ay) = \omega(x, y)$ for all $x, y \in \mathbb{F}^{2n}$, and conclude that $\det A = 1$ for every symplectic matrix.
5. Prove that
$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \;\middle|\; \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$$
and deduce that $\mathrm{SU}(2)$ is diffeomorphic to the unit sphere $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$.

1.3 The Matrix Exponential and Logarithm; One-Parameter Subgroups

The classical groups of the previous section were described statically, as solution sets of matrix equations. The subject acquires its dynamics from a single construction that turns a matrix into a motion: the exponential of a matrix X is a point of $\mathrm{GL}_n(\mathbb{F})$, and letting X vary linearly, through tX for $t \in \mathbb{R}$, sweeps out a curve through the identity whose velocity at time zero is X itself. This is the bridge between the linear world of $M_n(\mathbb{F})$, where one adds and takes brackets, and the curved world of the group, where one multiplies. Everything in the group theory of the following chapters is read off this bridge: the tangent space at the identity becomes a Lie algebra, and the exponential map carries its linear structure onto the group.

The construction copies the scalar power series $e^x = \sum_{k \geq 0} x^k/k!$ verbatim, replacing the scalar x by a matrix X and scalar powers by matrix powers. The first task is to make sense of the resulting infinite sum of matrices, and the tool for that is the operator norm, whose submultiplicativity tames every convergence question in this section.

Definition 1.3.1: Operator Norm on $M_n(\mathbb{F})$

The *operator norm* of $X \in M_n(\mathbb{F})$ is

$$\|X\| = \sup_{v \neq 0} \frac{\|Xv\|}{\|v\|} = \sup_{\|v\|=1} \|Xv\|$$

where $\|v\|$ is the Euclidean norm on $\mathbb{F}^n$. It makes $M_n(\mathbb{F})$ a normed vector space, and the supremum is finite and attained because the unit sphere of $\mathbb{F}^n$ is compact and $v \mapsto \|Xv\|$ is continuous.

Two properties of this norm carry the entire section. First, it dominates each matrix entry, so convergence in operator norm is the same as entrywise convergence and hence as convergence in $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$ with its usual topology; a sequence of matrices converges in operator norm if and only if each of its n^2 entry-sequences converges. Second, and this is the property with no scalar analogue, the norm is *submultiplicative*:

$$\|XY\| \leq \|X\|\,\|Y\| \qquad \text{for all } X, Y \in M_n(\mathbb{F})$$

This holds because $\|XYv\| \leq \|X\|\,\|Yv\| \leq \|X\|\,\|Y\|\,\|v\|$ for every v; iterating gives $\|X^k\| \leq \|X\|^k$, which is what forces the exponential series to converge as fast as its scalar shadow.

Definition 1.3.2: Matrix Exponential

For $X \in M_n(\mathbb{F})$ the *matrix exponential* is the sum of the series

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots$$

where $X^0 = I$ by convention. It defines a map $\exp\colon M_n(\mathbb{F}) \to M_n(\mathbb{F})$.

For the definition to be legitimate the series must converge, and it must converge to a matrix depending smoothly on X; both follow from the norm estimates just recorded.

Proposition 1.3.3: Convergence and Smoothness of the Exponential

For every $X \in M_n(\mathbb{F})$ the series $\sum_{k\geq 0} X^k/k!$ converges absolutely in $M_n(\mathbb{F})$, so $\exp(X)$ is a well-defined matrix, and it satisfies the bound $\|\exp(X)\| \leq e^{\|X\|}$. Moreover the map $\exp\colon M_n(\mathbb{F}) \to M_n(\mathbb{F})$ is smooth, and its value at 0 is $\exp(0) = I$.

***Proof*:**

The space $M_n(\mathbb{F}) \cong \mathbb{F}^{n^2}$ is a finite-dimensional normed vector space, hence complete, so a series converges as soon as it converges absolutely; it therefore suffices to bound the norms of the terms. Submultiplicativity gives $\|X^k\| \leq \|X\|^k$, so

$$\sum_{k=0}^{\infty} \left\| \frac{X^k}{k!} \right\| = \sum_{k=0}^{\infty} \frac{\|X^k\|}{k!} \leq \sum_{k=0}^{\infty} \frac{\|X\|^k}{k!} = e^{\|X\|}$$

The right-hand series is the convergent scalar exponential of the real number $\|X\|$, so the series of norms converges; the matrix series converges absolutely, and its sum $\exp(X)$ has norm at most $e^{\|X\|}$ by the triangle inequality applied to the partial sums.

For smoothness, each partial sum $S_N(X) = \sum_{k=0}^{N} X^k/k!$ is a polynomial map $M_n(\mathbb{F}) \to M_n(\mathbb{F})$, hence smooth. On any ball $\|X\| \leq r$ the tail is bounded uniformly by $\sum_{k>N} r^k/k!$, which tends to 0 as $N \to \infty$ independently of X; the same estimate applied to the series of any fixed partial derivative (itself an exponential-type series in the entries, dominated in the same way) shows that all partial derivatives converge uniformly on compact sets. A limit of smooth maps whose derivatives to every order converge uniformly on compact sets is smooth, so exp is smooth. Setting $X = 0$ leaves only the $k = 0$ term, $\exp(0) = I$, completing the proof.

The bound $\|\exp(X)\| \leq e^{\|X\|}$ is the running estimate of the section: it lets one differentiate the series term by term, rearrange it, and multiply two such series together, all justified by absolute convergence. The algebraic properties collected next are exactly those of the scalar exponential that survive the passage to matrices, and the one that does not survive, the additivity $e^{x+y} = e^x e^y$, is isolated afterward because its failure is the source of the bracket.

Proposition 1.3.4: Algebraic Properties of the Exponential

For all $X \in M_n(\mathbb{F})$ and all $s, t \in \mathbb{R}$ (or $\mathbb{C}$) the matrix exponential satisfies:

1. $\exp(0) = I$, and $t \mapsto \exp(tX)$ is a smooth curve in $M_n(\mathbb{F})$ with $\dfrac{d}{dt}\exp(tX) = X\exp(tX) = \exp(tX)X$.
2. $\exp((s+t)X) = \exp(sX)\exp(tX)$; in particular $\exp(X)$ is invertible with inverse $\exp(X)^{-1} = \exp(-X)$, so $\exp(X) \in \mathrm{GL}_n(\mathbb{F})$.
3. For every $g \in \mathrm{GL}_n(\mathbb{F})$, conjugation is intertwined: $\exp(gXg^{-1}) = g\,\exp(X)\,g^{-1}$.
4. The determinant of the exponential is the exponential of the trace: $\det\exp(X) = e^{\operatorname{tr}X}$.

***Proof*:**
We prove the four statements in turn; each rests on the absolute convergence of proposition 1.3.3, which permits term-by-term differentiation and the rearrangement of the double series obtained from a product.

1. That $\exp(0) = I$ is proposition 1.3.3. Differentiating the series $\exp(tX) = \sum_k t^k X^k/k!$ term by term, legitimate because the differentiated series $\sum_k t^{k-1}X^k/(k-1)!$ converges uniformly on compact t-intervals by the same domination as before, gives
$$\frac{d}{dt}\exp(tX) = \sum_{k=1}^{\infty} \frac{t^{k-1}X^k}{(k-1)!} = X\sum_{j=0}^{\infty}\frac{t^jX^j}{j!} = X\exp(tX)$$
where the third equality reindexes $j = k-1$ and factors one X to the left; factoring it to the right instead gives $\exp(tX)X$, and the two agree because X commutes with every power of itself.
2. Fix X and consider $f(t) = \exp((s+t)X)$ and $h(t) = \exp(sX)\exp(tX)$ as functions of t. By part (1) both satisfy the same linear differential equation $y'(t) = Xy(t)$ with the same initial value $y(0) = \exp(sX)$: for f this is the chain rule applied to part (1), and for h it is part (1) applied to the right-hand factor with $\exp(sX)$ constant. The initial value problem $y' = Xy$, $y(0) = \exp(sX)$ has a unique solution (its difference from any other solution has derivative X times itself and vanishes at 0, hence vanishes identically, a point proved in full in theorem 1.3.10), so $f = h$; that is $\exp((s+t)X) = \exp(sX)\exp(tX)$. Setting $t = -s$ and using $\exp(0) = I$ gives $\exp(sX)\exp(-sX) = I$, so $\exp(sX)$ is invertible with the stated inverse; taking $s = 1$ puts $\exp(X)$ in $\mathrm{GL}_n(\mathbb{F})$.
3. Since $(gXg^{-1})^k = gX^kg^{-1}$ for every k, and conjugation $A \mapsto gAg^{-1}$ is a continuous linear map, it commutes with the convergent sum:
$$\exp(gXg^{-1}) = \sum_{k=0}^{\infty}\frac{(gXg^{-1})^k}{k!} = \sum_{k=0}^{\infty}\frac{gX^kg^{-1}}{k!} = g\left(\sum_{k=0}^{\infty}\frac{X^k}{k!}\right)g^{-1} = g\,\exp(X)\,g^{-1}$$
4. Consider $\varphi(t) = \det\exp(tX)$, a smooth real (or complex) function of t. By Jacobi's formula

for the derivative of the determinant (established in proposition 1.2.2), and part (1),

$$\varphi'(t) = D_{\exp(tX)}\det\left(\frac{d}{dt}\exp(tX)\right) = \det\big(\exp(tX)\big)\operatorname{tr}\big(\exp(tX)^{-1}X\exp(tX)\big) = \varphi(t)\operatorname{tr}(X)$$

where the last equality uses that trace is invariant under conjugation, $\operatorname{tr}(\exp(tX)^{-1}X\exp(tX)) = \operatorname{tr}(X)$. Thus φ solves the scalar equation $\varphi' = (\operatorname{tr}X)\varphi$ with $\varphi(0) = \det I = 1$, whose unique solution is $\varphi(t) = e^{t\operatorname{tr}X}$. Setting $t = 1$ gives $\det\exp(X) = e^{\operatorname{tr}X}$, completing the proof.

Part (2) is the statement that exp lands in the group and, restricted to a single line $\mathbb{R}X$, is a homomorphism from the additive group $\mathbb{R}$ to $\mathrm{GL}_n(\mathbb{F})$; this is the germ of the one-parameter subgroups taken up at the end of the section. Part (4) already pays a concrete dividend: it identifies the trace-zero matrices as exactly those whose exponentials have determinant 1, so the exponential of the space of trace-zero matrices lands in $\mathrm{SL}_n(\mathbb{F})$, the first instance of the pattern by which the defining equation of a classical group is the linearization of the group condition. What part (2) does *not* say is that $\exp(X+Y) = \exp(X)\exp(Y)$ for two different matrices X, Y; that identity requires the two to commute, and its failure is the mechanism behind the entire theory.

Proposition 1.3.5: Exponential of a Sum of Commuting Matrices

Let $X, Y \in M_n(\mathbb{F})$. If $XY = YX$ then

$$\exp(X+Y) = \exp(X)\exp(Y) = \exp(Y)\exp(X)$$

The commutativity hypothesis cannot be dropped: there exist X, Y with $\exp(X+Y) \neq \exp(X)\exp(Y)$.

***Proof*:**
Step 1: the commuting case. Suppose $XY = YX$. Both series converge absolutely by proposition 1.3.3, so their product may be expanded as a double series and summed in any order, in particular grouped by the total degree $m = j + k$:

$$\exp(X)\exp(Y) = \left(\sum_{j=0}^{\infty}\frac{X^j}{j!}\right)\left(\sum_{k=0}^{\infty}\frac{Y^k}{k!}\right) = \sum_{m=0}^{\infty}\sum_{j+k=m}\frac{X^jY^k}{j!\,k!} = \sum_{m=0}^{\infty}\frac{1}{m!}\sum_{j=0}^{m}\binom{m}{j}X^jY^{m-j}$$

The inner sum is the binomial expansion of $(X+Y)^m$, which is valid *precisely because X and Y commute*: only then do the factors in $(X+Y)^m$ reorder into powers of X times powers of Y with binomial coefficients. Hence the last sum is $\sum_m (X+Y)^m/m! = \exp(X+Y)$. The same computation with the roles of X and Y interchanged gives $\exp(Y)\exp(X) = \exp(Y+X) = \exp(X+Y)$, so all three agree.

Step 2: a non-commuting counterexample. Take the nilpotent matrices

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

which do not commute, since $XY = \operatorname{diag}(1,0)$ while $YX = \operatorname{diag}(0,1)$. As $X^2 = Y^2 = 0$, the series truncate: $\exp(X) = I + X = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\exp(Y) = I + Y = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, so

$$\exp(X)\exp(Y) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

On the other hand $X + Y = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ satisfies $(X+Y)^2 = I$, so its exponential sums the even and odd parts to hyperbolic functions,

$$\exp(X+Y) = \cosh(1)\, I + \sinh(1)\,(X+Y) = \begin{pmatrix} \cosh 1 & \sinh 1 \\ \sinh 1 & \cosh 1 \end{pmatrix}$$

with $\cosh 1 \approx 1.543$ and $\sinh 1 \approx 1.175$. These two matrices are unequal, so $\exp(X+Y) \neq \exp(X)\exp(Y)$, as we sought to show.

The gap between $\exp(X+Y)$ and $\exp(X)\exp(Y)$ measures the failure of X and Y to commute, and unwinding it to second order will produce the commutator $[X,Y] = XY - YX$ in the next section; the Baker-Campbell-Hausdorff formula of Chapter 3 is the exact accounting of this gap to all orders. For now the counterexample serves as a warning label on part (2) of proposition 1.3.4: along a single line the exponential is a homomorphism, but the map on all of $M_n(\mathbb{F})$ is not.

Having built exp, one wants to invert it, both to recover X from $\exp(X)$ and to introduce coordinates on the group near the identity. The inverse is the matrix logarithm, defined by the series for $\log(1+x)$ with the same substitution of a matrix for the scalar. Because that scalar series has radius of convergence 1, the matrix version converges only near the identity, which is exactly the regime needed.

Proposition 1.3.6: The Matrix Logarithm

For $A \in M_n(\mathbb{F})$ with $\|A\| < 1$ the series

$$\log(I+A) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} A^k = A - \frac{A^2}{2} + \frac{A^3}{3} - \cdots$$

converges absolutely and defines a smooth map on the open unit ball $\{A \in M_n(\mathbb{F}) \mid \|A\| < 1\}$. It inverts the exponential where both are defined: $\exp\big(\log(I+A)\big) = I + A$ for $\|A\| < 1$, and $\log\big(\exp(X)\big) = X$ whenever $\|\exp(X) - I\| < 1$, in particular whenever $\|X\| < \log 2$.

***Proof*:**
Step 1: convergence and smoothness. For $\|A\| < 1$, submultiplicativity gives $\|A^k\| \leq \|A\|^k$, so

$$\sum_{k=1}^{\infty} \left\| \frac{(-1)^{k+1}}{k} A^k \right\| \leq \sum_{k=1}^{\infty} \frac{\|A\|^k}{k} = -\log(1 - \|A\|) < \infty$$

so the series converges absolutely, and the same domination on the ball $\|A\| \leq r < 1$ makes the convergence, together with that of every differentiated series, uniform on compact subsets; the limit is therefore smooth by the argument of proposition 1.3.3.

Step 2: the two composites are formal inverses. The scalar identities

$$\exp\big(\log(1+a)\big) = 1+a, \qquad \log\big(\exp(x)\big) = x$$

hold as identities of formal power series in a single indeterminate, because exp and log are inverse as formal power series over $\mathbb{Q}$. Both composites $\exp\circ\log$ and $\log\circ\exp$, applied to a *single* matrix A (respectively X), only ever combine powers of that one matrix, which commute with one another; so every rearrangement used to verify the formal identity is a rearrangement of an absolutely convergent series of commuting matrices, valid by proposition 1.3.5 and Step 1. Substituting the matrix for the indeterminate therefore transports the two formal identities to genuine matrix identities on their domains of convergence: $\exp(\log(I+A)) = I + A$ for $\|A\| < 1$, and $\log(\exp(X)) = X$ whenever $\|\exp(X) - I\| < 1$.

Step 3: the explicit domain. If $\|X\| < \log 2$ then, using $\exp(X) - I = \sum_{k\geq 1} X^k/k!$ and $\|X^k\| \leq \|X\|^k$,

$$\|\exp(X) - I\| \leq \sum_{k=1}^{\infty} \frac{\|X\|^k}{k!} = e^{\|X\|} - 1 < e^{\log 2} - 1 = 1$$

so the hypothesis $\|\exp(X) - I\| < 1$ of Step 2 is met and $\log(\exp(X)) = X$, completing the proof.

The logarithm inverts the exponential on overlapping neighborhoods, and the two maps are smooth; what remains is to package this into the statement that exp is a diffeomorphism near 0, which supplies the analytic charts on the group used throughout the book. The engine is the value of the differential of exp at the origin, and it is the simplest possible value.

Theorem 1.3.7: The Exponential is a Local Diffeomorphism at the Origin

The differential of $\exp\colon M_n(\mathbb{F}) \to M_n(\mathbb{F})$ at 0 is the identity, $D\exp_0 = \mathrm{id}_{M_n(\mathbb{F})}$. Consequently there are open neighborhoods U of 0 in $M_n(\mathbb{F})$ and V of I in $\mathrm{GL}_n(\mathbb{F})$ such that

$$\exp\colon U \to V \qquad \text{is a diffeomorphism, with inverse } \log\colon V \to U$$

***Proof*:**
The differential $D\exp_0$ acts on a tangent vector $X \in M_n(\mathbb{F})$ by differentiating along the line $t \mapsto tX$ through the origin:

$$D\exp_0(X) = \frac{d}{dt}\bigg|_{t=0} \exp(tX) = \frac{d}{dt}\bigg|_{t=0} \Big(I + tX + \tfrac{t^2}{2}X^2 + \cdots\Big) = X$$

the higher terms all carrying a factor t that vanishes at $t = 0$. Thus $D\exp_0 = \mathrm{id}$, which is invertible. By the inverse function theorem ([2]), exp restricts to a diffeomorphism from some open neighborhood U of 0 onto some open neighborhood $V = \exp(U)$ of $\exp(0) = I$. Shrinking U if necessary so that $\|X\| < \log 2$ on U, proposition 1.3.6 identifies the inverse diffeomorphism with log, since $\log(\exp(X)) = X$ there; and shrinking V so that $\|A - I\| < 1$ on V makes log defined on all of V with $\exp(\log(A)) = A$. This exhibits $\exp\colon U \to V$ and $\log\colon V \to U$ as mutually inverse diffeomorphisms, completing the proof.

This is the analytic foundation for everything that follows. Near the identity, log is a coordinate chart on $\mathrm{GL}_n(\mathbb{F})$ valued in the vector space $M_n(\mathbb{F})$, and exp reads those coordinates back onto the group; the group multiplication, transported through log into a neighborhood of 0, becomes the operation $X * Y = \log(\exp X \exp Y)$ that Baker-Campbell-Hausdorff will expand in brackets. That $D\exp_0 = \mathrm{id}$ is what makes the linearization exact to first order: to first order the group law near I is just addition in $M_n(\mathbb{F})$, and the correction terms are governed by the noncommutativity measured in proposition 1.3.5. The chart is local, valid only near I; the question of how far exp reaches globally, in particular whether it is onto, depends on the group and is settled group by group in later chapters (for the compact groups it is onto, established in Part III).

Before turning to the theorem that organizes all of this, it is worth computing the exponential in the cases where it can be summed by hand, both to make the definition concrete and to see the algebraic properties confirmed on examples.

Example 1.3.8: Rotations, Nilpotents, and $\det = e^{\mathrm{tr}}$

1. *A plane-rotation generator.* Let $X = \left(\begin{smallmatrix} 0 & -\theta \\ \theta & 0 \end{smallmatrix}\right)$ for $\theta \in \mathbb{R}$. Then $X^2 = -\theta^2 I$, so the powers cycle through $I, X, -\theta^2 I, -\theta^2 X, \theta^4 I, \dots$, and splitting the exponential series into even and odd powers gives

$$\exp(X) = \Big(\sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m}}{(2m)!}\Big) I + \Big(\sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!}\Big) \frac{X}{\theta} = \cos\theta\, I + \sin\theta\, \frac{X}{\theta} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

so $\exp(X) = R(\theta) \in \mathrm{SO}(2)$ is the rotation by angle θ. The generator X is skew-symmetric with trace 0, and indeed $\det\exp(X) = \cos^2\theta + \sin^2\theta = 1 = e^0 = e^{\mathrm{tr}X}$, as proposition 1.3.4(4) requires. This recovers, from the single generator $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the entire circle group $\mathrm{SO}(2)$ met in example 1.2.11.

2. *A nilpotent matrix.* Let $N = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, the 3×3 nilpotent Jordan block. Then $N^2 = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ and $N^3 = 0$, so the series terminates after three terms:

$$\exp(tN) = I + tN + \frac{t^2}{2}N^2 = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

a polynomial in t, not a transcendental function. The trace of N is 0 and $\det\exp(tN) = 1$, again matching part (4); the curve $t \mapsto \exp(tN)$ is a one-parameter subgroup consisting of unipotent upper-triangular matrices.

3. *A diagonal check of* $\det = e^{\mathrm{tr}}$. For $D = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ the powers are $D^k = \mathrm{diag}(\lambda_1^k, \dots, \lambda_n^k)$, so summing entrywise gives $\exp(D) = \mathrm{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})$. Its determinant is $\prod_i e^{\lambda_i} = e^{\sum_i \lambda_i} = e^{\mathrm{tr}D}$, confirming proposition 1.3.4(4) directly. Since every matrix over $\mathbb{C}$ is conjugate to an upper-triangular one, with the eigenvalues on the diagonal, the conjugation equivariance of part (3) and the conjugation-invariance of both det and tr propagate this identity from the diagonal case to all of $M_n(\mathbb{C})$, which is the second proof of $\det\exp X = e^{\mathrm{tr}X}$ promised in outline.

The three computations exhibit the two extreme behaviors of the exponential. When the generator has imaginary spectrum, as in (1), the exponential is periodic and traces a circle; when the generator is nilpotent, as in (2), the exponential is polynomial and the orbit runs off to infinity. Every matrix group interpolates between these, and the interpolation is organized by the observation that each of these curves $t \mapsto \exp(tX)$ is a homomorphism $\mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$. The final result of the section is that these are the *only* such homomorphisms: the exponential does not just produce one-parameter subgroups, it produces all of them.

Definition 1.3.9: One-Parameter Subgroup

A *one-parameter subgroup* of $\mathrm{GL}_n(\mathbb{F})$ is a smooth group homomorphism $\gamma\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$; that is, a smooth curve with

$$\gamma(s+t) = \gamma(s)\gamma(t) \quad \text{for all } s,t \in \mathbb{R}, \qquad \gamma(0) = I$$

By proposition 1.3.4(2), for each fixed X the curve $\gamma_X(t) = \exp(tX)$ is a one-parameter subgroup, with velocity $\gamma_X'(0) = X$ at the identity by theorem 1.3.7. The content of the next theorem is the converse: prescribing the initial velocity X determines the whole homomorphism, so the assignment $X \mapsto \exp(tX)$ is a bijection between $M_n(\mathbb{F})$ and the set of one-parameter subgroups.

Theorem 1.3.10: Classification of One-Parameter Subgroups

Every one-parameter subgroup $\gamma\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$ has the form

$$\gamma(t) = \exp(tX), \qquad X = \gamma'(0) \in M_n(\mathbb{F})$$

Conversely every $X \in M_n(\mathbb{F})$ arises this way from a unique one-parameter subgroup, namely $t \mapsto \exp(tX)$. Thus $X \mapsto (t \mapsto \exp(tX))$ is a bijection from $M_n(\mathbb{F})$ onto the set of one-parameter subgroups of $\mathrm{GL}_n(\mathbb{F})$, with inverse $\gamma \mapsto \gamma'(0)$.

***Proof*:**
Step 1: a one-parameter subgroup solves a linear differential equation. Let γ be a one-parameter subgroup and set $X = \gamma'(0)$. Differentiating the homomorphism relation $\gamma(s+t) = \gamma(s)\gamma(t)$ with respect to s and then setting $s = 0$ gives, since $\gamma(0) = I$,

$$\gamma'(t) = \left.\frac{d}{ds}\right|_{s=0} \gamma(s+t) = \left.\frac{d}{ds}\right|_{s=0} \big(\gamma(s)\gamma(t)\big) = \gamma'(0)\gamma(t) = X\gamma(t)$$

So γ solves the linear matrix initial value problem

$$\gamma'(t) = X\gamma(t), \qquad \gamma(0) = I \tag{1.2}$$

Step 2: the exponential solves the same problem. By proposition 1.3.4(1) the curve $\eta(t) = \exp(tX)$ satisfies $\eta'(t) = X\exp(tX) = X\eta(t)$ and $\eta(0) = I$, so η is also a solution of (1.2).

Step 3: uniqueness of the solution. It remains to show (1.2) has only one solution, so that $\gamma = \eta$. Suppose γ_1, γ_2 both solve it, and set $\delta(t) = \exp(-tX)\gamma_1(t) - \exp(-tX)\gamma_2(t) = \exp(-tX)\big(\gamma_1(t) - \gamma_2(t)\big)$. Using $\frac{d}{dt}\exp(-tX) = -X\exp(-tX)$ from proposition 1.3.4(1) and the product rule, the derivative of $u(t) = \exp(-tX)\gamma_i(t)$ is

$$u'(t) = -X\exp(-tX)\gamma_i(t) + \exp(-tX)\,X\gamma_i(t) = \exp(-tX)\big(-X + X\big)\gamma_i(t) = 0$$

where the middle step uses $\exp(-tX)X = X\exp(-tX)$, again from part (1). So each $\exp(-tX)\gamma_i(t)$ is constant, equal to its value $\exp(0)\gamma_i(0) = I$ at $t = 0$; hence $\exp(-tX)\gamma_i(t) = I$ for all t, giving $\gamma_i(t) = \exp(tX)$ for both i. In particular the solution is unique, so $\gamma(t) = \exp(tX)$.

Step 4: the bijection. Step 1 through Step 3 show every one-parameter subgroup is $t \mapsto \exp(tX)$ with $X = \gamma'(0)$, so the map $X \mapsto (t \mapsto \exp(tX))$ is surjective onto one-parameter subgroups. It is injective because its inverse is the well-defined operation $\gamma \mapsto \gamma'(0)$: from $\gamma(t) = \exp(tX)$ one recovers $\gamma'(0) = X$ by theorem 1.3.7. Hence the assignment is a bijection with the stated inverse, completing the proof.

The theorem is the first crossing of the bridge announced at the start: it identifies a purely group-theoretic object, the smooth homomorphisms $\mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$, with a purely linear one, the matrices $X = \gamma'(0)$. The matrix X is the *infinitesimal generator* of the flow γ, and the differential equation (1.2) is the statement that the group element moves, at each instant, by left translation of its current velocity X. This is the seed of the Lie algebra: the next section defines the Lie algebra of a matrix group G as the set of generators X whose one-parameter subgroups $\exp(tX)$ stay inside G, and the classification theorem is what guarantees this set captures all the infinitesimal motions of G. The uniqueness argument of Step 3, that $\exp(-tX)\gamma(t)$ is constant, recurs verbatim when the flow of a left-invariant vector field is constructed intrinsically in Chapter 2; here it is available by bare hands because everything lives in the linear space $M_n(\mathbb{F})$.

Exercise 1.3.1

1. Show that $\exp(X^\mathsf{T}) = \exp(X)^\mathsf{T}$ and $\exp(X^*) = \exp(X)^*$ for all $X \in M_n(\mathbb{F})$. Deduce that if X is skew-symmetric ($X^\mathsf{T} = -X$) then $\exp(X) \in \mathrm{SO}(n)$, and if X is skew-Hermitian ($X^* = -X$) then $\exp(X) \in \mathrm{U}(n)$.
2. Show that $\exp\colon M_2(\mathbb{R}) \to \mathrm{GL}_2(\mathbb{R})$ is not surjective by proving that no real X satisfies $\exp(X) = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$. (Hint: $\det \exp X = e^{\operatorname{tr} X} > 0$ forces the image into $\mathrm{GL}_2^+(\mathbb{R})$; then analyze the real Jordan form of a putative X with $\exp(X)$ having the single eigenvalue -1.)
3. Let $X = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$ be block-diagonal. Prove that $\exp(X) = \begin{pmatrix} \exp(A) & 0 \\ 0 & \exp(B) \end{pmatrix}$. Use this together with example 1.3.8(1) to write down $\exp(tX)$ for $X = \operatorname{diag}\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}\right) \in \mathfrak{so}(4)$, and identify its image as a one-parameter subgroup of $\mathrm{SO}(4)$.
4. For $X, Y \in M_n(\mathbb{F})$ define $\beta(t) = \exp(tX)\,Y\,\exp(-tX)$. Compute $\beta(0)$ and $\beta'(0)$, showing $\beta'(0) = XY - YX$. (This is the computation that will define the bracket in the next section; do it here by differentiating the product.)
5. Let $\gamma\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$ be a one-parameter subgroup with generator $X = \gamma'(0)$, and let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a closed subgroup. Prove that if $\gamma(t) \in G$ for all t in some open interval around 0, then $\gamma(t) \in G$ for all $t \in \mathbb{R}$.
6. Using the logarithm series of proposition 1.3.6, compute $\log(R(\theta))$ for the rotation $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ with $|\theta|$ small, and verify that it equals the generator $\begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$ of example 1.3.8(1), consistent with exp and log being mutually inverse near I.

1.4 The Lie Algebra of a Matrix Group

The exponential map of section 1.3 turns a single matrix X into a one-parameter family $t \mapsto \exp(tX)$ of group elements, a curve through the identity whose velocity at time zero is X itself. This section reverses the emphasis: rather than fix X and watch the curve, fix a matrix group G and ask which matrices X generate a one-parameter family that stays inside G for all time. The answer is a vector space, the *Lie algebra* of G, and it is the first appearance of the object around which the entire book is built. For each classical group of section 1.2 the space will be a concrete list of matrix conditions, and its dimension will match the manifold dimension already computed, so that the algebra recovers by pure linear algebra what the regular value theorem found by calculus.

The vector-space structure is only half the story. The set of matrices generating one-parameter subgroups of G carries a second operation, the commutator $[X, Y] = XY - YX$, and this bracket is what promotes the tangent space to a Lie algebra in the abstract sense. The bracket measures the failure of $\exp(X)$ and $\exp(Y)$ to commute, and its closure inside the tangent space is the algebraic shadow of the fact that G is closed under multiplication. By the end of the section the reader will have both the concrete matrix Lie algebras and the abstract definition they instantiate, together with the running example $\mathfrak{su}(2)$, whose bracket is the cross product on $\mathbb{R}^3$.

Definition 1.4.1: Lie Algebra of a Matrix Group

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group. The *Lie algebra* of G is the set

$$\mathfrak{g} = \{X \in M_n(\mathbb{F}) \mid \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}$$

of matrices whose associated one-parameter subgroup lies entirely in G.

The definition is engineered to be checkable: to decide whether X belongs to $\mathfrak{g}$ one differentiates the defining equations of G along the curve $\exp(tX)$ at $t = 0$. Before extracting those conditions for the classical groups, the first order of business is to see that $\mathfrak{g}$ is a real vector space at all, which is not obvious from the definition: it asks about the exponential of tX, and the exponential is neither additive nor linear.

Proposition 1.4.2: The Lie Algebra is a Real Vector Space

For any matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$, the Lie algebra $\mathfrak{g}$ of definition 1.4.1 is a real vector subspace of $M_n(\mathbb{F})$. It contains 0, and if $X, Y \in \mathfrak{g}$ and $\lambda \in \mathbb{R}$ then $\lambda X \in \mathfrak{g}$ and $X + Y \in \mathfrak{g}$.

***Proof*:**
The zero matrix lies in $\mathfrak{g}$ because $\exp(t \cdot 0) = I \in G$ for every t. For closure under real scalars, fix $X \in \mathfrak{g}$ and $\lambda \in \mathbb{R}$; then for every $t \in \mathbb{R}$ the product $t\lambda$ is again a real number, so

$$\exp\big(t(\lambda X)\big) = \exp\big((t\lambda)X\big) \in G$$

because $X \in \mathfrak{g}$. Hence $\lambda X \in \mathfrak{g}$. This uses only that the scalar is real; it is the reason $\mathfrak{g}$ is a real vector space even when $\mathbb{F} = \mathbb{C}$.

Closure under addition is where the exponential's nonlinearity must be confronted. The tool is the *Lie product formula*

$$\exp(X+Y) = \lim_{m\to\infty} \left(\exp\!\left(\tfrac{X}{m}\right)\exp\!\left(\tfrac{Y}{m}\right)\right)^m \tag{1.3}$$

valid for all $X, Y \in M_n(\mathbb{F})$, which expresses the exponential of a sum as a limit of products of exponentials, precisely the kind of expression a group is closed under.

Step 1: proof of the Lie product formula. Fix X, Y and set $A_m = \exp(X/m)\exp(Y/m)$. Expanding each factor to second order in the small parameter $1/m$ gives

$$\exp\!\left(\tfrac{X}{m}\right) = I + \tfrac{X}{m} + O\!\left(\tfrac{1}{m^2}\right), \qquad \exp\!\left(\tfrac{Y}{m}\right) = I + \tfrac{Y}{m} + O\!\left(\tfrac{1}{m^2}\right)$$

where the $O(1/m^2)$ terms are matrices of operator norm bounded by a constant (depending on $\|X\|$, $\|Y\|$) times $1/m^2$. Multiplying,

$$A_m = I + \frac{X+Y}{m} + O\!\left(\tfrac{1}{m^2}\right)$$

Since $\|A_m - I\| \to 0$, for m large the matrix A_m lies in the domain where the logarithm series of proposition 1.3.6 converges, and there $\log(I+B) = B + O(\|B\|^2)$; applying this with $B = A_m - I = \frac{X+Y}{m} + O(1/m^2)$ gives

$$\log A_m = \frac{X+Y}{m} + O\!\left(\tfrac{1}{m^2}\right)$$

Because log and exp are mutually inverse near I and 0 respectively (theorem 1.3.7), one has $A_m = \exp(\log A_m)$, and therefore

$$A_m^m = \exp(m \log A_m) = \exp\!\left(X + Y + O\!\left(\tfrac{1}{m}\right)\right)$$

As $m \to \infty$ the argument $X+Y+O(1/m)$ converges to $X+Y$ in $M_n(\mathbb{F})$, and exp is continuous, so $A_m^m \to \exp(X+Y)$. This is (1.3).

Step 2: additive closure. Suppose $X, Y \in \mathfrak{g}$ and fix $t \in \mathbb{R}$. Applying (1.3) to the matrices tX and tY,

$$\exp\big(t(X+Y)\big) = \exp(tX + tY) = \lim_{m\to\infty}\left(\exp\!\left(\tfrac{tX}{m}\right)\exp\!\left(\tfrac{tY}{m}\right)\right)^m$$

For each fixed m, the matrices $\exp(tX/m)$ and $\exp(tY/m)$ lie in G, because $X, Y \in \mathfrak{g}$ and t/m is a real number; their product raised to the mth power is a product of elements of G, hence lies in G, a subgroup. The one-parameter subgroup $\exp(t(X+Y))$ is thus a limit of elements of G. A matrix group G is defined by polynomial equations in the entries (the classical groups of section 1.2 are each $F^{-1}(c)$ for a continuous F), so G is closed in $\mathrm{GL}_n(\mathbb{F})$, and a limit of elements of G that remains invertible lies in G. Since $\exp(t(X+Y))$ is invertible with inverse $\exp(-t(X+Y))$, it lies in G. As t was arbitrary, $X+Y \in \mathfrak{g}$, completing the proof.

The role of the Lie product formula is worth dwelling on. Addition of matrices is a linear operation with no group-theoretic meaning, yet (1.3) realizes $\exp(X+Y)$ entirely through multiplication and limits, the two operations a closed subgroup respects. This is the mechanism that transfers the linear structure of $M_n(\mathbb{F})$ onto the tangent space $\mathfrak{g}$: the sum $X+Y$ is legitimate inside $\mathfrak{g}$ precisely because its exponential can be manufactured from the exponentials of X and Y without ever leaving G. The same interplay between the additive world of $\mathfrak{g}$ and the multiplicative world of G drives the bracket computation below.

For the classical groups the definition of $\mathfrak{g}$ unwinds into a linear condition on X, one per group, obtained by differentiating the defining relation. These conditions are the working description of the classical Lie algebras, and the point of the theorem is that they reproduce the dimensions of section 1.2 exactly.

Theorem 1.4.3: The Classical Lie Algebras

The Lie algebras of the classical groups are the following matrix spaces, with the indicated real dimensions.

1. $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$, the full matrix algebra; $\dim_{\mathbb{R}} \mathfrak{gl}_n(\mathbb{R}) = n^2$ and $\dim_{\mathbb{R}} \mathfrak{gl}_n(\mathbb{C}) = 2n^2$.
2. $\mathfrak{sl}_n(\mathbb{F}) = \{X \in M_n(\mathbb{F}) \mid \operatorname{tr} X = 0\}$, the traceless matrices; $\dim_{\mathbb{R}} \mathfrak{sl}_n(\mathbb{R}) = n^2 - 1$ and $\dim_{\mathbb{R}} \mathfrak{sl}_n(\mathbb{C}) = 2(n^2 - 1)$.
3. $\mathfrak{so}(n) = \left\{X \in M_n(\mathbb{R}) \;\middle|\; X^{\mathsf{T}} = -X\right\}$, the real skew-symmetric matrices; $\dim_{\mathbb{R}} \mathfrak{so}(n) = \binom{n}{2}$.
4. $\mathfrak{u}(n) = \{X \in M_n(\mathbb{C}) \mid X^* = -X\}$, the skew-Hermitian matrices; $\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2$.
5. $\mathfrak{su}(n) = \{X \in M_n(\mathbb{C}) \mid X^* = -X,\ \operatorname{tr} X = 0\}$, the traceless skew-Hermitian matrices; $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$.
6. $\mathfrak{sp}(2n, \mathbb{F}) = \left\{X \in M_{2n}(\mathbb{F}) \;\middle|\; X^{\mathsf{T}} J + JX = 0\right\}$, where $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$; $\dim_{\mathbb{R}} \mathfrak{sp}(2n, \mathbb{R}) = n(2n+1)$.

In each case the dimension of $\mathfrak{g}$ equals the manifold dimension of the corresponding group computed in section 1.2.

***Proof*:**
The argument is the same for each group: differentiate the defining relation of G along $t \mapsto \exp(tX)$ at $t = 0$, using $\frac{d}{dt}\big|_0 \exp(tX) = X$ and the derivative rules for the exponential from proposition 1.3.4.

1. *The general linear group.* Every $X \in M_n(\mathbb{F})$ has $\exp(tX) \in \mathrm{GL}_n(\mathbb{F})$ for all t, since $\exp(tX)$ is invertible with inverse $\exp(-tX)$ by proposition 1.3.4. Hence $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$, of real dimension n^2 over $\mathbb{R}$ and $2n^2$ over $\mathbb{C}$, matching $\dim \mathrm{GL}_n(\mathbb{F})$ from proposition 1.1.2.
2. *The special linear group.* The relation defining $\mathrm{SL}_n(\mathbb{F})$ is $\det = 1$. By the identity $\det \exp(tX) = e^{t \operatorname{tr} X}$ of proposition 1.3.4, the curve $\exp(tX)$ lies in $\mathrm{SL}_n(\mathbb{F})$ for all t if and only if $e^{t \operatorname{tr} X} = 1$ for all $t \in \mathbb{R}$, which holds if and only if $\operatorname{tr} X = 0$. Thus $\mathfrak{sl}_n(\mathbb{F})$ is the space of traceless matrices. The trace is one linear equation on $M_n(\mathbb{F})$, and it is surjective onto $\mathbb{F}$, so its kernel has real codimension 1 over $\mathbb{R}$ and real codimension 2 over $\mathbb{C}$, giving $n^2 - 1$ and $2(n^2 - 1)$, matching proposition 1.2.2.
3. *The orthogonal and special orthogonal groups.* The relation is $A^{\mathsf{T}} A = I$. Substituting $A = \exp(tX)$ and using $\exp(tX)^{\mathsf{T}} = \exp(tX^{\mathsf{T}})$, the curve satisfies $\exp(tX^{\mathsf{T}}) \exp(tX) = I$ for all t. Differentiating at $t = 0$, the product rule gives $X^{\mathsf{T}} + X = 0$, so X is skew-

symmetric. Conversely, if $X^\top = -X$ then $tX^\top$ and tX commute (they are negatives), so $\exp(tX^\top)\exp(tX) = \exp\big(t(X^\top + X)\big) = \exp(0) = I$ by proposition 1.3.5, and $\exp(tX) \in \mathrm{O}(n)$ for all t. A skew-symmetric matrix has zero diagonal and its below- diagonal entries are the negatives of those above, so its free parameters are the $\binom{n}{2}$ strictly-upper entries; hence $\dim \mathfrak{so}(n) = \binom{n}{2}$, matching proposition 1.2.4. The determinant-one constraint cutting $\mathrm{SO}(n)$ out of $\mathrm{O}(n)$ imposes nothing further on the algebra: every skew-symmetric X already has $\mathrm{tr}X = 0$, so $\det\exp(tX) = e^{t\,\mathrm{tr}X} = 1$ automatically, and $\mathfrak{so}(n)$ is at once the Lie algebra of $\mathrm{O}(n)$ and of $\mathrm{SO}(n)$, reflecting that the two groups share the identity component.

4. *The unitary group.* The relation is $A^*A = I$. With $A = \exp(tX)$ and $\exp(tX)^* = \exp(tX^*)$ (the conjugate transpose of $\exp(tX)$ is exp of the conjugate transpose, as the power series has real coefficients), the same differentiation gives $X^* + X = 0$, so X is skew-Hermitian; the converse is the commuting exponential identity as in (3). A skew-Hermitian matrix has purely imaginary diagonal entries (n real parameters) and $\binom{n}{2}$ free complex entries above the diagonal ($n(n-1)$ real parameters), the below-diagonal entries being determined by $X^* = -X$; the total is $n + n(n-1) = n^2$, matching proposition 1.2.6.

5. *The special unitary group.* Here X must be skew-Hermitian and additionally satisfy the special condition. By (2) the determinant constraint $\det\exp(tX) = 1$ forces $\mathrm{tr}X = 0$, so $\mathfrak{su}(n)$ is the space of traceless skew-Hermitian matrices. Tracelessness removes one real parameter from $\mathfrak{u}(n)$: a skew-Hermitian matrix has purely imaginary diagonal $\mathrm{diag}(i\theta_1, \dots, i\theta_n)$, and $\mathrm{tr}X = i(\theta_1 + \cdots + \theta_n) = 0$ is the single real equation $\sum_j \theta_j = 0$. Hence $\dim \mathfrak{su}(n) = n^2 - 1$, matching proposition 1.2.6.

6. *The symplectic group.* The relation is $A^\top JA = J$. With $A = \exp(tX)$, the condition $\exp(tX)^\top J\exp(tX) = J$ for all t, differentiated at $t = 0$, yields $X^\top J + JX = 0$; the converse again follows because $X^\top J + JX = 0$ makes $t \mapsto \exp(tX)^\top J\exp(tX)$ have vanishing derivative and value J at $t = 0$, hence constantly J. The condition $X^\top J = -JX$ says exactly that JX is symmetric: $(JX)^\top = X^\top J^\top = -X^\top J = JX$, using $J^\top = -J$. The map $X \mapsto JX$ is a linear bijection of $M_{2n}(\mathbb{F})$ (with inverse $Y \mapsto J^{-1}Y = -JY$, since $J^2 = -I$), so $\mathfrak{sp}(2n, \mathbb{F})$ is linearly isomorphic to the space of symmetric $2n \times 2n$ matrices, of dimension $\binom{2n+1}{2} = \frac{2n(2n+1)}{2} = n(2n + 1)$, matching proposition 1.2.9.

In every case the linearized condition reproduces the dimension found by the regular value theorem in section 1.2, completing the proof.

The coincidence of dimensions is not an accident but the concrete face of a general principle. The Lie algebra $\mathfrak{g}$ is the tangent space to G at the identity: differentiating a defining equation of G along a curve through I is exactly the operation the regular value theorem performs when it identifies the tangent space T_IG with $\ker DF_I$, and the two computations agree because they are the same computation. The exponential description of definition 1.4.1 and the tangent-space description of section 1.2 therefore define the same object; the next chapter will make the identification $\mathfrak{g} = T_IG$ the definition of the Lie algebra of an abstract Lie group. One reads off from theorem 1.4.3 that the skew-symmetric matrices form the infinitesimal rotations and the skew-Hermitian matrices the infinitesimal unitaries: exponentiating a small skew matrix produces a rotation close to the identity, the linear-algebra content of “a rotation is generated by an angular velocity”.

So far $\mathfrak{g}$ is only a vector space. The operation that makes it a Lie algebra is the commutator, and its presence is forced by the same closure under multiplication that gave additive closure. If $X, Y \in \mathfrak{g}$, then conjugating the one-parameter subgroup of Y by an element of the one-parameter subgroup of X stays in G, and differentiating this conjugation produces a new element of $\mathfrak{g}$, which turns out to be the commutator.

Proposition 1.4.4: The Lie Algebra is Closed under the Commutator

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g}$. For all $X, Y \in \mathfrak{g}$, the commutator

$$[X, Y] = XY - YX$$

again lies in $\mathfrak{g}$. The bracket $[-,-]$ is $\mathbb{R}$-bilinear, alternating (so that $[X, X] = 0$ and $[Y, X] = -[X, Y]$), and satisfies the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all $X, Y, Z \in \mathfrak{g}$.

***Proof*:**
Step 1: conjugation stays in the group. Fix $X, Y \in \mathfrak{g}$ and $s \in \mathbb{R}$, and consider the curve

$$c(s) = \exp(sX)\, Y \exp(-sX) \in M_n(\mathbb{F})$$

obtained by conjugating Y. For each real s the elements $\exp(sX)$ and $\exp(-sX) = \exp(sX)^{-1}$ lie in G, so conjugation by $\exp(sX)$ is an automorphism of G carrying the one-parameter subgroup $t \mapsto \exp(tY)$ to $t \mapsto \exp(sX)\exp(tY)\exp(-sX)$. This last curve lies in G, and by the conjugation equivariance of the exponential (proposition 1.3.4),

$$\exp(sX)\exp(tY)\exp(-sX) = \exp\big(t\, c(s)\big) \tag{1.4}$$

where $c(s) = \exp(sX)\, Y \exp(-sX)$. The left side lies in G for all t, so (1.4) shows $\exp(t\, c(s)) \in G$ for all t; that is, $c(s) \in \mathfrak{g}$ for every $s \in \mathbb{R}$.

Step 2: differentiate the conjugation. The set $\mathfrak{g}$ is a vector subspace of the finite-dimensional space $M_n(\mathbb{F})$ (proposition 1.4.2), hence closed, and $s \mapsto c(s)$ is a smooth curve lying in $\mathfrak{g}$. A closed subspace contains the derivative of any differentiable curve lying in it, so $c'(0) \in \mathfrak{g}$. Computing the derivative by the product rule, with $\frac{d}{ds}\exp(sX) = X\exp(sX)$ and $\frac{d}{ds}\exp(-sX) = -\exp(-sX)X$,

$$c'(s) = X\exp(sX)\, Y \exp(-sX) - \exp(sX)\, Y \exp(-sX)\, X$$

Evaluating at $s = 0$, where $\exp(0) = I$, gives

$$c'(0) = XY - YX = [X, Y]$$

Therefore $[X, Y] \in \mathfrak{g}$, which is the closure claim.

Step 3: the algebraic identities. Bilinearity of $(X, Y) \mapsto XY - YX$ is inherited from the bilinearity of matrix multiplication: $[aX + bX', Y] = a[X, Y] + b[X', Y]$ and likewise in the second slot, for $a, b \in \mathbb{R}$. The bracket is alternating because $[X, X] = XX - XX = 0$; expanding $[X+Y, X+Y] = 0$ and using bilinearity gives $[X, Y] + [Y, X] = 0$, so $[Y, X] = -[X, Y]$. For the Jacobi identity, expand each of the three double brackets. Writing out $[X,[Y,Z]] = X(YZ-ZY)-(YZ-ZY)X$ and the two cyclic analogues, the sum

$$[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]]$$

consists of the twelve triple products $XYZ, XZY, \ldots$, each appearing once with a plus sign and once with a minus sign; they cancel in pairs, leaving 0. This establishes all three identities, completing the proof.

The conjugation curve $c(s)$ in the proof is the first appearance of the adjoint action, which the next chapter will develop into the representations Ad and ad that encode the bracket intrinsically. What matters now is the structural conclusion: the tangent space $\mathfrak{g}$ is not just a vector space but is closed under an antisymmetric bilinear product satisfying the Jacobi identity. This package of data has a name, and it is the central algebraic object of the theory.

Definition 1.4.5: Abstract Lie Algebra

A *Lie algebra* over a field k is a k-vector space $\mathfrak{g}$ equipped with a bilinear map

$$[-,-]\colon \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g}, \qquad (X,Y)\mapsto [X,Y]$$

called the *bracket*, satisfying two axioms:

1. the bracket is *alternating*: $[X,X]=0$ for all $X\in\mathfrak{g}$ (equivalently, over any field of characteristic $\neq 2$, antisymmetric: $[Y,X]=-[X,Y]$);
2. the bracket satisfies the *Jacobi identity*:

   $$[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$$

   for all $X,Y,Z\in\mathfrak{g}$.

A *homomorphism* of Lie algebras is a k-linear map $\varphi\colon\mathfrak{g}\to\mathfrak{h}$ preserving the bracket, $\varphi([X,Y]) = [\varphi(X),\varphi(Y)]$.

proposition 1.4.4 says precisely that the Lie algebra $\mathfrak{g}$ of a matrix group, with the commutator bracket $[X,Y]=XY-YX$, is an abstract real Lie algebra in the sense of definition 1.4.5. The classical algebras $\mathfrak{gl}_n, \mathfrak{sl}_n, \mathfrak{so}(n), \mathfrak{u}(n), \mathfrak{su}(n)$, and $\mathfrak{sp}(2n,\mathbb{F})$ are all real Lie algebras under the commutator, and one checks in each case that the bracket preserves the defining condition: for instance if X,Y are skew-symmetric then $[X,Y]^\intercal = (XY-YX)^\intercal = Y^\intercal X^\intercal - X^\intercal Y^\intercal = YX-XY = -[X,Y]$, so $[X,Y]\in\mathfrak{so}(n)$, consistent with proposition 1.4.4. The full matrix algebra $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ under the commutator is the ambient Lie algebra of which all the others are subalgebras, the exact analogue of $\mathrm{GL}_n(\mathbb{F})$ being the ambient group.

The abstract definition frees the bracket from its origin as a commutator of matrices. In Chapter 2 the Lie algebra of a general Lie group is constructed intrinsically, as the space of left-invariant vector fields under the Lie bracket of vector fields, with no matrices in sight, and definition 1.4.5 is the target of that construction: the intrinsic bracket will satisfy exactly these two axioms. The matrix Lie algebras of this chapter are then recovered as the special case where the group sits inside some $\mathrm{GL}_n(\mathbb{F})$, and the intrinsic bracket restricts to the commutator. The reader who wants a preview should keep in mind that the abstract axioms are all that survive of the group structure at the infinitesimal level, and that the entire structure theory of Parts II and beyond is carried out with the axioms alone.

The bridge between the group and its algebra is the exponential map. The defining property of $\mathfrak{g}$ is that exp sends it into G, and the local diffeomorphism of section 1.3 pins down the dimension of the correspondence.

Corollary 1.4.6: The Exponential Bridge and the Dimension Match

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g}$. Then exp maps $\mathfrak{g}$ into G, and the tangent space to G at the identity contains $\mathfrak{g}$:

$$\mathfrak{g} \subseteq T_I G$$

Suppose in addition that G is presented as a regular-value preimage $G = F^{-1}(c)$ for a smooth map F, as each classical group of section 1.2 is. Then $\mathfrak{g} = T_I G$, the map exp restricts to a diffeomorphism from a neighborhood of 0 in $\mathfrak{g}$ onto a neighborhood of I in G, and in particular

$$\dim_{\mathbb{R}} \mathfrak{g} = \dim G$$

***Proof*:**
That $\exp(\mathfrak{g}) \subseteq G$ is immediate from definition 1.4.1: for $X \in \mathfrak{g}$, taking $t = 1$ gives $\exp X = \exp(1 \cdot X) \in G$. For the inclusion into the tangent space, let $X \in \mathfrak{g}$. The curve $t \mapsto \exp(tX)$ lies in G by definition 1.4.1, passes through I at $t = 0$, and has velocity X there by proposition 1.3.4; a velocity vector of a curve in G through I is by definition an element of $T_I G$, so $X \in T_I G$. Hence $\mathfrak{g} \subseteq T_I G$.

Suppose now $G = F^{-1}(c)$ with c a regular value of a smooth F, so that $T_I G = \ker DF_I$. The reverse inclusion $T_I G \subseteq \mathfrak{g}$ is the content of the discussion following theorem 1.4.3: differentiating the defining relation $F = c$ along a curve in G through I shows that the linearized condition cutting out $\mathfrak{g}$ is exactly $\ker DF_I$, so $\mathfrak{g} = \ker DF_I = T_I G$. With this equality in hand, by theorem 1.3.7 the map exp is a diffeomorphism from a neighborhood U of 0 in $M_n(\mathbb{F})$ onto a neighborhood of I in $\mathrm{GL}_n(\mathbb{F})$, with $D\exp_0 = \mathrm{id}$; restricting to $\mathfrak{g}$, the differential $D\exp_0 = \mathrm{id}$ carries $\mathfrak{g} = T_0\mathfrak{g}$ isomorphically onto $\mathfrak{g} = T_I G$, so the inverse function theorem applied within G makes the restriction $\exp\colon U \cap \mathfrak{g} \to G$ a diffeomorphism onto a neighborhood of I in G. Two manifolds admitting a diffeomorphism between open sets have equal dimension, so $\dim_{\mathbb{R}} \mathfrak{g} = \dim G$, completing the proof.

The corollary is the quantitative form of the slogan that the Lie algebra is an infinitesimal copy of the group: near the identity, G looks exactly like its tangent space $\mathfrak{g}$, with exp providing the chart. The inclusion $\mathfrak{g} \subseteq T_I G$ holds for every matrix group and is the easy half, read off by differentiating $\exp(tX)$; the reverse inclusion, which upgrades it to the equality $\mathfrak{g} = T_I G$ and thereby forces the dimensions to match, is what requires the group to be presented as a regular-value preimage, and it is established above exactly for the classical groups of section 1.2. For a matrix group given only as an abstract embedded submanifold, with no defining map F at hand, the equality $\mathfrak{g} = T_I G$ is still true but rests on the closed-subgroup theorem, proved in Chapter 3, which produces the exponential chart from the closedness of G alone. The dimension equality has already been verified group by group in theorem 1.4.3, but the corollary explains why it had to hold: exp is a local chart from $\mathfrak{g}$ to G, so the two spaces cannot have different dimensions. The global behavior of exp is more delicate, and whether it is surjective onto G depends on the group; section 1.5 will show exp is onto for the compact group $\mathrm{SU}(2)$, while for $\mathrm{GL}_n(\mathbb{R})$ it is not, as the negative-determinant matrices are never exponentials.

Everything comes together in the smallest nonabelian compact example, $\mathfrak{su}(2)$, the Lie algebra of the running-example group $\mathrm{SU}(2)$. Its bracket is the cross product, and this identification is the first instance of the dictionary the book exists to develop.

Example 1.4.7: The Lie Algebra $\mathfrak{su}(2)$ and the Cross Product

By theorem 1.4.3, $\mathfrak{su}(2)$ is the space of traceless skew-Hermitian 2×2 complex matrices. A traceless skew-Hermitian matrix has purely imaginary diagonal summing to zero and off-diagonal entries $b, -\bar{b}$, so it has the form

$$\begin{pmatrix} ia & b \\ -\bar{b} & -ia \end{pmatrix}, \qquad a \in \mathbb{R},\ b \in \mathbb{C}$$

a real three-dimensional space, consistent with $\dim \mathfrak{su}(2) = 2^2 - 1 = 3$. A convenient basis is built from the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

which are Hermitian and traceless; multiplying by $-\frac{i}{2}$ produces traceless skew-Hermitian matrices. Set

$$X_k = -\tfrac{i}{2}\,\sigma_k, \qquad k = 1,2,3$$

explicitly

$$X_1 = \tfrac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \qquad X_2 = \tfrac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad X_3 = \tfrac{1}{2}\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

Each X_k is traceless and satisfies $X_k^* = -X_k$, so $X_1, X_2, X_3 \in \mathfrak{su}(2)$, and they are linearly independent over $\mathbb{R}$, hence a basis.

The bracket relations are computed directly from the Pauli identity $\sigma_j\sigma_k = \delta_{jk}I + i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell$, where $\varepsilon_{jk\ell}$ is the sign of the permutation (j,k,ℓ). The commutator of Pauli matrices is therefore $[\sigma_j,\sigma_k] = 2i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell$, and so

$$[X_j, X_k] = \left(-\tfrac{i}{2}\right)^2 [\sigma_j,\sigma_k] = -\tfrac{1}{4}\cdot 2i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell = \sum_\ell \varepsilon_{jk\ell}\left(-\tfrac{i}{2}\sigma_\ell\right) = \sum_\ell \varepsilon_{jk\ell} X_\ell$$

In cyclic form,

$$[X_1, X_2] = X_3, \qquad [X_2, X_3] = X_1, \qquad [X_3, X_1] = X_2 \tag{1.5}$$

These are exactly the multiplication rules of the standard basis e_1, e_2, e_3 of $\mathbb{R}^3$ under the cross product: $e_1 \times e_2 = e_3$, $e_2 \times e_3 = e_1$, $e_3 \times e_1 = e_2$.

Define the linear map $\Phi\colon \mathfrak{su}(2) \to \mathbb{R}^3$ by $\Phi(X_k) = e_k$ on the basis and extending by linearity. It is a linear isomorphism of three-dimensional real vector spaces, and it preserves the bracket: on basis elements (1.5) matches the cross product term by term, and both operations are bilinear, so $\Phi([X,Y]) = \Phi(X)\times\Phi(Y)$ for all X, Y. Therefore

$$\Phi\colon \mathfrak{su}(2) \xrightarrow{\sim} (\mathbb{R}^3, \times)$$

is an isomorphism of Lie algebras, where $\mathbb{R}^3$ carries the cross product as its bracket. The cross product does satisfy the Jacobi identity, which is the familiar vector identity $a \times (b \times c) + b \times (c \times$

> $a) + c \times (a \times b) = 0$; the isomorphism Φ makes this identity and the Jacobi identity for $\mathfrak{su}(2)$ the same statement. This identification is the infinitesimal form of the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ taken up in section 1.5: the algebra $(\mathbb{R}^3, \times)$ is also $\mathfrak{so}(3)$ (the skew-symmetric 3×3 matrices act on $\mathbb{R}^3$ as $v \mapsto \omega \times v$), so $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ even though $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ are not isomorphic as groups.

The isomorphism $\mathfrak{su}(2) \cong \mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ is a compressed statement of the entire local theory of these groups: two nonisomorphic groups share a Lie algebra, so they must agree near the identity and differ only in their global topology. That the infinitesimal object cannot distinguish $\mathrm{SU}(2)$ from $\mathrm{SO}(3)$, while the groups themselves are distinguished by a factor of $\mathbb{Z}/2$, is the phenomenon section 1.5 analyzes in full, and it is the reason the passage from group to algebra loses information that only the topology of the group can supply.

Exercise 1.4.1

1. Write down the standard basis of the skew-symmetric matrices $\mathfrak{so}(3)$, namely the three matrices L_1, L_2, L_3 with $(L_k)_{ij} = -\varepsilon_{kij}$, and verify the bracket relations $[L_1, L_2] = L_3$, $[L_2, L_3] = L_1$, $[L_3, L_1] = L_2$. Conclude that $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$, and hence $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ as real Lie algebras.

2. Show that for the commutator bracket the trace gives a Lie algebra homomorphism $\mathrm{tr}\colon \mathfrak{gl}_n(\mathbb{F}) \to \mathbb{F}$ (with $\mathbb{F}$ abelian, so $[\lambda, \mu] = 0$), and deduce that $\mathfrak{sl}_n(\mathbb{F}) = \ker \mathrm{tr}$ is an ideal. Prove the direct sum decomposition $\mathfrak{gl}_n(\mathbb{F}) = \mathfrak{sl}_n(\mathbb{F}) \oplus \mathbb{F}I$ of vector spaces, and determine the bracket of a scalar matrix λI with an arbitrary $X \in \mathfrak{gl}_n(\mathbb{F})$.

3. Using $\det \exp X = e^{\mathrm{tr} X}$, show directly from definition 1.4.1 that a matrix $X \in M_n(\mathbb{R})$ lies in $\mathfrak{sl}_n(\mathbb{R})$ if and only if $\mathrm{tr}X = 0$, without invoking theorem 1.4.3. Give an explicit $X \in \mathfrak{gl}_2(\mathbb{R})$ with $\mathrm{tr}X \neq 0$ and verify that $\exp(tX) \notin \mathrm{SL}_2(\mathbb{R})$ for some t.

4. For $X \in \mathfrak{su}(2)$ with $\Phi(X) = v \in \mathbb{R}^3$ under the isomorphism of example 1.4.7, show that $X^2 = -\frac{1}{4}|v|^2 I$, where $|v|$ is the Euclidean norm. Deduce the closed form
$$\exp(X) = \cos\left(\tfrac{|v|}{2}\right) I + \frac{\sin\left(\frac{|v|}{2}\right)}{|v|/2} X$$
for $v \neq 0$, and confirm that $\exp(X) \in \mathrm{SU}(2)$.

5. Verify that $\mathfrak{sp}(2n, \mathbb{F})$ is closed under the commutator: if $X^{\mathsf{T}} J + JX = 0$ and $Y^{\mathsf{T}} J + JY = 0$, show $[X, Y]^{\mathsf{T}} J + J[X, Y] = 0$. This confirms $\mathfrak{sp}(2n, \mathbb{F})$ is a Lie subalgebra of $\mathfrak{gl}_{2n}(\mathbb{F})$, as proposition 1.4.4 predicts.

6. Show that $\mathfrak{u}(n)$ is a *real* but not a complex vector subspace of $M_n(\mathbb{C})$: exhibit a skew-Hermitian matrix X for which iX is not skew-Hermitian. Explain how this is consistent with proposition 1.4.2, which asserts only that the Lie algebra of a matrix group is a real vector space.

1.5 $SU(2)$, $SO(3)$, and the Double Cover $SU(2) \to SO(3)$

Every structural feature that the later chapters develop in generality is already present, in fully computable form, in the smallest noncommutative compact group, $\mathrm{SU}(2)$. This section studies it directly: as a manifold it is a 3-sphere, and its Lie algebra $\mathfrak{su}(2)$ is three-dimensional, small enough that the exponential map, the adjoint action, and the passage to SO(3) can all be written out with explicit matrices. The centerpiece is the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, a two-to-one surjective homomorphism that realizes SO(3) as the quotient $\mathrm{SU}(2)/\{\pm I\}$ and identifies SU(2) as the universal cover of the rotation group. This one map explains why a rotation of ordinary space cannot be tracked continuously by a single square root, why $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2$, and where the spin of quantum mechanics comes from. It is the first place in the book where the correspondence between a group and its Lie algebra does concrete work, and SU(2) becomes the running example that every part returns to: its maximal torus and Weyl group in the compact theory, its irreducible representations and their characters at the analytic summit, and its noncompact sibling $\mathrm{SL}_2(\mathbb{R})$ in the theory of real forms.

The starting point is the explicit description of SU(2) obtained at the end of the previous section, which realizes it as a sphere.

Proposition 1.5.1: SU(2) is the 3-Sphere

The special unitary group in dimension two is

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \;\middle|\; \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$$

and the map $(\alpha, \beta) \mapsto \left(\begin{smallmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{smallmatrix}\right)$ is a diffeomorphism from the unit sphere $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$ onto SU(2). In particular SU(2) is compact, connected, and simply connected.

***Proof*:**
A matrix $A \in \mathrm{U}(2)$ has orthonormal columns; writing the first column as $(\alpha, \beta)^\mathsf{T}$ with $|\alpha|^2+|\beta|^2 = 1$, the second column is a unit vector orthogonal to it, hence a unit scalar multiple $\lambda(-\overline{\beta}, \overline{\alpha})^\mathsf{T}$ with $|\lambda| = 1$. The determinant of the resulting matrix is $\lambda(|\alpha|^2 + |\beta|^2) = \lambda$, so the constraint $\det A = 1$ defining SU(2) forces $\lambda = 1$, giving the displayed form. Conversely every matrix of that form is special unitary, as a direct check of $A^*A = I$ and $\det A = 1$ shows. The parametrization $(\alpha, \beta) \mapsto A$ is thus a bijection from $\{(\alpha, \beta) \in \mathbb{C}^2 \mid |\alpha|^2 + |\beta|^2 = 1\} = S^3$ onto SU(2); it is the restriction of a real-linear injection $\mathbb{C}^2 \to M_2(\mathbb{C})$, hence a smooth embedding, and its inverse (read off α and β from the first column) is smooth, so it is a diffeomorphism.
The sphere S^3 is compact, being closed and bounded in $\mathbb{R}^4$, and it is connected and simply connected: S^3 minus a point is diffeomorphic to $\mathbb{R}^3$ by stereographic projection, in which every loop contracts, and any loop in S^3 misses a point because a continuous image of the interval cannot fill the three-sphere. These topological facts about spheres are established in [4]; they transfer to SU(2) through the diffeomorphism, completing the proof.

That SU(2) is a sphere is the geometric fact underlying everything that follows. Compactness is what will later force its representation theory to be completely reducible and its irreducible representations to be finite-dimensional; connectedness means it is generated by any neighborhood of the identity, so that its structure is determined by its Lie algebra; and simple connectedness is precisely the property that makes SU(2), rather than SO(3), the group corresponding to the shared Lie algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ under the equivalence of Chapter 3. The three-sphere is also the unit group of the quaternions: writing a quaternion as $q = \alpha + \beta j$ with $\alpha, \beta \in \mathbb{C}$ identifies $\{q \in \mathbb{H} \mid |q| = 1\}$ with the matrices above, so SU(2), S^3, and the unit quaternions are three names for one object.

The Lie algebra of SU(2) was identified in section 1.4 as the space of traceless skew-Hermitian 2×2 matrices. Its concrete shape is worth having in view before exponentiating.

Example 1.5.2: The Lie Algebra $\mathfrak{su}(2)$ in Coordinates

By theorem 1.4.3 the Lie algebra of SU(2) is

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) \mid X^* = -X,\ \mathrm{tr}X = 0\}$$

the traceless skew-Hermitian matrices. Skew-Hermiticity forces the diagonal entries to be purely imaginary and the off-diagonal entries to be conjugate-negatives, and tracelessness makes the two diagonal entries opposite, so a general element is

$$X = \begin{pmatrix} ic & -\bar{b} \\ b & -ic \end{pmatrix}, \qquad c \in \mathbb{R},\ b \in \mathbb{C}$$

depending on the three real parameters c and $\mathrm{Re}\, b, \mathrm{Im}\, b$; thus $\dim_{\mathbb{R}} \mathfrak{su}(2) = 3 = 2^2 - 1$, matching the dimension of SU(2) from proposition 1.2.6. A convenient basis is provided by the matrices

$$E_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad E_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad E_3 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

which are $\frac{1}{2i}$ times the Pauli matrices, and whose brackets are $[E_1, E_2] = E_3$, $[E_2, E_3] = E_1$, $[E_3, E_1] = E_2$, the cross-product relations of $\mathbb{R}^3$; this is the identification $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$ recorded in example 1.4.7.

The determinant provides a natural quadratic form on this space, and its shape is what makes SU(2) act by rotations below. For $X = \begin{pmatrix} ic & -\bar{b} \\ b & -ic \end{pmatrix} \in \mathfrak{su}(2)$,

$$\det X = (ic)(-ic) - (-\bar{b})(b) = c^2 + |b|^2 \tag{1.6}$$

a positive-definite quadratic form. Evaluating on the basis gives $\det E_j = \frac{1}{4}$ for each j, and since $\det(sX) = s^2 \det X$ for a scalar s, the form det is $\frac{1}{4}$ times the standard Euclidean squared norm in the coordinates dual to E_1, E_2, E_3; equivalently $\det X = -\frac{1}{2}\mathrm{tr}(X^2)$, because for traceless X the Cayley-Hamilton relation gives $X^2 = -(\det X)I$ and hence $\mathrm{tr}(X^2) = -2 \det X$. In either form the quadratic form is a scalar multiple of the trace form and is manifestly invariant under conjugation, the property that drives the double cover.

The exponential map sends $\mathfrak{su}(2)$ into SU(2), and because $\mathfrak{su}(2)$ is three-dimensional the exponential admits a closed formula in the spirit of Euler's formula $e^{i\theta} = \cos\theta + i \sin\theta$. The mechanism is that every $X \in \mathfrak{su}(2)$ satisfies a quadratic relation, so its powers cycle and the exponential series resums into cosine and sine.

Proposition 1.5.3: The Exponential is Onto $\mathrm{SU}(2)$

For $X \in \mathfrak{su}(2)$ set $|X| = \sqrt{\det X}$, so that $X^2 = -|X|^2 I$. Then

$$\exp X = \cos(|X|)\, I + \frac{\sin(|X|)}{|X|}\, X \tag{1.7}$$

interpreted as $\exp 0 = I$ when $X = 0$. The map $\exp\colon \mathfrak{su}(2) \to \mathrm{SU}(2)$ is surjective.

***Proof*:**
Step 1: the quadratic relation. By the Cayley-Hamilton theorem a traceless 2×2 matrix X satisfies $X^2 - (\operatorname{tr} X)X + (\det X)I = 0$, which with $\operatorname{tr} X = 0$ reads $X^2 = -(\det X)I$. For $X \in \mathfrak{su}(2)$ the determinant is $|X|^2 \geq 0$ by (1.6), so writing $r = |X|$ gives $X^2 = -r^2 I$.

Step 2: resumming the series. Assume $r > 0$. The relation $X^2 = -r^2 I$ gives $X^{2k} = (-1)^k r^{2k} I$ and $X^{2k+1} = (-1)^k r^{2k} X$ for all $k \geq 0$. Splitting the exponential series (absolutely convergent by definition 1.3.2) into even and odd powers,

$$\exp X = \sum_{k \geq 0} \frac{X^{2k}}{(2k)!} + \sum_{k \geq 0} \frac{X^{2k+1}}{(2k+1)!} = \left(\sum_{k \geq 0} \frac{(-1)^k r^{2k}}{(2k)!} \right) I + \left(\sum_{k \geq 0} \frac{(-1)^k r^{2k}}{(2k+1)!} \right) X$$

The first bracketed sum is $\cos r$; the second is $\frac{1}{r} \sum_k \frac{(-1)^k r^{2k+1}}{(2k+1)!} = \frac{\sin r}{r}$. This is (1.7). For $X = 0$ both sides equal I, and the right-hand side has the removable singularity $\frac{\sin r}{r} \to 1$ as $r \to 0$, so the formula holds uniformly with the stated convention.

Step 3: surjectivity. Let $A \in \mathrm{SU}(2)$; by proposition 1.5.1 write $A = \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix}$. Then $\operatorname{tr} A = \alpha + \overline{\alpha} = 2 \operatorname{Re} \alpha \in [-2, 2]$, so there is $r \in [0, \pi]$ with $\cos r = \operatorname{Re} \alpha$. If $A = I$ then $A = \exp 0$; if $A = -I$ then $\operatorname{Re} \alpha = -1$, $r = \pi$, and $A = \exp(\pi Y)$ for any $Y \in \mathfrak{su}(2)$ with $|Y| = 1$, since (1.7) gives $\exp(\pi Y) = \cos \pi\, I + \sin \pi\, Y = -I$. Otherwise $\operatorname{Re} \alpha \in (-1, 1)$, so $\sin r > 0$, and the matrix

$$X = \frac{r}{\sin r} \big(A - \operatorname{Re}(\alpha)\, I \big)$$

lies in $\mathfrak{su}(2)$: the matrix $A - \operatorname{Re}(\alpha) I$ is traceless because $\operatorname{tr} A = 2 \operatorname{Re} \alpha$, and skew-Hermitian because its diagonal entries are $\pm i \operatorname{Im} \alpha$ (purely imaginary) while its off-diagonal entries $-\overline{\beta}, \beta$ are conjugate-negatives; scaling by the real factor $r / \sin r$ preserves both properties. This X has $|X| = r$, since $\det X = (r/\sin r)^2 \det(A - \operatorname{Re}(\alpha) I) = (r/\sin r)^2 (\operatorname{Im} \alpha)^2 + (r/\sin r)^2 |\beta|^2$ and $\sin^2 r = 1 - \operatorname{Re}(\alpha)^2 = (\operatorname{Im} \alpha)^2 + |\beta|^2$ (using $|\alpha|^2 + |\beta|^2 = 1$), giving $\det X = r^2$. Substituting into (1.7),

$$\exp X = \cos(r)\, I + \frac{\sin r}{r} \cdot \frac{r}{\sin r} \big(A - \operatorname{Re}(\alpha) I \big) = \operatorname{Re}(\alpha)\, I + A - \operatorname{Re}(\alpha)\, I = A$$

so $A = \exp X$. Every element of $\mathrm{SU}(2)$ is therefore an exponential, as we sought to show.

Formula (1.7) is the noncommutative avatar of Euler's formula: as t runs over $\mathbb{R}$, the curve $t \mapsto \exp(tX)$

traces a great circle on the sphere SU(2), periodic with period $2\pi/|X|$, and its image is the one-parameter subgroup generated by X from theorem 1.3.10. Surjectivity of the exponential is special to compact connected groups and is proved in general in Part III; for SU(2) it is visible by hand because the group is a sphere and every point lies on a great circle through the identity. The formula also makes the covering map computable: a rotation angle on the SU(2) side is $|X|$, and it corresponds to a rotation angle of $2|X|$ on the SO(3) side, the source of the factor of two in the double cover.

Example 1.5.4: One-Parameter Subgroups Through the Poles

Take $X = E_1 = \frac{1}{2}\,\mathrm{diag}(i, -i)$, so $|E_1| = \sqrt{\det E_1} = \frac{1}{2}$ and $tX = \frac{t}{2}\,\mathrm{diag}(i, -i)$ has $|tX| = |t|/2$. Formula (1.7) gives

$$\exp(tE_1) = \cos\Big(\frac{t}{2}\Big) I + 2\sin\Big(\frac{t}{2}\Big) E_1 = \begin{pmatrix} e^{it/2} & 0 \\ 0 & e^{-it/2} \end{pmatrix}$$

directly recovering the diagonal exponential $\exp \mathrm{diag}(a, -a) = \mathrm{diag}(e^a, e^{-a})$ with $a = it/2$. This one-parameter subgroup is the standard maximal torus of SU(2), the diagonal circle, which the compact theory of Part III returns to. It has period 4π in t, not 2π: the parameter must run to 4π before $\exp(tE_1)$ returns to I, whereas the corresponding rotation of $\mathbb{R}^3$ has period 2π. The doubling is again the signature of the cover.

The group SO(3) is 3-dimensional, connected, and compact, with Lie algebra of the same dimension. Its Lie algebra $\mathfrak{so}(3)$, the real skew-symmetric 3×3 matrices, is spanned by the infinitesimal rotations about the three coordinate axes,

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

whose brackets are $[L_1, L_2] = L_3$, $[L_2, L_3] = L_1$, $[L_3, L_1] = L_2$, the identical cross-product relations found for E_1, E_2, E_3 in example 1.5.2. The linear map $L_i \mapsto E_i$ is therefore an isomorphism of Lie algebras, so

$$\mathfrak{su}(2) \;\cong\; \mathfrak{so}(3) \;\cong\; (\mathbb{R}^3, \times) \tag{1.8}$$

all three being the three-dimensional real Lie algebra with the cross-product bracket. Concretely $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ via $v \mapsto (w \mapsto v \times w)$: the matrix L_v of "cross with v" acts on w as $v \times w$, and $L_{v\times v'} = [L_v, L_{v'}]$ recovers the Jacobi identity for the cross product. The isomorphism (1.8) predicts, by the correspondence of Chapter 3, a covering homomorphism between the associated connected groups. The rest of the section constructs it explicitly.

The construction is the adjoint action: SU(2) acts on its own Lie algebra by conjugation, and because $\mathfrak{su}(2)$ carries the invariant quadratic form (1.6), this action lands in the orthogonal group of that form, which is a copy of SO(3).

Theorem 1.5.5: The Double Cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$

For $g \in \mathrm{SU}(2)$ let $\varphi(g)$ be the conjugation map

$$\varphi(g)\colon \mathfrak{su}(2) \to \mathfrak{su}(2), \qquad \varphi(g)(X) = gXg^{-1}$$

Identifying $\mathfrak{su}(2)$ with $\mathbb{R}^3$ via the basis E_1, E_2, E_3, with the Euclidean structure given by the form det of (1.6), the assignment $g \mapsto \varphi(g)$ is a homomorphism

$$\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$$

It is surjective, and its kernel is exactly the center $\{\pm I\}$. Hence φ induces an isomorphism

$$\mathrm{SU}(2)/\{\pm I\} \xrightarrow{\sim} \mathrm{SO}(3)$$

and is a two-to-one covering map.

***Proof*:**
Step 1: $\varphi(g)$ is a well-defined orthogonal transformation. For $g \in \mathrm{SU}(2)$ and $X \in \mathfrak{su}(2)$ the matrix gXg^{-1} is again traceless (trace is conjugation-invariant) and skew-Hermitian, since $g^{-1} = g^*$ gives $(gXg^*)^* = gX^*g^* = -gXg^*$; so $\varphi(g)$ maps $\mathfrak{su}(2)$ to itself, linearly, with inverse $\varphi(g^{-1})$. It preserves the form (1.6) because the determinant is conjugation-invariant, $\det(\varphi(g)X) = \det(gXg^{-1}) = \det X$. Thus $\varphi(g)$ is a linear isometry of the Euclidean space $(\mathfrak{su}(2), \det) \cong \mathbb{R}^3$, that is $\varphi(g) \in \mathrm{O}(3)$. That φ is a homomorphism is immediate: $\varphi(gh)(X) = (gh)X(gh)^{-1} = g(hXh^{-1})g^{-1} = \varphi(g)\varphi(h)(X)$, and $\varphi(I) = \mathrm{id}$.

Step 2: the image lies in $\mathrm{SO}(3)$. The group $\mathrm{SU}(2)$ is connected by proposition 1.5.1, and $\varphi\colon \mathrm{SU}(2) \to \mathrm{O}(3)$ is continuous (its matrix entries are quadratic polynomials in the entries of g), so the image $\varphi(\mathrm{SU}(2))$ is a connected subset of $\mathrm{O}(3)$. The determinant $\det \circ \varphi\colon \mathrm{SU}(2) \to \{\pm 1\}$ is continuous on a connected space, hence constant, and it takes the value $\det \varphi(I) = \det(\mathrm{id}) = 1$; therefore $\det \varphi(g) = 1$ for all g, so $\varphi(\mathrm{SU}(2)) \subseteq \mathrm{SO}(3)$ and $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$.

Step 3: the kernel is $\{\pm I\}$. Suppose $\varphi(g) = \mathrm{id}$, that is $gXg^{-1} = X$ for all $X \in \mathfrak{su}(2)$, so g commutes with every traceless skew-Hermitian matrix. The matrices E_1, E_2, E_3 span $\mathfrak{su}(2)$ over $\mathbb{R}$, and together with I they span $M_2(\mathbb{C})$ over $\mathbb{C}$ (they are, up to scalars, the Pauli matrices, which with I form a basis of $M_2(\mathbb{C})$); so g commutes with all of $M_2(\mathbb{C})$. An element commuting with every matrix is a scalar, $g = \lambda I$, and $g \in \mathrm{SU}(2)$ forces $\lambda^2 = \det g = 1$, hence $\lambda = \pm 1$. Thus $\ker \varphi = \{\pm I\}$, and since $\pm I$ are the only scalars in $\mathrm{SU}(2)$, this is exactly the center.

Step 4: surjectivity. Fix a unit vector $u \in \mathfrak{su}(2)$, so $u^2 = -I$ by Step 1 of proposition 1.5.3. The one-parameter group $s \mapsto \varphi(\exp(su))$ in $\mathrm{SO}(3)$ has derivative at $s = 0$ equal to the operator $\mathrm{ad}(u)(X) = uX - Xu = [u, X]$ on $\mathfrak{su}(2)$, since $\frac{d}{ds}\big|_0 \exp(su)X\exp(-su) = uX - Xu$ by proposition 1.3.4; hence $\varphi(\exp(su)) = \exp(s\ \mathrm{ad}(u))$ as maps of $\mathfrak{su}(2)$. Under the identification (1.8) the operator $\mathrm{ad}(u)$ is rotation generator about the axis u; because the structure constants of E_1, E_2, E_3 in the orthonormal frame of the form (1.6) are twice those of the geometric generators L_i (each E_j has $\det E_j = \frac{1}{4}$, i.e. Euclidean length $\frac{1}{2}$, while the L_i have length 1), the flow $\exp(s\ \mathrm{ad}(u))$ is the rotation of $\mathbb{R}^3$ by angle $2s$ about the axis u. Consequently, given $R \in \mathrm{SO}(3)$, Euler's theorem writes R as rotation by some angle θ about some unit axis $u \in S^2 \subset \mathfrak{su}(2)$, and then $g = \exp(\frac{\theta}{2}u)$ satisfies $\varphi(g) = \exp(\frac{\theta}{2}\,\mathrm{ad}(u)) =$ rotation by θ about $u = R$. As R was arbitrary, φ is surjective.

Step 5: the covering. By Steps 2 through 4, $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ is a surjective homomorphism with kernel $\{\pm I\}$, so the first isomorphism theorem for groups ([1]) gives a group isomorphism $\mathrm{SU}(2)/\{\pm I\} \xrightarrow{\sim} \mathrm{SO}(3)$. Both sides are compact three-dimensional manifolds, and the induced map is a continuous bijection between compact Hausdorff spaces, hence a homeomorphism; its differential at the identity is the Lie algebra isomorphism (1.8) (Step 4 computed $d\varphi_I = \frac{1}{2}\,\mathrm{ad}$ under the frames, a linear isomorphism), so it is a diffeomorphism. Finally φ is two-to-one with fibers $\{g, -g\} = g \cdot \ker\varphi$, and since $d\varphi_I$ is an isomorphism φ is a local diffeomorphism at I, spread to every point by left translation; a two-to-one local diffeomorphism between compact manifolds with discrete fibers is a two-sheeted covering map, completing the proof.

The factor of two is the arithmetic heart of the cover, and it is worth isolating. For a unit vector $u \in \mathfrak{su}(2)$ the element $\exp(\frac{\theta}{2}u)$ on the SU(2) side maps under φ to rotation by the full angle θ on the SO(3) side: the group parameter is half the spatial rotation angle. This is exactly the discrepancy seen in example 1.5.4, where $\exp(tE_1)$ had period 4π while its image rotation of $\mathbb{R}^3$ had period 2π. Passing to $-I$ at the halfway point $t = 2\pi$ is the mechanism: the two elements $\pm g$ of a fiber are the two lifts of a single rotation, and no continuous choice of lift exists over all of SO(3).

The theorem says that SO(3) is SU(2) with antipodal points of the sphere S^3 glued together, which is the standard model of three-dimensional real projective space. So $\mathrm{SO}(3) \cong \mathbb{R}P^3$ as a manifold: the rotation group of ordinary space is real projective three-space. The two preimages $\pm g$ of a rotation $R = \varphi(g)$ are the two ways to represent that rotation by a unit quaternion, differing by a sign, and the impossibility of choosing the sign continuously over all of SO(3) is the failure of SO(3) to be simply connected, computed next.

Corollary 1.5.6: The Fundamental Group of SO(3)

The map $\varphi\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ is the universal cover of SO(3), and

$$\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2$$

The generator is any loop in SO(3) that lifts to a path in SU(2) from I to $-I$, such as the image under φ of $t \mapsto \exp(tE_1)$, $t \in [0, 2\pi]$; this loop is not contractible, but traversing it twice is.

***Proof*:**
By proposition 1.5.1 the total space $\mathrm{SU}(2) \cong S^3$ is simply connected, and by theorem 1.5.5 the map φ is a covering with fiber the two-element group $\{\pm I\}$. A covering map with simply-connected total space is the universal cover of its base, and for a universal cover arising from a central subgroup (here $\{\pm I\}$) the fundamental group of the base is the deck group, which is the kernel: $\pi_1(\mathrm{SO}(3)) \cong \ker\varphi = \{\pm I\} \cong \mathbb{Z}/2$. These covering-space facts are proved in [4].
For the explicit generator, the path $\gamma(t) = \exp(tE_1)$ for $t \in [0, 2\pi]$ runs in SU(2) from $\gamma(0) = I$ to $\gamma(2\pi) = \mathrm{diag}(e^{i\pi}, e^{-i\pi}) = -I$ by example 1.5.4, so $\varphi \circ \gamma$ is a loop in SO(3) (it returns to $\varphi(-I) = \varphi(I) = \mathrm{id}$) whose lift γ has distinct endpoints $I \neq -I$. A loop whose lift to the universal cover is not closed is noncontractible, so $\varphi \circ \gamma$ generates $\pi_1(\mathrm{SO}(3))$. Doubling it, $t \in [0, 4\pi]$,

produces a lift $\gamma(t)$ from I back to $\gamma(4\pi) = I$, a closed lift, so the doubled loop is contractible; the generator therefore has order two, completing the proof.

The order-two element of $\pi_1(\mathrm{SO}(3))$ is the mathematics behind the physical fact that a continuous 2π rotation of a rigid body is not homotopic to no rotation, while a 4π rotation is: the "belt trick", in which a 360° twist in a belt cannot be undone without moving the ends but a 720° twist can, is a demonstration of a noncontractible loop in SO(3) together with its contractible square. In quantum mechanics the same two-to-one relationship is why the state of a spin-$\frac{1}{2}$ particle acquires a sign under a 2π rotation and returns to itself only after 4π: the physical symmetry group is SO(3), but the group acting on wavefunctions is its double cover SU(2), and the sign is the nontrivial element of the kernel. The representations that see this sign, the ones that do not descend to SO(3), are the half-integer-spin representations of SU(2) classified in Part III.

This section launches the running example that threads the rest of the book. The group SU(2) reappears in each part, always as the smallest place to see a new phenomenon in full: its diagonal circle is the maximal torus whose conjugates cover the group, and the reflection swapping $\mathrm{diag}(z, \overline{z})$ to $\mathrm{diag}(\overline{z}, z)$ is the Weyl group $\mathbb{Z}/2$, both taken up in the compact theory of maximal tori. Its irreducible representations are the symmetric powers of the standard two-dimensional representation, one in each dimension, and their characters are the functions $\chi_n(\theta) = \sin((n+1)\theta)/\sin\theta$ that the Weyl character formula produces as its first nontrivial instance. Its noncompact sibling $\mathrm{SL}_2(\mathbb{R})$, sharing the complexified Lie algebra $\mathfrak{sl}_2(\mathbb{C})$ but replacing the sphere by a group with a hyperbolic plane inside it, is the running example of the theory of real forms and the Iwasawa decomposition in Part IV. Every one of these returns to the concrete matrix computations of this section.

Exercise 1.5.1

1. Using (1.7), compute $\exp(\theta E_3)$ explicitly as a 2×2 complex matrix, and verify that $\exp(2\pi E_3) = -I$ while $\exp(4\pi E_3) = I$. Interpret the period 4π in terms of the double cover.
2. Show that the map sending the quaternion $q = a+bi+cj+dk$ (with $a, b, c, d \in \mathbb{R}$) to the matrix $\left(\begin{smallmatrix} a+bi & c+di \\ -c+di & a-bi \end{smallmatrix}\right)$ restricts to an isomorphism from the group of unit quaternions $\{q \mid |q| = 1\}$ onto SU(2), and that under this identification $\mathfrak{su}(2)$ is the space of purely imaginary quaternions $bi + cj + dk$.
3. Fix $g = \exp(\frac{\theta}{2}E_3)$. Compute $\varphi(g)$ as a 3×3 matrix in the basis E_1, E_2, E_3 by evaluating gE_1g^{-1}, gE_2g^{-1}, and gE_3g^{-1}, and confirm that $\varphi(g)$ is the rotation of $\mathbb{R}^3$ by angle θ about the third axis.
4. Prove that the center of SU(2) is exactly $\{\pm I\}$, and deduce from theorem 1.5.5 that SO(3) has trivial center.
5. Show that there is no continuous group homomorphism $\psi\colon \mathrm{SO}(3) \to \mathrm{SU}(2)$ with $\varphi\circ\psi = \mathrm{id}_{\mathrm{SO}(3)}$; that is, the double cover does not split. Deduce that SU(2) and $\mathrm{SO}(3)\times\{\pm I\}$ are not isomorphic as Lie groups.
6. Explain why SU(2) and SO(3) are not isomorphic as Lie groups despite having isomorphic Lie algebras, and reconcile this with the principle that a Lie algebra determines its simply-connected group uniquely.

2

Lie Groups and Their Lie Algebras

The previous chapter met the classical groups at the surface, as sets of matrices cut out by explicit equations, and produced the two objects on which the theory turns, the exponential map and the Lie algebra, by differentiating matrix-valued functions. Nothing in those constructions truly required the ambient matrices. The tangent space at the identity, the one-parameter subgroups, the bracket: each was a manifestation of the group law and the smooth structure alone, and each survives when the matrices are removed and only a manifold carrying a compatible group law remains. That manifold is a Lie group in the sense of this chapter, and the task now is to rebuild the machinery of chapter 1 intrinsically, on any Lie group, with no embedding into GL_n presupposed.

The mechanism that replaces the matrix computations is left-invariance. A single tangent vector at the identity spreads, by left translation, to a vector field defined over the whole group; the vector fields arising this way are closed under the Lie bracket of vector fields, and this bracket makes the tangent space at the identity a Lie algebra $\mathfrak{g}$, the intrinsic counterpart of the matrix Lie algebra of chapter 1. The flows of these left-invariant fields are the one-parameter subgroups, and assembling them at time one defines the exponential map $\exp\colon \mathfrak{g} \to G$. Differentiating the action of the group on itself by conjugation produces the adjoint representations Ad and ad, and recovers the bracket in the form $[X, Y] = \mathrm{ad}(X)Y$. For a matrix group every one of these constructions returns the answer already computed by hand: the intrinsic exponential is the matrix exponential, $\mathrm{Ad}(g)$ is conjugation $X \mapsto gXg^{-1}$, and $\mathrm{ad}(X)$ is the commutator $Y \mapsto XY - YX$. The abstract theory is thus anchored at every step to the concrete picture of the first chapter, and the reader who keeps $\mathrm{SU}(2)$ in view loses nothing by the ascent to generality.

2.1 Lie Groups: Definition, Examples, and the Category of Lie Groups

Chapter 1 studied groups of matrices: closed subgroups of $\mathrm{GL}_n(\mathbb{F})$, cut out by polynomial equations, whose smooth structure came for free from the ambient vector space $M_n(\mathbb{F})$. Every construction there was anchored in coordinates. This chapter rebuilds the same theory intrinsically. A Lie group will be defined as a manifold carrying a compatible group structure, with no reference to an ambient $\mathrm{GL}_n(\mathbb{F})$, and the tangent space at the identity will replace the concrete matrix Lie algebra. The payoff is generality: the theorems apply to groups that are not presented as matrix groups, and the constructions become functorial. The cost is that the smoothness which was automatic for matrix groups must now be built into the definition. This first section pays that cost, sets up the category of Lie groups, and records the first structural consequence of merging manifold and group, namely that a Lie group looks the same near every one of its points.

The unifying theme of the section is that the group structure of a Lie group is rigid in a way that an abstract group's is not: multiplication is a smooth map, so it can be differentiated, and the single operation of left translation by a group element is simultaneously a group-theoretic bijection and a diffeomorphism of manifolds. This dual nature is the source of every result below. Left translation transports the neighborhood of the identity to the neighborhood of any point, forcing the manifold to be homogeneous; it trivializes the tangent bundle, forcing parallelizability; and it preserves orientation, forcing orientability. Each of these is a geometric fact about G read off from a purely algebraic operation.

Definition 2.1.1: Lie Group

A *Lie group* is a smooth manifold G that is also a group, such that the multiplication and inversion maps

$$m\colon G\times G\to G,\quad (g,h)\mapsto gh,\qquad \iota\colon G\to G,\quad g\mapsto g^{-1}$$

are smooth, where $G\times G$ carries the product smooth structure. The dimension of G as a Lie group is its dimension as a manifold.

The two smoothness requirements can be collapsed into one. Smoothness of m and ι is equivalent to smoothness of the single map

$$\mu\colon G\times G\to G,\qquad (g,h)\mapsto gh^{-1}$$

Indeed, if m and ι are smooth then $\mu = m\circ(\mathrm{id}_G\times\iota)$ is a composite of smooth maps, hence smooth. Conversely, given μ smooth, inversion is recovered as $\iota(g)=\mu(e,g)$, the restriction of μ to the smooth submanifold $\{e\}\times G$, so ι is smooth; and then $m(g,h)=\mu(g,\iota(h))=\mu(g,h^{-1})$ is again a composite of smooth maps. The one-map formulation is the more economical to verify in examples, and it is the version used to check that a subset of a Lie group inheriting the two operations is itself a Lie group.

The manifold structure and the group structure are not independent data tacked together; the smoothness axiom forces them to interact. The first and most basic interaction is that the group elements act on G by diffeomorphisms. This is the intrinsic replacement for the observation, made throughout Chapter 1, that multiplication by a fixed matrix is a smooth invertible self-map of $\mathrm{GL}_n(\mathbb{F})$ with smooth inverse, visible directly from proposition 1.1.2.

Proposition 2.1.2: Translations and Conjugations are Diffeomorphisms

Let G be a Lie group and $g \in G$. The *left translation*, *right translation*, and *conjugation* maps

$$L_g\colon G \to G, \quad h \mapsto gh, \qquad R_g\colon G \to G, \quad h \mapsto hg, \qquad C_g\colon G \to G, \quad h \mapsto ghg^{-1}$$

are diffeomorphisms of G, with inverses $L_g^{-1} = L_{g^{-1}}$, $R_g^{-1} = R_{g^{-1}}$, and $C_g^{-1} = C_{g^{-1}}$. Consequently:

1. G is *homogeneous*: for any two points $g, h \in G$ the diffeomorphism $L_{hg^{-1}}$ carries g to h, so G looks the same near every point.
2. G is *parallelizable*: its tangent bundle TG is trivial, $TG \cong G \times T_eG$.
3. G is *orientable*.

***Proof*:**
Step 1: the translations are diffeomorphisms. Fix $g \in G$. Left translation L_g is the composite

$$G \xrightarrow{h \mapsto (g,h)} G \times G \xrightarrow{m} G$$

of the smooth map $h \mapsto (g, h)$ (smooth because its two components, the constant g and the identity, are smooth) with the smooth multiplication m of definition 2.1.1, so L_g is smooth. The group axioms give $L_g \circ L_{g^{-1}} = L_{gg^{-1}} = L_e = \mathrm{id}_G$ and likewise $L_{g^{-1}} \circ L_g = \mathrm{id}_G$, so L_g is a bijection with inverse $L_{g^{-1}}$, which is smooth by the same argument. Hence L_g is a diffeomorphism. The identical argument, using $h \mapsto (h, g)$ and the relation $R_g \circ R_{g^{-1}} = R_{g^{-1}g} = \mathrm{id}_G$, shows R_g is a diffeomorphism with inverse $R_{g^{-1}}$. Note that $C_g = L_g \circ R_{g^{-1}}$ is a composite of two diffeomorphisms, hence a diffeomorphism, with inverse $C_{g^{-1}} = L_{g^{-1}} \circ R_g$.

Step 2: homogeneity. Given $g, h \in G$, set $k = hg^{-1}$. Then L_k is a diffeomorphism by Step 1, and $L_k(g) = (hg^{-1})g = h$. A diffeomorphism carrying g to h identifies a neighborhood of g with a neighborhood of h compatibly with the smooth structure, so every point of G has a neighborhood diffeomorphic to a neighborhood of any other point. This is the assertion that G is homogeneous.

Step 3: parallelizability. For each $g \in G$ the differential of left translation at the identity,

$$(dL_g)_e\colon T_eG \to T_gG$$

is a linear isomorphism, since L_g is a diffeomorphism carrying e to g. Define

$$\Psi\colon G \times T_eG \to TG, \qquad (g, v) \mapsto (dL_g)_e(v)$$

The map Ψ is a bijection: for each g it restricts to the linear isomorphism $(dL_g)_e\colon \{g\} \times T_eG \to T_gG$ onto the fiber over g. That Ψ and its inverse are smooth follows from the smoothness of m: the assignment $(g, h) \mapsto (dL_g)_h$ is smooth because L_g is built from m, so Ψ is a smooth bundle isomorphism covering the identity of G. Its existence is exactly the statement that TG is trivial, $TG \cong G \times T_eG$, so G is parallelizable.

Step 4: orientability. A parallelizable manifold is orientable. Fix any basis $v_1, \ldots, v_n$ of T_eG, where $n = \dim G$. By Step 3 the vector fields $g \mapsto (dL_g)_e(v_i)$ are smooth and, at each point g, their values $(dL_g)_e(v_1), \ldots, (dL_g)_e(v_n)$ form a basis of T_gG, since $(dL_g)_e$ is a linear isomorphism.

> This global smooth frame determines an orientation of G: declare a basis of T_gG positively oriented when it has positive determinant relative to the frame. The frame is smooth, so the orientation is consistent across overlaps, and G is orientable, completing the proof.

The proposition is short, but it already separates Lie groups from arbitrary manifolds. Homogeneity says G has no distinguished point apart from the identity, and even the identity is distinguished only by the group structure, not by the geometry: near e and near any other g the manifold is indistinguishable. This is why so much of Lie theory is local analysis at the identity, the tangent space T_eG being the germ from which the whole group is reconstructed. Parallelizability is the geometric engine of the entire book. The trivialization $TG \cong G \times T_eG$ says that a tangent vector at e can be translated to a well-defined tangent vector at every point, producing a global vector field; these translated fields are the left-invariant vector fields of section 2.2, and their bracket is what makes T_eG a Lie algebra. Orientability then follows for free, which already rules out some manifolds as Lie groups: no even-dimensional real projective space $\mathbb{RP}^{2k}$ can be a Lie group, since it is not orientable.

The abstraction is only as convincing as the examples it captures. The matrix groups of Chapter 1 are the first source, and every group named there is a Lie group in the sense of definition 2.1.1. The next example collects these together with the smallest new examples, the circle and its products, which are not naturally presented as matrix groups at all.

Example 2.1.3: First Examples of Lie Groups

The following are Lie groups.

1. *The general linear group* $\mathrm{GL}_n(\mathbb{F})$, for $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. proposition 1.1.2 establishes exactly the content of definition 2.1.1: $\mathrm{GL}_n(\mathbb{F})$ is an open submanifold of $M_n(\mathbb{F})$, and multiplication and inversion are smooth. Its dimension is n^2 over $\mathbb{R}$ and $2n^2$ over $\mathbb{C}$. This is the ambient Lie group inside which the classical groups sit.

2. *The circle* $S^1 = \mathrm{U}(1)$. Regard $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ as the unit circle in $\mathbb{C}$, a one-dimensional smooth manifold, with group operation the multiplication of complex numbers. In the angle coordinate $z = e^{i\theta}$ the product is $e^{i\theta}e^{i\varphi} = e^{i(\theta+\varphi)}$ and the inverse is $e^{-i\theta}$, both smooth in θ, so S^1 is a one-dimensional abelian Lie group. As a group of 1×1 unitary matrices it is exactly $\mathrm{U}(1)$ of definition 1.2.5, and example 1.2.11 already identified $\mathrm{SO}(2) \cong \mathrm{U}(1) \cong S^1$.

3. *The tori* $\mathbb{T}^n = (S^1)^n$. The n-fold product of circles is a compact connected abelian Lie group of dimension n, with coordinatewise multiplication. Equivalently $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$, the quotient of the additive group $\mathbb{R}^n$ by the lattice $\mathbb{Z}^n$, with the quotient smooth structure. These are the maximal tori that control the structure theory of compact groups in Part III.

4. *Direct products.* If G and H are Lie groups, the product manifold $G \times H$ with componentwise multiplication $(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2)$ is a Lie group of dimension $\dim G + \dim H$; multiplication and inversion are smooth because they are so in each factor. The tori are the special case $G = H = S^1$ iterated.

5. *Discrete groups.* Any group Γ equipped with the discrete topology is a zero-dimensional Lie group: a 0-manifold is a countable set of points, and every map out of a discrete space is smooth, so m and ι are automatically smooth. Finite groups and the additive group $\mathbb{Z}$ are Lie groups in this way, of dimension 0.

6. *The classical groups.* The groups $\mathrm{SL}_n(\mathbb{F})$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and $\mathrm{Sp}(2n, \mathbb{F})$ of section 1.2 are each a smooth embedded submanifold of some $\mathrm{GL}_N(\mathbb{F})$, cut out as a regular-value preimage, on which the ambient multiplication and inversion restrict to smooth maps.

Each is therefore a Lie group. They are moreover *Lie subgroups* of $\mathrm{GL}_N(\mathbb{F})$: embedded submanifolds that are also subgroups. That every closed subgroup of a Lie group is automatically an embedded Lie subgroup, with no separate verification of the manifold structure required, is Cartan's closed subgroup theorem, proved in Chapter 3; here the manifold structure was already supplied by the regular value theorem in section 1.2.

7. *The running example* $\mathrm{SU}(2)$. By proposition 1.5.1, $\mathrm{SU}(2)$ is diffeomorphic to the 3-sphere S^3, and it is a compact, connected, simply connected Lie group of dimension 3. It is the smallest nonabelian compact Lie group and the thread this book follows from here.

The circle deserves a second look, because it exhibits a feature invisible in the matrix examples: a Lie group need not come with a preferred embedding into $\mathrm{GL}_n(\mathbb{F})$. The description $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ presents the torus as a quotient, with no matrices at all, and the two descriptions of S^1, as unit complex numbers and as $\mathbb{R}/\mathbb{Z}$, are related by the smooth surjective homomorphism $\mathbb{R} \to S^1$, $t \mapsto e^{2\pi it}$, whose kernel is $\mathbb{Z}$. This homomorphism is the exponential map of the circle in disguise, and the fact that S^1 is a quotient of its own tangent line by a discrete subgroup is the one-dimensional shadow of the covering-group theory of Chapter 4. That the same abstract Lie group admits several concrete presentations is precisely the freedom the intrinsic definition buys.

To organize maps between Lie groups one asks for the structure-preserving maps: those respecting both the group law and the smooth structure. Requiring smoothness of a group homomorphism is a strong condition, and Chapter 3 will show it is even automatic for continuous homomorphisms, so the class of morphisms is more robust than it first appears.

Definition 2.1.4: Homomorphism of Lie Groups

Let G and H be Lie groups. A *homomorphism of Lie groups* is a map $\varphi\colon G \to H$ that is both a group homomorphism, $\varphi(g_1g_2) = \varphi(g_1)\varphi(g_2)$ for all $g_1, g_2 \in G$, and a smooth map of manifolds. It is an *isomorphism of Lie groups* if it is in addition a bijection whose inverse $\varphi^{-1}\colon H \to G$ is also a homomorphism of Lie groups, that is, a group isomorphism that is a diffeomorphism.

A homomorphism of Lie groups automatically commutes with the translation structure of proposition 2.1.2: from $\varphi(gh) = \varphi(g)\varphi(h)$ one reads $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$, so φ intertwines left translation on G with left translation on H. This compatibility is what will force the differential $d\varphi_e$ at the identity to determine φ on all of a connected G, the theme of section 2.4. The determinant furnishes the running example of a homomorphism: $\det\colon \mathrm{GL}_n(\mathbb{F}) \to \mathbb{F}^\times$ is a group homomorphism, and it is smooth because the determinant is a polynomial in the matrix entries, so it is a homomorphism of Lie groups whose kernel is the Lie subgroup $\mathrm{SL}_n(\mathbb{F})$ of definition 1.2.1. The exponential $\mathbb{R} \to S^1$, $t \mapsto e^{2\pi it}$, and the inclusion $\mathrm{SU}(2) \hookrightarrow \mathrm{GL}_2(\mathbb{C})$ are two further homomorphisms already in hand.

With objects and morphisms fixed, the class of Lie groups assembles into a category. Naming it is not idle bookkeeping: the entire program of the book is to compare this category with the category of Lie algebras, and the functor Lie constructed in section 2.4 is the bridge between them.

Proposition 2.1.5: The Category of Lie Groups

Lie groups and their homomorphisms form a category **LieGrp**: the composite of two Lie group homomorphisms is a Lie group homomorphism, the identity map of a Lie group is a homomorphism, and composition is associative. An isomorphism in this category, in the categorical sense of a morphism with a two-sided inverse morphism, is exactly an isomorphism of Lie groups in the sense of definition 2.1.4.

***Proof*:**
If $\varphi\colon G \to H$ and $\psi\colon H \to K$ are Lie group homomorphisms, then $\psi\circ\varphi$ is a group homomorphism (the composite of group homomorphisms) and a smooth map (the composite of smooth maps), hence a Lie group homomorphism. For each Lie group G the identity map id_G is a group homomorphism and is smooth, so it is a Lie group homomorphism, and it is a two-sided identity for composition. Associativity of composition holds because composition of set maps is associative. Thus **LieGrp** is a category.

For the identification of isomorphisms: a morphism $\varphi\colon G \to H$ with a two-sided inverse morphism $\varphi^{-1}\colon H \to G$ in **LieGrp** is a bijection whose inverse is a Lie group homomorphism, hence a group isomorphism that is a diffeomorphism, which is a Lie group isomorphism by definition 2.1.4. Conversely a Lie group isomorphism is by definition a homomorphism admitting an inverse homomorphism, which is a two-sided inverse morphism in **LieGrp**, completing the proof.

One structural feature of a Lie group has no analogue for a bare topological group and organizes much of what follows: the piece of G containing the identity is itself a group. Let G^0 denote the connected component of G containing the identity e, the *identity component*. Because G is a manifold it is locally connected, so its connected components are open; in particular G^0 is an open subset of G. It is a subgroup: the map $(g,h) \mapsto gh^{-1}$ is continuous and sends $G^0 \times G^0$, a connected set containing (e,e), into a connected subset of G containing $ee^{-1} = e$; a connected set containing e lies in the component G^0, so $gh^{-1} \in G^0$ for all $g,h \in G^0$, and G^0 is a subgroup. It is normal: for any $g \in G$ the conjugation C_g of proposition 2.1.2 is a homeomorphism fixing e, so it carries the component G^0 onto the component of $C_g(e) = e$, which is again G^0; thus $gG^0g^{-1} = G^0$. Being open and a subgroup, G^0 has open cosets, so its cosets partition G into the connected components, and the quotient G/G^0 is a discrete group, the *component group*. For $\mathrm{GL}_n(\mathbb{R})$ this recovers the two components of proposition 1.1.3, with $G^0 = \mathrm{GL}_n^+(\mathbb{R})$ and component group $\mathbb{Z}/2$; for the connected classical groups of proposition 1.2.10 one has $G^0 = G$. The subgroup and quotient theory that turns G^0 and G/G^0 into the first tools for dissecting a Lie group is developed in full in Chapter 3.

Definition 2.1.6: Identity Component

The *identity component* G^0 of a Lie group G is the connected component of G containing the identity element e. It is an open normal subgroup of G, and the quotient G/G^0 is the *component group*, a discrete group whose elements are the connected components of G.

The identity component isolates the part of G accessible to the infinitesimal methods of this book. Everything the Lie algebra sees lies in G^0: the exponential image, the one-parameter subgroups, and the neighborhood of the identity all sit inside the connected piece, and section 2.3 will show that a connected Lie group is generated by any neighborhood of e, so G^0 is generated by exponentials. The passage from G^0 to G is the passage from the differential-geometric to the discrete: it is governed by the finite or countable component group, not by the Lie algebra, and this is why the theorems of Parts II and III are stated for connected groups. Two Lie groups with the same Lie algebra can differ only in their component groups and their fundamental groups, the discrete invariants the tangent space cannot detect, exactly as SU(2) and SO(3) share the Lie algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ of example 1.4.7 while differing by the factor $\mathbb{Z}/2$ of theorem 1.5.5.

Exercise 2.1.1

1. Let G and H be Lie groups. Prove in full that the product manifold $G \times H$, with the componentwise group operation, is a Lie group of dimension $\dim G + \dim H$, by exhibiting the map $\big((g_1, h_1), (g_2, h_2)\big) \mapsto (g_1 g_2^{-1}, h_1 h_2^{-1})$ as smooth and invoking the one-map criterion following definition 2.1.1. Identify the tangent space $T_{(e,e)}(G \times H)$ with $T_eG \oplus T_eH$.

2. Let G be a Lie group with identity e and inversion $\iota\colon G \to G$. Using that $m(g, \iota(g)) = e$ is constant, differentiate at e to prove that the differential of inversion at the identity is minus the identity,
$$d\iota_e = -\mathrm{id}_{T_eG}$$
(Hint: the differential of m at (e, e) is $dm_{(e,e)}(v, w) = v + w$ under the identification $T_{(e,e)}(G \times G) \cong T_eG \oplus T_eG$; establish this first from $m(g, e) = g$ and $m(e, h) = h$.)

3. Let $\varphi\colon G \to H$ be a homomorphism of Lie groups. Prove that φ has constant rank, that is, the rank of the differential $d\varphi_g\colon T_gG \to T_{\varphi(g)}H$ is the same for every $g \in G$. (Hint: use the intertwining relation $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$ and that translations are diffeomorphisms, proposition 2.1.2, to relate $d\varphi_g$ to $d\varphi_e$ by isomorphisms.)

4. Prove that $\mathbb{T}^n = (S^1)^n$ is a compact, connected, abelian Lie group of dimension n, and that the map $\mathbb{R}^n \to \mathbb{T}^n$, $(t_1, \ldots, t_n) \mapsto (e^{2\pi i t_1}, \ldots, e^{2\pi i t_n})$, is a surjective homomorphism of Lie groups with kernel $\mathbb{Z}^n$. Conclude that $\mathbb{T}^n \cong \mathbb{R}^n/\mathbb{Z}^n$ as groups.

5. Let G be a Lie group. Prove that the identity component G^0 is open in G, and that it is closed. Deduce that the connected components of G are exactly the cosets gG^0, that each is open and closed, and that G^0 is the smallest open subgroup of G. (Hint: an open subgroup is closed because its complement is a union of its nontrivial cosets, each of which is open.)

6. Using proposition 2.1.2, explain why the even-dimensional real projective spaces $\mathbb{RP}^{2k}$ and the Klein bottle carry no Lie group structure, while the odd-dimensional spheres S^1 and S^3 do. Which single property from proposition 2.1.2 does the obstruction rest on, and why is that property alone not enough to guarantee a manifold is a Lie group?

2.2 Left-Invariant Vector Fields and the Lie Algebra

Chapter 1 built the Lie algebra of a matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$ by hand, as the space of matrices X whose one-parameter subgroups $t \mapsto \exp(tX)$ stay inside G, and identified it with the tangent space T_IG. That construction leaned on the ambient matrix ring at every step: the exponential series, the commutator $XY - YX$, the trace. A general Lie group carries no ambient ring, so the same object must be built from the manifold alone. The raw material is the tangent space T_eG at the identity, a vector space of the right dimension but with no bracket in sight. The task of this section is to manufacture the bracket intrinsically, from nothing but the group multiplication and the differential calculus of [2].

The mechanism is left translation. A single tangent vector $v \in T_eG$ can be transported to every point g of the group by pushing it forward along the diffeomorphism $L_g \colon h \mapsto gh$, producing a vector field on all of G that is, by construction, insensitive to the group's own left multiplication. Such *left-invariant* vector fields form a vector space canonically isomorphic to T_eG, and, because the Lie bracket of two of them is again one, they form a Lie subalgebra of all vector fields. Transporting that bracket back to T_eG turns the identity tangent space into a Lie algebra in the abstract sense of definition 1.4.5, with no choice of coordinates, no exponential, and no matrices. This is the definition of the Lie algebra of a Lie group that the rest of the book uses, and its first job will be to reproduce, for $G = \mathrm{GL}_n(\mathbb{F})$, exactly the commutator of Chapter 1.

Definition 2.2.1: Left-Invariant Vector Field

Let G be a Lie group with identity e, and for $g \in G$ let $L_g \colon G \to G$, $L_g(h) = gh$, denote left translation, a diffeomorphism by proposition 2.1.2. A vector field X on G is *left-invariant* if it is L_g-related to itself for every $g \in G$, that is,

$$(L_g)_* X = X \qquad \text{for all } g \in G$$

Written out at a point, this is the condition

$$d(L_g)_h\big(X_h\big) = X_{gh} \qquad \text{for all } g, h \in G$$

Setting $h = e$ shows that a left-invariant field is determined by its value at the identity through

$$X_g = d(L_g)_e\big(X_e\big) \tag{2.1}$$

The set of left-invariant vector fields on G is written $\mathfrak{g}$.

The defining equation says that the flow of X commutes with left multiplication: transporting a tangent vector from h to gh by the differential of L_g lands on the value the field already assigns at gh. A left-invariant field is thus rigid data, pinned down everywhere by its single value X_e at the identity through (2.1). This is the first sign that $\mathfrak{g}$ will be no larger than T_eG. The reverse direction, that *every* $v \in T_eG$ arises this way from a genuine smooth vector field, is the substance of the isomorphism below, and it is where the smoothness of the group operations enters.

Example 2.2.2: Left-Invariant Fields on the Circle

Take $G = S^1$, realized as the unit complex numbers under multiplication, with identity $e = 1$. The tangent space T_1S^1 is one-dimensional, spanned by the velocity $v = i$ of the curve $\theta \mapsto e^{i\theta}$ at $\theta = 0$. Left translation by $g = e^{i\alpha}$ is $L_g(e^{i\theta}) = e^{i(\alpha+\theta)}$, whose differential rotates tangent vectors by α; pushing $v = i$ forward from 1 to g gives the velocity ig of the curve $\theta \mapsto e^{i(\alpha+\theta)}$ at its start. In the angular coordinate θ, this is the constant coordinate field

$$X = \frac{\partial}{\partial\theta}$$

the unit tangent that circulates the circle at constant angular speed. It is nowhere zero, so a single left-invariant field trivializes the tangent bundle of S^1; the general form of this phenomenon, parallelizability, appears in (2.3) below. Every left-invariant field on S^1 is a constant real multiple of $\partial/\partial\theta$, matching $\dim S^1 = 1$.

The isomorphism between left-invariant fields and T_eG is the load-bearing statement of the section. It has two parts. The evaluation map $X \mapsto X_e$ is manifestly linear and injective by (2.1); the work is in the inverse, showing that the field $g \mapsto d(L_g)_e v$ produced from a tangent vector v is smooth, not just a set-theoretic assignment of vectors. Once the isomorphism is in place, the bracket of vector fields descends to T_eG, and the abstract Lie algebra materializes.

Theorem 2.2.3: Left-Invariant Fields and the Bracket on T_eG

Let G be a Lie group.

1. The evaluation map
$$\varepsilon\colon \mathfrak{g} \to T_eG, \qquad X \mapsto X_e$$
is a linear isomorphism from the space of left-invariant vector fields to the tangent space at the identity. Its inverse sends $v \in T_eG$ to the field $\widetilde{v}$ defined by $\widetilde{v}_g = d(L_g)_e v$, which is smooth.

2. If X and Y are left-invariant, so is their Lie bracket $[X, Y]$. Hence $\mathfrak{g}$ is a Lie subalgebra of the Lie algebra of all vector fields on G.

3. Transporting the bracket through ε makes T_eG a Lie algebra: for $u, v \in T_eG$, set
$$[u, v] := \big[\widetilde{u}, \widetilde{v}\big]_e$$
This bracket is bilinear, alternating, and satisfies the Jacobi identity, so T_eG is a Lie algebra in the sense of definition 1.4.5.

***Proof*:**
Step 1: evaluation is a linear isomorphism. The map ε is linear because the value of a sum or scalar multiple of vector fields at e is the corresponding sum or scalar multiple of values. It is injective: if $X_e = 0$ then (2.1) gives $X_g = d(L_g)_e(0) = 0$ for every g, so X is the zero field. For surjectivity, fix $v \in T_eG$ and define $\widetilde{v}_g = d(L_g)_e v$. This assignment satisfies the left-invariance identity: for $g, h \in G$, the chain rule applied to $L_{gh} = L_g \circ L_h$ gives $d(L_{gh})_e = d(L_g)_h \circ d(L_h)_e$, whence
$$d(L_g)_h\big(\widetilde{v}_h\big) = d(L_g)_h\big(d(L_h)_e v\big) = d(L_{gh})_e v = \widetilde{v}_{gh}$$
so $\widetilde{v}$ is left-invariant provided it is a smooth vector field. Smoothness is the one point that uses more than the chain rule, and it is where the smoothness of multiplication enters. Let $m\colon G \times G \to G$ be the smooth multiplication. The value $\widetilde{v}_g = d(L_g)_e v$ is the image of v under the partial differential of m in its second argument at (g, e); a smooth map has smooth partial differentials, so $g \mapsto \widetilde{v}_g$ is a smooth section of TG. Concretely, choose a smooth curve $c\colon (-\epsilon, \epsilon) \to G$ with $c(0) = e$ and $c'(0) = v$; then
$$\widetilde{v}_g = \frac{d}{dt}\bigg|_0 m\big(g, c(t)\big) = \frac{d}{dt}\bigg|_0 g\, c(t)$$
and the right side depends smoothly on g because m is smooth and the t-derivative of the smooth family $g \mapsto m(g, c(\cdot))$ is smooth in g (see [2] for smooth dependence of derivatives on parameters). Thus $\widetilde{v} \in \mathfrak{g}$ and $\varepsilon(\widetilde{v}) = \widetilde{v}_e = d(L_e)_e v = v$, so ε is surjective with inverse $v \mapsto \widetilde{v}$, hence a linear isomorphism.

Step 2: the bracket of left-invariant fields is left-invariant. The input is the naturality of the Lie bracket under smooth maps, from [2]: if $\varphi\colon M \to N$ is smooth, X_1 on M is φ-related to Y_1 on N, and X_2 is φ-related to Y_2, then $[X_1, X_2]$ is φ-related to $[Y_1, Y_2]$. Apply this with $\varphi = L_g$ and $M = N = G$: a left-invariant field X is by definition L_g-related to itself, and likewise Y, so $[X, Y]$ is L_g-related to $[X, Y]$. As this holds for every $g \in G$, the bracket $[X, Y]$ is left-invariant. The bracket of vector fields is already bilinear, alternating, and satisfies the Jacobi identity on all vector fields ([2]); restricting to the subspace $\mathfrak{g}$, closed under the bracket by what was just shown, exhibits $\mathfrak{g}$ as a Lie subalgebra.

Step 3: transport to T_eG. By Step 1 the map ε is a linear isomorphism, so $[u, v] := [\widetilde{u}, \widetilde{v}]_e$ is a well-defined bilinear operation on T_eG, the image of the bracket on $\mathfrak{g}$ under ε. Transporting a Lie bracket through a linear isomorphism preserves each of its axioms: bilinearity, the alternating property $[u, u] = [\widetilde{u}, \widetilde{u}]_e = 0$, and the Jacobi identity, since each is a multilinear identity that ε^{-1} carries from $\mathfrak{g}$ to T_eG and ε carries back. Therefore $(T_eG, [-, -])$ satisfies the two axioms of definition 1.4.5, completing the proof.

The theorem is the intrinsic replacement for the entire vector-space and bracket apparatus of Chapter 1. Evaluation at e collapses the infinite-dimensional space of left-invariant fields onto the finite-dimensional T_eG with no loss, and the bracket of fields, which a priori knows about the whole manifold, is completely recorded by its value at a single point. That value is the intrinsic bracket. The construction is canonical: it required choosing neither coordinates nor a curve representing each tangent vector (the curve c in Step 1 was an auxiliary device whose t-derivative is coordinate-free), so the Lie-algebra structure on T_eG belongs to the group and to nothing else.

Definition 2.2.4: The Lie Algebra of a Lie Group

Let G be a Lie group. Its *Lie algebra* is the tangent space at the identity,

$$\mathfrak{g} = \mathrm{Lie}(G) := T_eG$$

equipped with the bracket $[u, v] = [\widetilde{u}, \widetilde{v}]_e$ transported from left-invariant vector fields by theorem 2.2.3. Via the isomorphism ε of that theorem, $\mathrm{Lie}(G)$ is identified with the Lie algebra of left-invariant vector fields, and the two descriptions are used interchangeably.

Because ε is a linear isomorphism onto T_eG, the Lie algebra has the dimension of the group:

$$\dim \mathfrak{g} = \dim T_eG = \dim G \tag{2.2}$$

Two notations for the same object are now in play, and both are useful. As a tangent space, $\mathfrak{g} = T_eG$ is where velocities of curves through e live and where the exponential map of section 2.3 will land its derivative. As a space of left-invariant fields, $\mathfrak{g}$ is where the bracket is visibly a Lie bracket of vector fields, with its Jacobi identity inherited rather than recomputed. The identification ε lets one pass freely between the pointwise picture and the global-field picture, and an element of $\mathfrak{g}$ should be read as both at once: a tangent vector X_e and the entire field X it generates.

A basis of $\mathfrak{g}$ has a geometric consequence worth recording immediately. If $v_1, \ldots, v_n$ is a basis of T_eG, then the left-invariant fields $\widetilde{v}_1, \ldots, \widetilde{v}_n$ are, at every point g, the images $d(L_g)_e v_1, \ldots, d(L_g)_e v_n$ of a basis under the isomorphism $d(L_g)_e$, hence a basis of T_gG. They therefore form a *global frame*

$$T_gG = \mathrm{span}_{\mathbb{R}}\big\{(\widetilde{v}_1)_g, \ldots, (\widetilde{v}_n)_g\big\} \quad \text{for every } g \in G \tag{2.3}$$

which trivializes the tangent bundle, $TG \cong G \times \mathbb{R}^n$. Every Lie group is thus *parallelizable*, the property already asserted in proposition 2.1.2, and (2.3) is its proof: left translation spreads any frame at e to a frame everywhere. This is a strong restriction on which manifolds can be groups. The circle S^1, the torus $\mathbb{T}^n$, and every Lie group are parallelizable, whereas the even-dimensional sphere S^2 is not: the hairy-ball theorem states that every continuous vector field on S^2 vanishes at some point, so S^2 admits no nowhere-zero field, let alone a global frame, and therefore carries no Lie group structure.

The next property makes the exponential map of the following section possible at all. A one-parameter subgroup will be the integral curve of a left-invariant field through the identity, and for it to be defined for every real time, the field must be complete. On a general manifold an integral curve can run off to infinity in finite time; left-invariance forbids this, because the flow near the identity can be translated to reproduce the flow near any point, extending every integral curve indefinitely.

Proposition 2.2.5: Left-Invariant Fields Are Complete

Every left-invariant vector field X on a Lie group G is complete: for each $g \in G$ the maximal integral curve of X starting at g is defined on all of $\mathbb{R}$. Consequently X generates a global flow $\theta\colon \mathbb{R} \times G \to G$, and $\dim \mathfrak{g} = \dim G$.

***Proof*:**
Step 1: integral curves are translates of one another. Let $\gamma\colon (a,b) \to G$ be an integral curve of X, so $\gamma'(t) = X_{\gamma(t)}$, and fix $g \in G$. The left-translated curve $\sigma(t) = g\,\gamma(t) = L_g(\gamma(t))$ is again an integral curve of X. Indeed, differentiating and using the chain rule,

$$\sigma'(t) = d(L_g)_{\gamma(t)}\big(\gamma'(t)\big) = d(L_g)_{\gamma(t)}\big(X_{\gamma(t)}\big) = X_{g\gamma(t)} = X_{\sigma(t)}$$

where the third equality is the left-invariance of X from definition 2.2.1. Thus L_g carries integral curves of X to integral curves of X, defined on the same time interval.

Step 2: extend the flow near e to all time. By the existence and uniqueness theorem for integral curves [2], there is an $\epsilon > 0$ and an integral curve $\gamma\colon (-\epsilon, \epsilon) \to G$ with $\gamma(0) = e$. Suppose, for contradiction, that the maximal integral curve γ_g of X with $\gamma_g(0) = g$ had a finite upper endpoint T for its domain. Choose a time t_0 with $T - \epsilon < t_0 < T$, and set $h = \gamma_g(t_0)$. By Step 1 applied to the base curve γ, the translate $t \mapsto h\,\gamma(t) = L_h(\gamma(t))$ is an integral curve of X defined on $(-\epsilon, \epsilon)$ with value h at $t = 0$. By uniqueness of integral curves it agrees with $\gamma_g(t_0 + t)$ wherever both are defined, and it extends γ_g to times up to $t_0 + \epsilon > T$, contradicting the maximality of T. The same argument rules out a finite lower endpoint. Hence γ_g is defined on all of $\mathbb{R}$, so X is complete.

Step 3: the global flow and the dimension. A complete vector field generates a global flow $\theta\colon \mathbb{R} \times G \to G$, $\theta(t, g) = \gamma_g(t)$, smooth by the smooth dependence of solutions on initial conditions [2]. Finally $\dim \mathfrak{g} = \dim T_eG = \dim G$ because ε of theorem 2.2.3 is a linear isomorphism, restating (2.2), which completes the proof.

The completeness of every left-invariant field is what will let the exponential map be defined on all of $\mathfrak{g}$, not just on a neighborhood of the origin: for each $X \in \mathfrak{g}$ the integral curve through the identity runs for all time, and its value at $t = 1$ is $\exp X$ in section 2.3. On a general manifold no such global object exists, because a vector field's integral curves may escape in finite time; the group structure removes this obstruction entirely. The proof extracts the mechanism cleanly: the flow of a left-invariant field is determined near the identity and then copied to every other point by left translation, so there is no room for finite-time blowup. This homogeneity of the flow is the analytic counterpart of the algebraic fact that a group looks the same at every point.

It remains to check that the intrinsic construction reproduces the concrete Lie algebra of Chapter 1 when $G = \mathrm{GL}_n(\mathbb{F})$. This is the compatibility that justifies calling the abstract T_eG the Lie algebra: it must specialize to the matrix commutator, or the two chapters would be describing different objects.

Proposition 2.2.6: The Intrinsic Lie Algebra of $\mathrm{GL}_n(\mathbb{F})$

Let $G = \mathrm{GL}_n(\mathbb{F})$, an open subset of $M_n(\mathbb{F})$ by proposition 1.1.2, so that its tangent space at the identity is canonically

$$T_I\mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F}) = \mathfrak{gl}_n(\mathbb{F})$$

the space of theorem 1.4.3. For $X \in M_n(\mathbb{F})$ the associated left-invariant vector field $\widetilde{X}$ is given at $A \in \mathrm{GL}_n(\mathbb{F})$ by

$$\widetilde{X}_A = AX \tag{2.4}$$

that is, $\widetilde{X}$ is the linear vector field $A \mapsto AX$ on $\mathrm{GL}_n(\mathbb{F})$.

***Proof*:**
Because $\mathrm{GL}_n(\mathbb{F})$ is open in the vector space $M_n(\mathbb{F})$, its tangent space at any point A is canonically $M_n(\mathbb{F})$ itself, and a tangent vector at A is the velocity of a curve of matrices through A. Left translation $L_A(B) = AB$ is the restriction to $\mathrm{GL}_n(\mathbb{F})$ of the linear map $B \mapsto AB$ on $M_n(\mathbb{F})$. The differential of a linear map is the map itself under this canonical identification, so

$$d(L_A)_I \colon M_n(\mathbb{F}) \to M_n(\mathbb{F}), \qquad Z \mapsto AZ$$

Applying (2.1) with $g = A$ and $X_e = X \in T_I\mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F})$ gives $\widetilde{X}_A = d(L_A)_I(X) = AX$, which is (2.4). This is smooth in A, confirming directly that $\widetilde{X}$ is a genuine vector field, as theorem 2.2.3 guarantees in general, completing the proof.

The left-invariant field of X on $\mathrm{GL}_n(\mathbb{F})$ is the matrix version of the constant field on the circle in example 2.2.2: it assigns to each group element A the tangent vector AX, spreading the single matrix $X = \widetilde{X}_I$ over the whole group by left multiplication. With (2.4) in hand the intrinsic bracket can be identified with the commutator. The Lie bracket of two vector fields on an open set of $M_n(\mathbb{F})$ is computed from their directional derivatives: writing the fields as $M_n(\mathbb{F})$-valued functions $A \mapsto AX$ and $A \mapsto AY$, the bracket is

$$[\widetilde{X}, \widetilde{Y}]_A = D(\widetilde{Y})_A(\widetilde{X}_A) - D(\widetilde{X})_A(\widetilde{Y}_A)$$

where $D(\widetilde{Y})_A$ is the derivative of $A \mapsto AY$, namely the linear map $Z \mapsto ZY$. Substituting $\widetilde{X}_A = AX$ and $\widetilde{Y}_A = AY$,

$$[\widetilde{X}, \widetilde{Y}]_A = (AX)Y - (AY)X = A(XY - YX)$$

Evaluating at $A = I$ gives $[\widetilde{X}, \widetilde{Y}]_I = XY - YX$. Through the identification ε of theorem 2.2.3, the intrinsic bracket on $T_I\mathrm{GL}_n(\mathbb{F}) = \mathfrak{gl}_n(\mathbb{F})$ is therefore the commutator

$$[X, Y] = XY - YX \tag{2.5}$$

recovering exactly the matrix bracket of proposition 1.4.4. The value $[\widetilde{X}, \widetilde{Y}]_A = A(XY - YX)$ at every A, not just at I, is itself the left-invariant field of the commutator, a computation that reconfirms Step 2 of theorem 2.2.3 in this case. The full, coordinate-free explanation of why the intrinsic bracket agrees with the commutator, expressed through the adjoint representation, is deferred to section 2.5, where it appears as theorem 2.5.4; the direct computation above is enough for now to see that the intrinsic Lie algebra of $\mathrm{GL}_n(\mathbb{F})$ is $\mathfrak{gl}_n(\mathbb{F})$ with its commutator, and a matrix subgroup $G \subseteq \mathrm{GL}_n(\mathbb{F})$ inherits the bracket of its ambient $\mathfrak{gl}_n(\mathbb{F})$.

For the running example, the identification closes cleanly. The intrinsic Lie algebra of $\mathrm{SU}(2)$ is $T_I\mathrm{SU}(2)$, and this tangent space is precisely the space of traceless skew-Hermitian matrices of example 1.4.7: a tangent vector at I is the velocity $c'(0)$ of a curve c in $\mathrm{SU}(2)$ through I, and differentiating the relations $c(t)^*c(t) = I$ and $\det c(t) = 1$ at $t = 0$ returns $X^* + X = 0$ and $\mathrm{tr} X = 0$. By (2.5) the intrinsic bracket on $T_I\mathrm{SU}(2)$ is the commutator, so the intrinsic Lie algebra of $\mathrm{SU}(2)$ is $\mathfrak{su}(2)$ with the bracket of example 1.4.7, and in particular it is the cross product on $\mathbb{R}^3$. The abstract machine of this section, applied to $\mathrm{SU}(2)$, returns the concrete algebra Chapter 1 computed by hand.

Exercise 2.2.1

1. On the n-torus $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$, with group operation addition of coordinates modulo 1 and identity $e = 0$, show that the constant coordinate fields $\partial/\partial\theta_1, \dots, \partial/\partial\theta_n$ are left-invariant, and that they form a basis of the Lie algebra $\mathrm{Lie}(\mathbb{T}^n)$. Compute all brackets $[\partial/\partial\theta_i, \partial/\partial\theta_j]$ and conclude that $\mathrm{Lie}(\mathbb{T}^n)$ is abelian, that is, $[u, v] = 0$ for all u, v.

2. Let G be a Lie group of dimension n and let $v_1, \dots, v_n$ be a basis of T_eG. Prove in detail that the left-invariant fields $\widetilde{v}_1, \dots, \widetilde{v}_n$ form a global frame, using that $d(L_g)_e$ is a linear isomorphism for every g. Deduce that TG is trivial, and explain why this shows the even-dimensional spheres S^{2k} with $k \geq 1$ carry no Lie group structure.

3. For $G = \mathrm{GL}_n(\mathbb{F})$ and $X \in M_n(\mathbb{F})$, verify by a direct differential computation that the field $\widetilde{X}_A = AX$ of proposition 2.2.6 is left-invariant: show that $d(L_B)_A(\widetilde{X}_A) = \widetilde{X}_{BA}$ for all $A, B \in \mathrm{GL}_n(\mathbb{F})$. Then take $X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ in $\mathfrak{gl}_2(\mathbb{R})$ and write out $\widetilde{X}_A$ for a general $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$.

4. A vector field X on G is *right-invariant* if $(R_g)_*X = X$ for all g, where $R_g(h) = hg$. State and prove the right-invariant analogue of part (1) of theorem 2.2.3: evaluation at e is a linear isomorphism from right-invariant fields to T_eG, with inverse $v \mapsto \widehat{v}$ where $\widehat{v}_g = d(R_g)_e v$. For $G = \mathrm{GL}_n(\mathbb{F})$, compute $\widehat{X}_A$ and contrast it with the left-invariant field of proposition 2.2.6.

5. Show that a connected Lie group G is abelian if and only if every left-invariant vector field is also right-invariant. (Left and right translations always commute by associativity; for the harder direction the exponential map of section 2.3 is used to conclude that the elements moved by these flows are central and generate G, so a clear argument reducing the claim to the commuting of left and right translations is what is asked for.)

6. Using only the left-invariant fields $\widetilde{X}_A = AX$ on $\mathrm{GL}_2(\mathbb{R})$ and the coordinate expression for the Lie bracket of vector fields, recompute $[\widetilde{X}, \widetilde{Y}]_I$ for $X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, and confirm it equals the commutator $XY - YX$ of proposition 1.4.4.

2.3 One-Parameter Subgroups and the Exponential Map

chapter 1 produced the exponential of a matrix group by summing a power series and produced the one-parameter subgroups of $\mathrm{GL}_n(\mathbb{F})$ as the curves $t \mapsto \exp(tX)$; the classification theorem theorem 1.3.10 showed these are all of them. Neither the series nor the classification made any use of the ambient matrices beyond the linear differential equation $\gamma' = X\gamma$ that a homomorphism $\mathbb{R} \to \mathrm{GL}_n(\mathbb{F})$ must satisfy. On a general Lie group there is no power series, but there is still a differential equation: a smooth homomorphism $\gamma\colon \mathbb{R} \to G$ is determined by its velocity $X = \gamma'(0)$, because X generates a left-invariant vector field whose flow line through the identity is γ. This section builds the intrinsic exponential map out of that flow, transferring every property of the matrix exponential to the abstract setting, and then verifies that for a matrix group the two exponentials coincide.

The linchpin is the observation, recorded in theorem 2.2.3 and proposition 2.2.5, that a tangent vector $X \in \mathfrak{g} = T_eG$ determines a left-invariant vector field X^L whose flow exists for all time. Because the field is left-invariant, its flow interacts with the group law in the simplest possible way: translating a flow line by a group element produces another flow line. That single fact turns the integral curve through the identity into a homomorphism, and it is what the first theorem establishes.

Theorem 2.3.1: One-Parameter Subgroups and the Lie Algebra

Let G be a Lie group with Lie algebra $\mathfrak{g} = T_eG$. A *one-parameter subgroup* of G is a smooth homomorphism $\gamma\colon \mathbb{R} \to G$, that is, a smooth curve with $\gamma(s+t) = \gamma(s)\gamma(t)$ for all $s, t \in \mathbb{R}$ (so in particular $\gamma(0) = e$). For each $X \in \mathfrak{g}$ let X^L be the left-invariant vector field with $X^L_e = X$ (definition 2.2.1) and let γ_X be the integral curve of X^L through e. Then:

1. γ_X is a one-parameter subgroup of G, defined for all $t \in \mathbb{R}$, with $\gamma_X'(0) = X$.
2. Every one-parameter subgroup γ of G equals γ_X for $X = \gamma'(0)$.

Consequently $X \mapsto \gamma_X$ is a bijection from $\mathfrak{g}$ onto the set of one-parameter subgroups of G, with inverse $\gamma \mapsto \gamma'(0)$.

***Proof*:**
The two statements are proved in turn.

1. *The integral curve is a homomorphism.* By proposition 2.2.5 the field X^L is complete, so its integral curve $\gamma_X : \mathbb{R} \to G$ through $\gamma_X(0) = e$ is defined for all $t \in \mathbb{R}$; it satisfies

$$\gamma_X'(t) = X^L_{\gamma_X(t)}, \qquad \gamma_X(0) = e \tag{2.6}$$

and $\gamma_X'(0) = X^L_e = X$ is the velocity at the identity.

Fix $s \in \mathbb{R}$ and consider the two curves

$$\alpha(t) = \gamma_X(s+t), \qquad \beta(t) = \gamma_X(s)\gamma_X(t) = L_{\gamma_X(s)}\big(\gamma_X(t)\big)$$

The claim is $\alpha = \beta$; since s is arbitrary, this claim is exactly the homomorphism property. Both curves start at the same point: $\alpha(0) = \gamma_X(s)$ and $\beta(0) = \gamma_X(s)\gamma_X(0) = \gamma_X(s)e = \gamma_X(s)$. Both satisfy (2.6). For α this is the chain rule applied to the time-translation

$t \mapsto s+t$: $\alpha'(t) = \gamma_X'(s+t) = X^L_{\gamma_X(s+t)} = X^L_{\alpha(t)}$. For β this is where left-invariance enters. Writing $g = \gamma_X(s)$ and using that L_g is a diffeomorphism (proposition 2.1.2),

$$\beta'(t) = \frac{d}{dt} L_g\big(\gamma_X(t)\big) = (dL_g)_{\gamma_X(t)}\big(\gamma_X'(t)\big) = (dL_g)_{\gamma_X(t)}\big(X^L_{\gamma_X(t)}\big)$$

The defining property of a left-invariant field is that it is L_g-related to itself, $(dL_g)_h(X^L_h) = X^L_{gh}$ for all g, h (definition 2.2.1); applying this with $h = \gamma_X(t)$ turns the last expression into $X^L_{g\gamma_X(t)} = X^L_{\beta(t)}$. Thus β also solves (2.6), now with initial point $\gamma_X(s)$.

Both α and β are integral curves of X^L through the common initial point $\gamma_X(s)$ at $t = 0$. By the uniqueness of integral curves of a smooth vector field ([2]), $\alpha = \beta$ for all t; that is, $\gamma_X(s+t) = \gamma_X(s)\gamma_X(t)$ for all $s, t \in \mathbb{R}$. Hence γ_X is a smooth homomorphism $\mathbb{R} \to G$, a one-parameter subgroup, with $\gamma_X'(0) = X$.

2. *Every one-parameter subgroup is an integral curve.* Let $\gamma\colon \mathbb{R} \to G$ be a one-parameter subgroup and set $X = \gamma'(0) \in T_eG = \mathfrak{g}$. Differentiating $\gamma(s+t) = \gamma(s)\gamma(t) = L_{\gamma(s)}(\gamma(t))$ with respect to t and evaluating at $t = 0$, where $\gamma(0) = e$ and $\gamma'(0) = X$,

$$\gamma'(s) = \frac{d}{dt}\bigg|_{t=0} L_{\gamma(s)}\big(\gamma(t)\big) = (dL_{\gamma(s)})_e\big(\gamma'(0)\big) = (dL_{\gamma(s)})_e(X) = X^L_{\gamma(s)}$$

the last equality being the definition of the left-invariant field X^L generated by X. So γ solves (2.6) with $\gamma(0) = e$, the same problem γ_X solves. Uniqueness of integral curves gives $\gamma = \gamma_X$.

Parts (1) and (2) show $X \mapsto \gamma_X$ is a surjection onto the one-parameter subgroups with a two-sided inverse $\gamma \mapsto \gamma'(0)$: part (1) gives $\gamma_X'(0) = X$, and part (2) gives $\gamma_{\gamma'(0)} = \gamma$. Hence the assignment is a bijection with the stated inverse, completing the proof.

The theorem is the intrinsic form of the classification theorem 1.3.10, and the parallel is exact. There, a one-parameter subgroup of $\mathrm{GL}_n(\mathbb{F})$ solved the linear equation $\gamma' = X\gamma$ and was pinned down by its initial velocity; here it solves the equation $\gamma' = X^L_\gamma$ of an integral curve and is pinned down the same way. The matrix version is the special case in which the left-invariant field X^L is the explicit vector field $A \mapsto AX$ and the flow can be summed as a power series. What has been removed is any need for that series: the group law forces the integral curve to be a homomorphism, and completeness forces it to run for all time, so the correspondence between generators and one-parameter subgroups holds on any Lie group. Evaluating each subgroup at a single time now packages the whole correspondence into one map.

Definition 2.3.2: The Exponential Map

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The *exponential map* $\exp\colon \mathfrak{g} \to G$ sends $X \in \mathfrak{g}$ to the value at time 1 of its one-parameter subgroup:

$$\exp(X) = \gamma_X(1)$$

where γ_X is the one-parameter subgroup of theorem 2.3.1 with $\gamma_X'(0) = X$.

The definition compresses the entire family $\{\gamma_X\}_{X\in\mathfrak{g}}$ into a single map on $\mathfrak{g}$, and the family is recovered from it by rescaling. For $X \in \mathfrak{g}$ and $t \in \mathbb{R}$, the curve $s \mapsto \gamma_X(ts)$ is a one-parameter subgroup with velocity $t\gamma_X'(0) = tX$ at $s = 0$; by the uniqueness in theorem 2.3.1 it is γ_{tX}, so $\gamma_{tX}(s) = \gamma_X(ts)$. Taking $s = 1$ gives

$$\exp(tX) = \gamma_{tX}(1) = \gamma_X(t) \tag{2.7}$$

so that $t \mapsto \exp(tX)$ is exactly the one-parameter subgroup γ_X, and exp traces out the whole flow as t varies. This identity is the workhorse of every property below: it turns statements about the family of flows into statements about the single map exp.

The next proposition collects the properties inherited from (2.7). They are the intrinsic counterparts of the algebraic properties of the matrix exponential in proposition 1.3.4, and the proofs are shorter because the analytic labor has already been done in building the flow.

Proposition 2.3.3: Properties of the Exponential Map

Let G be a Lie group with Lie algebra $\mathfrak{g}$ and exponential map $\exp\colon \mathfrak{g} \to G$. For all $X \in \mathfrak{g}$ and all $s, t \in \mathbb{R}$:

1. $\exp(0) = e$.
2. $\exp\big((s+t)X\big) = \exp(sX)\exp(tX)$, and consequently $\exp(-X) = \exp(X)^{-1}$; the curve $t \mapsto \exp(tX)$ is the one-parameter subgroup γ_X with velocity $\frac{d}{dt}\big|_0 \exp(tX) = X$.
3. $\exp\colon \mathfrak{g} \to G$ is smooth.

***Proof*:**
Part (1). The one-parameter subgroup generated by $X = 0$ is the constant curve $\gamma_0 \equiv e$, since the zero vector field has the constant integral curve through e; hence $\exp(0) = \gamma_0(1) = e$.

Part (2). By (2.7), $t \mapsto \exp(tX) = \gamma_X(t)$ is the one-parameter subgroup γ_X, so the homomorphism property $\gamma_X(s+t) = \gamma_X(s)\gamma_X(t)$ of theorem 2.3.1 reads

$$\exp\big((s+t)X\big) = \exp(sX)\exp(tX)$$

Setting $t = -s$ and using part (1) gives $\exp(sX)\exp(-sX) = \exp(0) = e$, so $\exp(-sX) = \exp(sX)^{-1}$; taking $s = 1$ gives $\exp(-X) = \exp(X)^{-1}$. The velocity statement is $\frac{d}{dt}\big|_0 \exp(tX) = \gamma_X'(0) = X$.

Part (3). Smoothness of exp is the smooth dependence of a flow on its initial data, applied to a single well-chosen field. Consider the Lie group G together with its Lie algebra $\mathfrak{g}$, and form on the product manifold $\mathfrak{g} \times G$ the vector field V whose value at (X, g) is $(0, X_g^L)$: at each point it is the zero vector in the $\mathfrak{g}$-direction and the left-invariant field X^L evaluated at g in the G-direction. The assignment $(X, g) \mapsto X_g^L$ is smooth in both variables, because $X_g^L = (dL_g)_e(X)$ depends smoothly on g through the smooth multiplication of G and linearly, hence smoothly, on X; so V is a smooth vector field on $\mathfrak{g} \times G$. Its integral curve through (X, e) is $t \mapsto (X, \gamma_X(t))$, since the first coordinate is constant and the second solves (2.6). This field is complete, as each X^L is complete by proposition 2.2.5. The flow of a complete smooth vector field is smooth jointly in time and initial condition ([2]), so the time-1 flow $(X, e) \mapsto (X, \gamma_X(1)) = (X, \exp X)$ is smooth; projecting to G shows $X \mapsto \exp(X)$ is smooth, completing the proof.

Part (2) is the reason the map deserves the name “exponential”: restricted to a single line $\mathbb{R}X$ in $\mathfrak{g}$ it converts addition of real multiples of X into multiplication in G, exactly as the scalar exponential converts addition into multiplication. The restriction to a line is essential. The full map exp is not a homomorphism from $(\mathfrak{g}, +)$ to G unless G is abelian, because $\exp(X)\exp(Y)$ and $\exp(X+Y)$ disagree already at second order when X and Y fail to commute, precisely the failure of $\exp(A+B) = \exp(A)\exp(B)$ recorded for matrices in proposition 1.3.5. What survives universally is the one-parameter law of part (2), and it is enough to make exp a chart near the identity.

The value of the differential of exp at the origin is what promotes the one-parameter law into a local coordinate system. As with the matrix exponential in theorem 1.3.7, this differential is the identity, and the reason is the same one-line computation: differentiating $t \mapsto \exp(tX)$ at $t = 0$ returns X.

Theorem 2.3.4: The Exponential is a Local Diffeomorphism

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Identifying $T_0\mathfrak{g}$ with $\mathfrak{g}$ and T_eG with $\mathfrak{g}$, the differential of exp at the origin is the identity,

$$d(\exp)_0 = \mathrm{id}_{\mathfrak{g}}$$

Consequently there are open neighborhoods U of 0 in $\mathfrak{g}$ and V of e in G such that $\exp\colon U \to V$ is a diffeomorphism.

***Proof*:**
The differential $d(\exp)_0$ acts on a tangent vector $X \in T_0\mathfrak{g} = \mathfrak{g}$ by differentiating exp along any curve through 0 with velocity X; the line $t \mapsto tX$ is such a curve, so by (2.7) and proposition 2.3.3(2),

$$d(\exp)_0(X) = \left.\frac{d}{dt}\right|_{t=0} \exp(tX) = \gamma_X'(0) = X$$

under the identification $T_eG = \mathfrak{g}$. Thus $d(\exp)_0 = \mathrm{id}_{\mathfrak{g}}$, which is invertible. Since exp is smooth (proposition 2.3.3) with invertible differential at 0, the inverse function theorem ([2]) provides open neighborhoods U of 0 and $V = \exp(U)$ of $\exp(0) = e$ on which exp is a diffeomorphism, completing the proof.

The theorem installs exp as a chart on G centered at the identity, with $\mathfrak{g}$ as the model vector space, so that near e the group is described by the linear data of its Lie algebra. This is the intrinsic version of the exponential chart of theorem 1.3.7, and it is what makes the Lie algebra a faithful record of the group near the identity. A first structural consequence is that a connected group hides nothing beyond what this chart already shows: the whole group is assembled from arbitrarily small pieces near e.

Corollary 2.3.5: Connected Groups Are Generated Near the Identity

Let G be a connected Lie group. Then G is generated by the image $\exp(\mathfrak{g})$; equivalently, G is generated by any neighborhood of the identity. Explicitly, every $g \in G$ is a finite product $g = \exp(X_1)\exp(X_2)\cdots\exp(X_k)$ with $X_1, \ldots, X_k \in \mathfrak{g}$.

***Proof*:**
Let V be the neighborhood of e on which exp is a diffeomorphism from a neighborhood U of 0, as in theorem 2.3.4; then $V \subseteq \exp(\mathfrak{g})$, so it suffices to show that any neighborhood W of e generates G. Let $H = \bigcup_{k \geq 1} W_{\text{sym}}^k$ be the subgroup generated by $W_{\text{sym}} = W \cap W^{-1}$, which is a symmetric neighborhood of e (the inversion map is a homeomorphism by proposition 2.1.2, so W^{-1} is open and W_{sym} is an open neighborhood of e) and W_{sym}^k denotes the set of k-fold products. Then H is a subgroup of G.

The subgroup H is open: for $h \in H$, the translate $hW_{\text{sym}} = L_h(W_{\text{sym}})$ is an open neighborhood of h contained in H, since L_h is a diffeomorphism (proposition 2.1.2) and $hW_{\text{sym}} \subseteq H$ as a product of elements of H. An open subgroup is also closed: its complement $G \setminus H = \bigsqcup_{gH \neq H} gH$ is a union of cosets, each open (being a translate of the open set H), hence $G \setminus H$ is open and H is closed. Thus H is a nonempty subset of the connected space G that is both open and closed, so $H = G$. Every $g \in G$ therefore lies in some $W_{\text{sym}}^k \subseteq \exp(\mathfrak{g})^k$, giving the product representation with each factor in $\exp(\mathfrak{g})$, as we sought to show.

The corollary explains why the Lie algebra is so powerful a tool for connected groups: knowing $\mathfrak{g}$ determines the local structure of G near e through the exponential chart, and connectedness then propagates that local structure to all of G by taking products. A homomorphism of connected groups is thereby determined by its behavior near the identity, and ultimately by its differential at e, a theme the next section develops in full. The corollary does not assert that exp is surjective; the elements of G are products of exponentials, not necessarily single exponentials, and the gap between the two is real, as the exercises make concrete.

Everything so far has been intrinsic, defined through flows of left-invariant fields with no reference to matrices. To close the loop with chapter 1, the intrinsic exponential must be shown to agree with the matrix exponential on a matrix group. The verification is a matter of identifying the left-invariant field on $\mathrm{GL}_n(\mathbb{F})$ and recognizing its flow.

Proposition 2.3.6: The Intrinsic Exponential is the Matrix Exponential

Let $G = \mathrm{GL}_n(\mathbb{F})$, whose Lie algebra is $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ identified with $T_I\mathrm{GL}_n(\mathbb{F})$. For every $X \in \mathfrak{gl}_n(\mathbb{F})$ the intrinsic exponential of definition 2.3.2 agrees with the matrix exponential of definition 1.3.2:

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

The same holds on every matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$: the intrinsic exponential of G is the restriction of the matrix exponential to $\mathfrak{g}$.

***Proof*:**
Step 1: the left-invariant field on $\mathrm{GL}_n(\mathbb{F})$. The group $\mathrm{GL}_n(\mathbb{F})$ is an open subset of the vector space $M_n(\mathbb{F})$ (proposition 1.1.2), so its tangent space at any point A is canonically $M_n(\mathbb{F})$, and left translation $L_A(B) = AB$ is the restriction of a *linear* map $M_n(\mathbb{F}) \to M_n(\mathbb{F})$. The differential of a linear map is the map itself, so $(dL_A)_I(X) = AX$ for every $X \in M_n(\mathbb{F})$. Hence the left-invariant field X^L generated by $X \in T_I\mathrm{GL}_n(\mathbb{F}) = M_n(\mathbb{F})$ is

$$X_A^L = (dL_A)_I(X) = AX \tag{2.8}$$

the vector field $A \mapsto AX$ on $\mathrm{GL}_n(\mathbb{F})$.

Step 2: the integral curve is the matrix exponential. By (2.8) the integral curve γ_X of X^L through I solves the initial value problem

$$\gamma_X'(t) = X^L_{\gamma_X(t)} = \gamma_X(t)\,X, \qquad \gamma_X(0) = I$$

which is exactly the linear matrix differential equation (1.2) governing one-parameter subgroups of $\mathrm{GL}_n(\mathbb{F})$, written on the right rather than the left. Its unique solution is the matrix exponential $t \mapsto \sum_{k\geq 0} t^k X^k/k!$: by proposition 1.3.4(1) the matrix curve $\eta(t) = \sum_k t^k X^k/k!$ satisfies $\eta'(t) = \eta(t)X$ and $\eta(0) = I$; uniqueness for this right-multiplication equation follows by the identical argument that theorem 1.3.10 runs for the left form $\gamma' = X\gamma$ (if η_1, η_2 both solve it, then $(\eta_1\eta_2^{-1})' = 0$, so $\eta_1\eta_2^{-1}$ is constant, equal to I at $t = 0$). Therefore $\gamma_X(t) = \sum_k t^k X^k/k!$, and evaluating at $t = 1$,

$$\exp(X) = \gamma_X(1) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

the matrix exponential of definition 1.3.2.

Step 3: passage to a subgroup. Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{F})$, and let $X \in \mathfrak{g}$. The one-parameter subgroup of G generated by X is a one-parameter subgroup of the ambient $\mathrm{GL}_n(\mathbb{F})$ generated by the same X, because the inclusion $G \hookrightarrow \mathrm{GL}_n(\mathbb{F})$ is a homomorphism of Lie groups carrying the left-invariant field of X on G to that on $\mathrm{GL}_n(\mathbb{F})$: concretely, an integral curve of X^L in G through I is also an integral curve of the ambient field $A \mapsto AX$ through I, so by uniqueness the two one-parameter subgroups coincide as curves in $\mathrm{GL}_n(\mathbb{F})$. Evaluating at $t = 1$, the intrinsic exponential of G at X equals the intrinsic exponential of $\mathrm{GL}_n(\mathbb{F})$ at X, which by Steps 1 and 2 is the matrix exponential. Hence the intrinsic exponential of G is the restriction of the matrix exponential to $\mathfrak{g}$, completing the proof.

The proposition closes the circle opened in chapter 1: the exponential built there by summing a series and the exponential built here by flowing a left-invariant field are one and the same. The two constructions look unrelated, one a computation in a normed algebra and the other the integration of a vector field on a manifold, but both solve the identical differential equation, and the uniqueness of its solution forces them to agree. From now on the notation exp carries no ambiguity: on a matrix group it is the matrix exponential, and the abstract properties proved above specialize to the concrete properties of chapter 1. The running example makes the agreement fully explicit.

Example 2.3.7: The Exponential on $\mathfrak{su}(2)$

For $G = \mathrm{SU}(2)$ the intrinsic Lie algebra is $\mathfrak{su}(2) = T_I\mathrm{SU}(2)$, the traceless skew-Hermitian 2×2 matrices (example 1.4.7). By proposition 2.3.6 the intrinsic exponential is the matrix exponential restricted to $\mathfrak{su}(2)$, so the closed form already computed in chapter 1 is the intrinsic answer. Writing $X \in \mathfrak{su}(2)$ with $\Phi(X) = v \in \mathbb{R}^3$ under the isomorphism $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$ of example 1.4.7, the computation $X^2 = -\frac{1}{4}|v|^2 I$ splits the exponential series into its even and odd parts exactly as cos and sin split, giving, for $v \neq 0$,

$$\exp(X) = \cos\left(\tfrac{|v|}{2}\right) I + \frac{\sin\left(\frac{|v|}{2}\right)}{|v|/2}\, X \tag{2.9}$$

This is the map of proposition 1.5.3, which established that exp is *onto* $\mathrm{SU}(2)$: every element of $\mathrm{SU}(2)$ has the form (2.9) for a suitable X, so $\exp\colon \mathfrak{su}(2) \to \mathrm{SU}(2)$ is surjective. The surjectivity is a feature of $\mathrm{SU}(2)$ as a compact connected group, not a general fact; it sharpens the product

> representation of corollary 2.3.5 from a finite product of exponentials down to a single one. The intrinsic theorem 2.3.1 recovers, as its content on $\mathrm{SU}(2)$, the concrete one-parameter subgroups computed in example 1.5.4: the curve $t \mapsto \exp(tX)$ with $\Phi(X) = v$ is a great circle on the sphere $\mathrm{SU}(2) \cong S^3$, closing up with period $4\pi/|v|$ in t.

The closed form (2.9) shows the abstract theory returning the concrete chapter 1 formula with nothing lost. The period $4\pi/|v|$, twice the $\mathrm{SO}(3)$-side period $2\pi/|v|$, is the same factor of 2 that separates $\mathrm{SU}(2)$ from $\mathrm{SO}(3)$ in the double cover theorem 1.5.5: a one-parameter subgroup of $\mathrm{SU}(2)$ must wind twice before it closes, because a rotation by 2π lifts to $-I$ rather than I upstairs. The exponential map is where this doubling first becomes visible in coordinates, and it foreshadows the role exp will play in Chapter 3 in translating between a group and its covers through their common Lie algebra.

Exercise 2.3.1

1. Show that $\exp\colon \mathfrak{g} \to G$ is a group homomorphism from $(\mathfrak{g}, +)$ to G if and only if $\exp(X)\exp(Y) = \exp(Y)\exp(X)$ for all $X, Y \in \mathfrak{g}$. Using the matrices $X = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ in $\mathfrak{gl}_2(\mathbb{R})$, verify that $\exp(X)\exp(Y) \neq \exp(X+Y)$, so that exp is not a homomorphism on $\mathrm{GL}_2(\mathbb{R})$.

2. Prove that if G is a connected abelian Lie group then $\exp\colon \mathfrak{g} \to G$ is a surjective group homomorphism. (You may use corollary 2.3.5 for surjectivity after establishing the homomorphism property; for the homomorphism property, show first that for commuting X, Y the curve $t \mapsto \exp(tX)\exp(tY)$ is a one-parameter subgroup with velocity $X + Y$.)

3. Exhibit a connected Lie group G on which exp is *not* surjective, by showing that no $X \in \mathfrak{sl}_2(\mathbb{R})$ satisfies $\exp(X) = \begin{pmatrix} -2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$, even though this matrix lies in the connected group $\mathrm{SL}_2(\mathbb{R})$. (Hint: analyze the eigenvalues of $\exp(X)$ for a traceless real X according to the sign of $\det X$.)

4. Show that $\exp\colon \mathfrak{g} \to G$ need not be injective by computing its fibers for $G = S^1 = \{z \in \mathbb{C} \big| |z| = 1\}$ with $\mathfrak{g} = i\mathbb{R}$. Determine the kernel of exp as a homomorphism $(\mathfrak{g}, +) \to S^1$ and identify it with a lattice in $\mathfrak{g}$.

5. Prove the reparametrization identity (2.7), that $\exp(tX) = \gamma_X(t)$, directly from the uniqueness in theorem 2.3.1, by checking that $s \mapsto \gamma_X(ts)$ is the one-parameter subgroup generated by tX.

6. Combining example 2.3.7 with corollary 2.3.5, explain why every element of $\mathrm{SU}(2)$ is a *single* exponential $\exp(X)$, whereas corollary 2.3.5 guarantees only a finite product of exponentials in general. State the property of $\mathrm{SU}(2)$ responsible, and contrast with $\mathrm{SL}_2(\mathbb{R})$ from exercise 2.3.1 (3).

2.4 The Differential of a Homomorphism and the Naturality of the Exponential

The previous three sections built, for a single Lie group G, the dictionary that assigns to G its Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ and the exponential map $\exp\colon \mathfrak{g} \to G$. This section makes the dictionary functorial. A homomorphism $\varphi\colon G \to H$ of Lie groups is a map of the group structure, and being smooth it also carries tangent data; its differential at the identity is a linear map $d\varphi_e\colon \mathfrak{g} \to \mathfrak{h}$, and the central fact is that this linear map is compatible with everything the previous sections built. It respects the bracket, so it is a homomorphism of Lie algebras, and it commutes with the exponential, so the diagram relating $\exp_G$ and $\exp_H$ closes. These two compatibilities together say that passing from a group to its Lie algebra is a functor, and that exp is a natural transformation.

The payoff is immediate and structural. Because exp generates a connected group from a neighborhood of the identity, a homomorphism out of a connected group is pinned down completely by the single linear map $d\varphi_e$: the infinitesimal data determines the global map. This is the first half of the correspondence the book exists to develop, and it reduces questions about maps of groups to questions about maps of vector spaces equipped with a bracket. The section closes by reading the classical determinant homomorphism through this lens, recovering the identity $\det \exp X = e^{\operatorname{tr} X}$ of proposition 1.3.4 as an instance of naturality.

Definition 2.4.1: The Differential of a Homomorphism

Let $\varphi\colon G \to H$ be a homomorphism of Lie groups, with Lie algebras $\mathfrak{g} = T_eG$ and $\mathfrak{h} = T_eH$. The *differential* of φ is the linear map

$$d\varphi_e\colon \mathfrak{g} \longrightarrow \mathfrak{h}$$

given by the tangent map of φ at the identity, sending a tangent vector $X \in T_eG$, represented by a curve $t \mapsto c(t)$ with $c(0) = e$ and $c'(0) = X$, to the velocity $d\varphi_e(X) = \frac{d}{dt}\big|_0 \varphi(c(t)) \in T_eH$ of its image.

The differential is well defined precisely because φ preserves the identity: $\varphi(e_G) = e_H$, so φ maps a curve through e_G to a curve through e_H, and the tangent map at that point lands in $\mathfrak{h} = T_{e_H}H$ as claimed. It is a linear map by the chain rule of [2], being the restriction of the smooth map φ's tangent map $d\varphi\colon TG \to TH$ to the fibre over the identity. So far $d\varphi_e$ is only a linear map between vector spaces; the content of the section is that it interacts correctly with the two extra structures those vector spaces carry, the bracket and the exponential. For a homomorphism of matrix groups the differential is computed by differentiating φ along $t \mapsto \exp(tX)$, and for the classical inclusion homomorphisms it is nothing but the corresponding inclusion of matrix Lie algebras from theorem 1.4.3. A concrete instance appears within the worked example below, where $\varphi = \det$ has differential the trace.

The mechanism that makes $d\varphi_e$ respect the bracket is the notion of φ-relatedness of vector fields. Recall from [2] that two vector fields ξ on G and η on H are *φ-related* when $d\varphi_g(\xi_g) = \eta_{\varphi(g)}$ for every $g \in G$, and that the bracket of vector fields is natural under this relation: if ξ_1 is φ-related to η_1 and ξ_2 to η_2, then $[\xi_1, \xi_2]$ is φ-related to $[\eta_1, \eta_2]$. The Lie algebra $\mathfrak{g}$ was built in section 2.2 as the left-invariant vector fields, and the key observation is that the left-invariant field with value X at e is φ-related to the left-invariant field with value $d\varphi_e X$.

Proposition 2.4.2: The Differential is a Lie Algebra Homomorphism

Let $\varphi\colon G \to H$ be a homomorphism of Lie groups. Then the differential $d\varphi_e\colon \mathfrak{g} \to \mathfrak{h}$ is a homomorphism of Lie algebras:

$$d\varphi_e[X,Y] = [\, d\varphi_e X,\ d\varphi_e Y\,]$$

for all $X, Y \in \mathfrak{g}$, where the bracket on the left is that of $\mathfrak{g}$ and the bracket on the right is that of $\mathfrak{h}$.

***Proof*:**

Write L_g for left translation by g, and for a vector $X \in \mathfrak{g} = T_eG$ let $\widetilde{X}$ denote the left-invariant vector field on G with $\widetilde{X}_e = X$, characterized by $\widetilde{X}_g = d(L_g)_e X$ as in definition 2.2.1. By theorem 2.2.3, the bracket on $\mathfrak{g}$ is defined by transporting the vector-field bracket, so $\widetilde{[X,Y]} = [\widetilde{X}, \widetilde{Y}]$; the same holds on H with fields denoted $\widehat{(\cdot)}$.

Step 1: the invariant fields are φ-related. Fix $X \in \mathfrak{g}$ and set $X' = d\varphi_e X \in \mathfrak{h}$. The claim is that $\widetilde{X}$ on G is φ-related to $\widehat{X'}$ on H, that is

$$d\varphi_g\big(\widetilde{X}_g\big) = \widehat{X'}_{\varphi(g)} \qquad \text{for all } g \in G \tag{2.10}$$

Because φ is a homomorphism, $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$: both sides send $h \mapsto \varphi(gh) = \varphi(g)\varphi(h)$. Taking differentials at e and applying the chain rule of [2],

$$d\varphi_g \circ d(L_g)_e = d(L_{\varphi(g)})_e \circ d\varphi_e$$

as maps $T_eG \to T_{\varphi(g)}H$. Evaluating both sides on X and using $\widetilde{X}_g = d(L_g)_e X$ on the left and $\widehat{X'}_{\varphi(g)} = d(L_{\varphi(g)})_e X'$ with $X' = d\varphi_e X$ on the right gives exactly (2.10). Thus $\widetilde{X}$ is φ-related to $\widehat{d\varphi_e X}$ for every X.

Step 2: apply naturality of the bracket. Let $X, Y \in \mathfrak{g}$. By Step 1, $\widetilde{X}$ is φ-related to $\widehat{d\varphi_e X}$ and $\widetilde{Y}$ is φ-related to $\widehat{d\varphi_e Y}$. The naturality of the vector-field bracket under φ-relatedness ([2]) then gives that $[\widetilde{X}, \widetilde{Y}]$ is φ-related to $[\,\widehat{d\varphi_e X},\ \widehat{d\varphi_e Y}\,]$. Evaluating the φ-relatedness at $g = e$, where $\varphi(e) = e$, yields

$$d\varphi_e\big([\widetilde{X}, \widetilde{Y}]_e\big) = [\,\widehat{d\varphi_e X},\ \widehat{d\varphi_e Y}\,]_e$$

The left side is $d\varphi_e[X,Y]$, since $[\widetilde{X}, \widetilde{Y}]_e = \widetilde{[X,Y]}_e = [X,Y]$ by the definition of the bracket on $\mathfrak{g}$. The right side is $[\, d\varphi_e X,\ d\varphi_e Y\,]$ by the same definition on $\mathfrak{h}$, since $\widehat{Z}_e = Z$ for any $Z \in \mathfrak{h}$. Therefore $d\varphi_e[X,Y] = [\, d\varphi_e X,\ d\varphi_e Y\,]$, as we sought to show.

The proof isolates the one place where the group structure enters: (2.10) holds because φ intertwines left translations, which is a direct consequence of φ being a homomorphism. Everything after that is the general naturality of the bracket for smooth maps, imported from [2]. This is why the differential of a homomorphism respects the bracket while the differential of an arbitrary smooth map need not: only a homomorphism carries left-invariant fields to left-invariant fields. The result says that the assignment $\varphi \mapsto d\varphi_e$ converts the analytic datum of a smooth group homomorphism into the purely algebraic datum of a Lie algebra homomorphism, and it is the reason the structure theory of Parts II and III can be conducted at the level of Lie algebras and then transported back to groups.

The compatibility with the exponential is the second half, and it is the statement most often invoked in computation. The exponential of G is built from one-parameter subgroups, and φ carries one-parameter subgroups of G to one-parameter subgroups of H; tracking the velocity through this transport is exactly what the differential records. The result is a square that commutes: applying φ after $\exp_G$ equals applying $\exp_H$ after $d\varphi_e$.

Theorem 2.4.3: Naturality of the Exponential Map

Let $\varphi\colon G \to H$ be a homomorphism of Lie groups with differential $d\varphi_e\colon \mathfrak{g} \to \mathfrak{h}$. Then for every $X \in \mathfrak{g}$,

$$\varphi\big(\exp_G X\big) = \exp_H\big(d\varphi_e X\big)$$

equivalently, the square

$$\begin{tikzcd} \mathfrak{g} \arrow[r, "d\varphi_e"] \arrow[d, "\exp_G"'] & \mathfrak{h} \arrow[d, "\exp_H"] \\ G \arrow[r, "\varphi"'] & H \end{tikzcd}$$

commutes.

***Proof*:**
Fix $X \in \mathfrak{g}$ and consider the curve

$$\sigma\colon \mathbb{R} \to H, \qquad \sigma(t) = \varphi\big(\exp_G(tX)\big)$$

Step 1: σ is a one-parameter subgroup of H. By proposition 2.3.3 the map $t \mapsto \exp_G(tX)$ is the one-parameter subgroup of G with velocity X, so $\exp_G((s+t)X) = \exp_G(sX)\exp_G(tX)$ and $\exp_G(0) = e$. Since φ is a homomorphism,

$$\sigma(s+t) = \varphi\big(\exp_G(sX)\exp_G(tX)\big) = \varphi\big(\exp_G(sX)\big)\varphi\big(\exp_G(tX)\big) = \sigma(s)\sigma(t)$$

and $\sigma(0) = \varphi(e_G) = e_H$. Being a composite of smooth maps, σ is smooth. Hence σ is a one-parameter subgroup of H.

Step 2: identify its velocity. Differentiating σ at $t = 0$ by the chain rule of [2],

$$\sigma'(0) = \frac{d}{dt}\Big|_0 \varphi\big(\exp_G(tX)\big) = d\varphi_e\left(\frac{d}{dt}\Big|_0 \exp_G(tX)\right) = d\varphi_e(X)$$

where the last equality uses that $t \mapsto \exp_G(tX)$ has velocity X at $t = 0$, again by proposition 2.3.3.

Step 3: conclude by uniqueness. By theorem 2.3.1, a one-parameter subgroup of H is determined by its velocity at $t = 0$, and the one with velocity $d\varphi_e X$ is precisely $t \mapsto \exp_H(t\, d\varphi_e X)$ by proposition 2.3.3. Steps 1 and 2 show σ is a one-parameter subgroup with this same velocity, so

$$\varphi\big(\exp_G(tX)\big) = \sigma(t) = \exp_H\big(t\, d\varphi_e X\big) \qquad \text{for all } t \in \mathbb{R}$$

Setting $t = 1$ gives $\varphi(\exp_G X) = \exp_H(d\varphi_e X)$, completing the proof.

The theorem is the computational engine of the correspondence. It converts the exponential of H into the exponential of G across any homomorphism, so that to evaluate φ on an element of the form $\exp_G X$, which is to say on any element near the identity, one need only know the single linear map $d\varphi_e$ and the exponential of the target. For matrix groups, where by proposition 2.3.6 the intrinsic exponential is the matrix exponential, naturality reads: $\varphi(\exp X) = \exp(d\varphi_e X)$ with both exponentials the matrix power series, and the classical identities of Chapter 1 fall out as special cases, as the worked example demonstrates. The commuting square also explains the terminology of exp as a natural transformation: it is a family of maps $\exp_G \colon \mathfrak{g} \to G$, one per group, that commutes with every morphism, which is exactly the condition for a natural transformation. To be precise, both endpoints are functors into smooth manifolds: the source is Lie followed by the forgetful functor $\mathbf{LieAlg}_{\mathbb{R}} \to \mathbf{Man}$ sending $\mathfrak{g}$ to its underlying vector space, written informally $G \mapsto \mathfrak{g}$, and the target is the forgetful functor $\mathbf{LieGrp} \to \mathbf{Man}$, written $G \mapsto G$; the square above is the naturality condition relating them.

The most consequential corollary follows by combining naturality with the fact that exp generates a connected group. Because a connected Lie group is covered by products of exponentials, and naturality determines φ on each exponential from $d\varphi_e$, the entire homomorphism is fixed by its differential.

Corollary 2.4.4: A Homomorphism is Determined by its Differential

Let G be a *connected* Lie group and H any Lie group. If $\varphi, \psi \colon G \to H$ are homomorphisms of Lie groups with $d\varphi_e = d\psi_e$, then $\varphi = \psi$. Consequently the map $\varphi \mapsto d\varphi_e$ from homomorphisms $G \to H$ to Lie algebra homomorphisms $\mathfrak{g} \to \mathfrak{h}$ is injective when G is connected.

***Proof*:**
Suppose $d\varphi_e = d\psi_e$. For every $X \in \mathfrak{g}$, naturality (theorem 2.4.3) applied to each homomorphism gives

$$\varphi(\exp_G X) = \exp_H(d\varphi_e X) = \exp_H(d\psi_e X) = \psi(\exp_G X)$$

so φ and ψ agree on the image $\exp_G(\mathfrak{g})$. By theorem 2.3.4, $\exp_G$ is a local diffeomorphism at 0, so its image contains an open neighborhood U of e in G, which we may take symmetric ($U = U^{-1}$) by replacing it with $U \cap U^{-1}$; and by corollary 2.3.5 a connected group G is generated by any neighborhood of e, so $G = \langle U \rangle = \bigcup_{k \geq 1} U^k$ where $U^k = \{u_1 \cdots u_k : u_i \in U\}$.

It remains to propagate agreement on U to all of G. Let $W = \{g \in G \mid \varphi(g) = \psi(g)\}$ be the set on which the two homomorphisms agree. Since φ, ψ are homomorphisms, W is a subgroup of G: if $g, g' \in W$ then $\varphi(gg'^{-1}) = \varphi(g)\varphi(g')^{-1} = \psi(g)\psi(g')^{-1} = \psi(gg'^{-1})$. The subgroup W contains U, hence contains every product $u_1 \cdots u_k$ of elements of U, hence contains $\bigcup_{k \geq 1} U^k = G$. Therefore $W = G$, that is $\varphi = \psi$. The final claim restates this: two homomorphisms with equal image under $\varphi \mapsto d\varphi_e$, that is with $d\varphi_e = d\psi_e$, are themselves equal, so the map is injective, completing the proof.

The corollary is the reason the Lie algebra deserves to be called the infinitesimal group. For a connected group there is at most one homomorphism with a prescribed differential, so the passage $\varphi \mapsto d\varphi_e$ loses no information about φ; the only question left open is existence, whether a given Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{h}$ arises from some group homomorphism. Existence requires more, and it fails in general: the differential can be an isomorphism of Lie algebras without φ being an isomorphism of groups, because global topology intervenes. The exercises exhibit the standard witness, the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, whose differential is the isomorphism $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ of example 1.4.7; and the sharp existence statement, that when G is simply connected every Lie algebra homomorphism does integrate, is Lie's second theorem, proved in Chapter 3. Injectivity here plus that existence theorem there is the full equivalence.

Both compatibilities can now be packaged categorically. The assignment $G \mapsto \mathfrak{g}$ on objects and $\varphi \mapsto d\varphi_e$ on morphisms is a functor, and verifying functoriality amounts to the chain rule together with the bracket-preservation just proved.

Proposition 2.4.5: The Lie Functor

Let **LieGrp** be the category of Lie groups and Lie group homomorphisms, and $\mathbf{LieAlg}_{\mathbb{R}}$ the category of finite-dimensional real Lie algebras and Lie algebra homomorphisms. The assignment

$$\mathrm{Lie}\colon \mathbf{LieGrp} \to \mathbf{LieAlg}_{\mathbb{R}}, \qquad G \mapsto \mathfrak{g} = T_eG, \qquad (\varphi\colon G \to H) \mapsto (d\varphi_e\colon \mathfrak{g} \to \mathfrak{h})$$

is a functor. That is, $d(\mathrm{id}_G)_e = \mathrm{id}_{\mathfrak{g}}$, and for composable homomorphisms $\varphi\colon G \to H$ and $\psi\colon H \to K$,

$$d(\psi \circ \varphi)_e = d\psi_e \circ d\varphi_e$$

***Proof*:**
Each object $\mathfrak{g} = T_eG$ is a finite-dimensional real Lie algebra by theorem 2.2.3 and proposition 2.2.5, and each morphism $d\varphi_e$ is a Lie algebra homomorphism by proposition 2.4.2, so Lie lands in $\mathbf{LieAlg}_{\mathbb{R}}$ on both objects and morphisms. For the identity, the differential of the identity map $\mathrm{id}_G\colon G \to G$ at e is the identity linear map on T_eG, since id_G sends every curve to itself and hence every velocity to itself; thus $d(\mathrm{id}_G)_e = \mathrm{id}_{\mathfrak{g}}$. For composition, the chain rule of [2] applied to the composite $\psi \circ \varphi$ at the identity, using $\varphi(e_G) = e_H$, gives

$$d(\psi \circ \varphi)_e = d\psi_{\varphi(e_G)} \circ d\varphi_e = d\psi_{e_H} \circ d\varphi_e$$

which is $d\psi_e \circ d\varphi_e$. Both functor axioms hold, completing the proof.

The functor Lie is the formal container for the entire correspondence. corollary 2.4.4 says Lie is faithful on the full subcategory of connected Lie groups, since it is injective on each morphism set out of a connected group. The reverse direction, reconstructing a group homomorphism, and even a group, from Lie algebra data, is the substance of Lie's theorems. When G is simply connected, every Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{h}$ is $d\varphi_e$ for a unique φ, so Lie becomes fully faithful there; and every finite-dimensional real Lie algebra is $\mathrm{Lie}(G)$ for some simply connected G, so Lie is essentially surjective onto $\mathbf{LieAlg}_{\mathbb{R}}$. These two facts upgrade Lie to an equivalence between the category of simply connected Lie groups and the category of finite-dimensional real Lie algebras, which is Lie's third theorem. The equivalence, together with the integration of Lie algebra homomorphisms, is proved in Chapter 3; the present section supplies the functor itself and its faithfulness on connected groups, which is one half of the equivalence.

The determinant homomorphism is the cleanest place to watch naturality recover a Chapter 1 formula. Its differential is the trace, and feeding this through the naturality square with the multiplicative group of the field as target reproduces $\det \exp X = e^{\operatorname{tr} X}$ with the scalar exponential doing the work on the target side.

Example 2.4.6: The Determinant and the Trace

Take $G = \mathrm{GL}_n(\mathbb{F})$ and $H = \mathbb{F}^\times$, the multiplicative group of nonzero scalars, a one-dimensional Lie group with Lie algebra $\mathfrak{h} = T_1\mathbb{F}^\times = \mathbb{F}$ and exponential map the ordinary scalar exponential $\exp_H = \exp\colon \mathbb{F} \to \mathbb{F}^\times$, $z \mapsto e^z$. The determinant

$$\det\colon \mathrm{GL}_n(\mathbb{F}) \longrightarrow \mathbb{F}^\times$$

is a smooth homomorphism, since $\det(AB) = \det A \det B$ and the determinant is polynomial in the entries.

The differential $d(\det)_I\colon \mathfrak{gl}_n(\mathbb{F}) \to \mathbb{F}$ is the trace. To see this, differentiate det along the curve $t \mapsto \exp(tX) = I + tX + O(t^2)$ through I. Expand the determinant of $I + tX$ as the sum over permutations σ of $\pm \prod_i (I+tX)_{i\sigma(i)}$. Any $\sigma \neq \mathrm{id}$ moves at least two indices, so its term has at least two off-diagonal factors $tX_{i\sigma(i)}$, each carrying a factor t, and hence contributes only at order t^2; only the identity permutation contributes a linear term, namely $\prod_i(1+tX_{ii}) = 1+t\sum_i X_{ii}+O(t^2)$, so

$$\det\bigl(I + tX + O(t^2)\bigr) = 1 + t\,\mathrm{tr}X + O(t^2)$$

whence

$$d(\det)_I(X) = \frac{d}{dt}\Big|_0 \det\exp(tX) = \mathrm{tr}X$$

Thus $d(\det)_I = \mathrm{tr}\colon \mathfrak{gl}_n(\mathbb{F}) \to \mathbb{F}$, which is indeed a Lie algebra homomorphism to the abelian $\mathbb{F}$: $\mathrm{tr}[X,Y] = \mathrm{tr}(XY) - \mathrm{tr}(YX) = 0 = [\mathrm{tr}X, \mathrm{tr}Y]$, consistent with proposition 2.4.2.

Now apply naturality (theorem 2.4.3) to $\varphi = \det$. The square reads $\det(\exp_G X) = \exp_H(d(\det)_I X)$, and with $\exp_G$ the matrix exponential (proposition 2.3.6), $\exp_H$ the scalar exponential, and $d(\det)_I = \mathrm{tr}$, this is

$$\det\exp X = e^{\mathrm{tr}X}$$

the identity of proposition 1.3.4. The Chapter 1 computation, which proved $\det\exp X = e^{\mathrm{tr}X}$ by triangularizing X and multiplying eigenvalue exponentials, is here revealed as a single instance of naturality: the determinant is a homomorphism, so it must intertwine the two exponentials, and its differential can only be the trace.

The example is the template for reading any classical homomorphism through its differential. The inclusion $\mathrm{SU}(2) \hookrightarrow \mathrm{SU}(3)$ has differential the inclusion $\mathfrak{su}(2) \hookrightarrow \mathfrak{su}(3)$; the projection of a product $G \times H \to G$ has differential the projection $\mathfrak{g} \oplus \mathfrak{h} \to \mathfrak{g}$; and, as the exercises develop, conjugation $C_g\colon x \mapsto gxg^{-1}$ has a differential that is a homomorphism of $\mathfrak{g}$ to itself, the value at g of the adjoint representation constructed in section 2.5. In every case the differential is easier to compute than the map, and naturality recovers the map on exponentials from it.

Exercise 2.4.1

1. For a fixed g in a Lie group G, let $C_g\colon G \to G$ be conjugation, $C_g(x) = gxg^{-1}$. Verify that C_g is a Lie group homomorphism (indeed an automorphism), and show that its differential $d(C_g)_e\colon \mathfrak{g} \to \mathfrak{g}$ is a Lie algebra automorphism. For a matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$, compute $d(C_g)_e(X)$ explicitly in terms of matrix multiplication, and use theorem 2.4.3 to write $g\exp(X)g^{-1}$ as a single exponential. (This differential, as a function of g, is the adjoint representation $\mathrm{Ad}(g)$ constructed in section 2.5.)

2. The double cover $p\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ of theorem 1.5.5 is a homomorphism of Lie groups.

Explain why its differential $dp_e \colon \mathfrak{su}(2) \to \mathfrak{so}(3)$ is an isomorphism of Lie algebras, even though p is not injective as a map of groups. Reconcile this with corollary 2.4.4, and state what additional global datum distinguishes SU(2) from SO(3).

3. Classify the connected Lie groups of dimension 1 up to isomorphism: show there are exactly two, namely $\mathbb{R}$ (under addition) and the circle S^1. Use that a one-dimensional Lie algebra is abelian, that exp is a homomorphism onto the identity component for an abelian group, and corollary 2.4.4.

4. Determine all Lie group homomorphisms $S^1 \to S^1$, where $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$. Identify $\mathrm{Lie}(S^1)$ with $\mathbb{R}$ so that $\exp(t) = e^{2\pi i t}$, compute the differential of the candidate map $z \mapsto z^n$, and use corollary 2.4.4 to show these are all of them.

5. Let $\det \colon \mathrm{U}(n) \to S^1$ be the determinant, a homomorphism onto the circle. Compute its differential $d(\det)_I \colon \mathfrak{u}(n) \to \mathrm{Lie}(S^1)$, and verify theorem 2.4.3 directly on a skew-Hermitian $X \in \mathfrak{u}(n)$ by comparing $\det \exp X$ with the scalar exponential of $d(\det)_I(X)$.

6. Let $\varphi \colon G \to H$ be a homomorphism of Lie groups with G connected. Show that if $d\varphi_e = 0$ then φ is the constant map $g \mapsto e_H$. Deduce that if $d\varphi_e$ is injective then φ is a local diffeomorphism onto its image near e, and give an example (from exercise 2.4.1 (2)) where $d\varphi_e$ is injective but φ is not globally injective.

2.5 The Adjoint Representations Ad and ad; the Bracket Intrinsically

The intrinsic Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ of definition 2.2.4 was built by transporting the bracket of vector fields from left-invariant fields to the tangent space T_eG, an operation with no visible connection to the commutator $XY - YX$ that governed the matrix Lie algebras of chapter 1. This section supplies the missing link. The group acts on itself by conjugation, and differentiating that action twice produces two representations, Ad on the group level and ad on the algebra level, whose interplay expresses the bracket. The central result identifies the left-invariant-field bracket of theorem 2.2.3 with ad, and hence, for a matrix group, with the commutator of proposition 1.4.4. When the dust settles the bracket has three faces that turn out to be one: the commutator from Chapter 1, the left-invariant-field bracket from section 2.2, and the operator ad introduced here.

The starting point is the conjugation automorphism. For each $g \in G$ the map $C_g \colon G \to G$, $C_g(h) = ghg^{-1}$, is a Lie group automorphism fixing the identity, so its differential at e is a linear automorphism of $T_eG = \mathfrak{g}$. Collecting these differentials as g varies gives a representation of G on its own Lie algebra.

Definition 2.5.1: The Adjoint Representation Ad

Let G be a Lie group with Lie algebra $\mathfrak{g} = T_eG$. For $g \in G$ let $C_g \colon G \to G$ be the conjugation automorphism $C_g(h) = ghg^{-1}$, a diffeomorphism fixing e (proposition 2.1.2). Its differential at e is a linear isomorphism of $\mathfrak{g}$, denoted

$$\mathrm{Ad}(g) = d(C_g)_e \colon \mathfrak{g} \to \mathfrak{g}$$

The resulting map $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ is the *adjoint representation* of G.

That Ad deserves the name representation is the assertion that it is a Lie group homomorphism into $\mathrm{GL}(\mathfrak{g})$, the group of linear automorphisms of the finite-dimensional vector space $\mathfrak{g}$. This is where the definition earns its keep, and the proof is a differentiation of the identity $C_{gh} = C_g \circ C_h$.

Proposition 2.5.2: Ad is a Lie Group Homomorphism

The map $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ of definition 2.5.1 is a homomorphism of Lie groups: it is smooth, and

$$\mathrm{Ad}(gh) = \mathrm{Ad}(g)\ \mathrm{Ad}(h), \qquad \mathrm{Ad}(e) = \mathrm{id}_{\mathfrak{g}}$$

for all $g, h \in G$.

***Proof*:**
The conjugation maps compose by $C_{gh} = C_g \circ C_h$, since $C_{gh}(x) = (gh)x(gh)^{-1} = g(hxh^{-1})g^{-1} = C_g(C_h(x))$, and $C_e = \mathrm{id}_G$. Both C_g and C_h fix e, so the chain rule for differentials applies at e:

$$\mathrm{Ad}(gh) = d(C_{gh})_e = d(C_g \circ C_h)_e = d(C_g)_e \circ d(C_h)_e = \mathrm{Ad}(g)\ \mathrm{Ad}(h)$$

and $\mathrm{Ad}(e) = d(\mathrm{id}_G)_e = \mathrm{id}_{\mathfrak{g}}$. In particular $\mathrm{Ad}(g)\,\mathrm{Ad}(g^{-1}) = \mathrm{Ad}(e) = \mathrm{id}$, so each $\mathrm{Ad}(g)$ is invertible and Ad lands in $\mathrm{GL}(\mathfrak{g})$. Smoothness is a consequence of the smoothness of the group operations: the conjugation map $(g,h) \mapsto ghg^{-1}$ is smooth on $G \times G$ as a composite of multiplication and inversion, and $\mathrm{Ad}(g)$ is the partial differential in the h-variable at $h = e$, which depends smoothly on g because differentiation of a smooth family of maps produces a smooth family of differentials ([2]). Thus Ad is smooth, completing the proof.

The homomorphism Ad packages the conjugation action of G on itself into its infinitesimal effect on the tangent space at e. It sees only the germ of conjugation at the identity, which is exactly the data a Lie algebra is built to remember. Because $\mathrm{GL}(\mathfrak{g})$ is itself a Lie group, with Lie algebra the endomorphism algebra $\mathfrak{gl}(\mathfrak{g}) = \mathrm{End}(\mathfrak{g})$, the machinery of section 2.4 applies to Ad: it has a differential at e, and that differential is the second of the two representations.

Definition 2.5.3: The Adjoint Representation ad

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The *adjoint representation* of $\mathfrak{g}$ is the differential of Ad at the identity,

$$\mathrm{ad} = d(\mathrm{Ad})_e \colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}) = \mathrm{End}(\mathfrak{g})$$

Explicitly, for $X \in \mathfrak{g}$ the endomorphism $\mathrm{ad}(X)$ of $\mathfrak{g}$ is

$$\mathrm{ad}(X) = \left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(\exp tX)$$

the velocity at $t = 0$ of the curve $t \mapsto \mathrm{Ad}(\exp tX)$ in $\mathrm{GL}(\mathfrak{g})$.

The explicit formula is the naturality of exp from theorem 2.4.3 read for the homomorphism Ad: the one-parameter subgroup $t \mapsto \exp(tX)$ of G maps under Ad to the one-parameter subgroup $t \mapsto \mathrm{Ad}(\exp tX)$ of $\mathrm{GL}(\mathfrak{g})$, whose generator is $d(\mathrm{Ad})_e(X) = \mathrm{ad}(X)$. Since $\mathrm{GL}(\mathfrak{g})$ is a matrix group, its exponential is the matrix exponential (proposition 2.3.6), so the naturality square specializes to

$$\mathrm{Ad}(\exp tX) = \exp\bigl(t\,\mathrm{ad}(X)\bigr) \tag{2.11}$$

an identity of endomorphisms of $\mathfrak{g}$ that will reappear in the exercises as the statement $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}X}$. Differentiating (2.11) at $t = 0$ recovers the definition of $\mathrm{ad}(X)$, and integrating it recovers Ad from ad on the identity component.

By proposition 2.4.2 the differential of a Lie group homomorphism is a Lie algebra homomorphism, so ad preserves brackets: $\mathrm{ad}([X, Y]) = [\mathrm{ad}(X), \mathrm{ad}(Y)]$, where the bracket on the right is the commutator of endomorphisms in $\mathrm{End}(\mathfrak{g})$. This is the Jacobi identity in disguise, a point developed in the exercises. What remains, and it is the conceptual hinge of the chapter, is to compute ad in terms of the bracket already available on $\mathfrak{g}$, and to show the two agree.

The bracket on $\mathfrak{g}$ of definition 2.2.4 was defined by carrying the bracket of vector fields across the isomorphism $X \mapsto \tilde{X}$ of theorem 2.2.3 between T_eG and the left-invariant fields. The bracket induced by ad is a second candidate $\mathrm{ad}(X)Y$. The next theorem shows these coincide, which is what licenses writing $[X, Y] = \mathrm{ad}(X)Y$ without ambiguity and, through the matrix case below, ties the whole apparatus back to the commutator.

Theorem 2.5.4: The Bracket is the Adjoint Action

Let G be a Lie group with Lie algebra $\mathfrak{g} = T_eG$, and let $[-, -]$ denote the bracket of definition 2.2.4, transported from the Lie bracket of left-invariant vector fields via theorem 2.2.3. Then for all $X, Y \in \mathfrak{g}$,

$$[X, Y] = \mathrm{ad}(X)Y = \left.\frac{d}{dt}\right|_{t=0} \mathrm{Ad}(\exp tX)\, Y$$

The abstract bracket satisfying the axioms of definition 1.4.5 is therefore the infinitesimal conjugation action ad, and the two brackets on $\mathfrak{g}$ coincide.

***Proof*:**
Write $\tilde{X}$ for the left-invariant vector field with $\tilde{X}_e = X$, so that by definition of the bracket on $\mathfrak{g}$,

$$[X, Y] = [\tilde{X}, \tilde{Y}]_e \tag{2.12}$$

the right side being the value at e of the Lie bracket of the two left-invariant fields. The proof computes $[\tilde{X}, \tilde{Y}]_e$ by the flow characterization of the vector-field bracket and matches it with $\mathrm{ad}(X)Y$.

Step 1: the flow of a left-invariant field is right translation. Let $\gamma_X(t) = \exp(tX)$ be the integral curve of $\tilde{X}$ through e, which is the one-parameter subgroup with velocity X (proposition 2.3.3). Because $\tilde{X}$ is left-invariant, its integral curve through an arbitrary point g is $t \mapsto g\exp(tX) = L_g(\exp tX)$: applying the diffeomorphism L_g carries the integral curve through e to one through g, and L_g sends the field $\tilde{X}$ to itself. Hence the flow of $\tilde{X}$ at time t is right translation by $\exp(tX)$,

$$\Phi_t^{\tilde{X}} = R_{\exp(tX)}, \qquad g \mapsto g\exp(tX) \tag{2.13}$$

These flows exist for all $t \in \mathbb{R}$ because left-invariant fields are complete (proposition 2.2.5).

Step 2: the bracket as a Lie derivative. The Lie bracket of two vector fields is the Lie derivative of one along the flow of the other: for any smooth vector fields $\tilde{X}, \tilde{Y}$,

$$[\tilde{X}, \tilde{Y}]_g = \frac{d}{dt}\bigg|_{t=0} \big(d\Phi_{-t}^{\tilde{X}}\big)_{\Phi_t^{\tilde{X}}(g)}\big(\tilde{Y}_{\Phi_t^{\tilde{X}}(g)}\big)$$

the standard formula $[\tilde{X}, \tilde{Y}] = \mathcal{L}_{\tilde{X}}\tilde{Y}$ for the bracket in terms of the pushforward of $\tilde{Y}$ by the backward flow of $\tilde{X}$ ([2]). Evaluate at $g = e$ and substitute the flow (2.13), so $\Phi_t^{\tilde{X}}(e) = \exp(tX)$ and $\Phi_{-t}^{\tilde{X}} = R_{\exp(-tX)}$:

$$[\tilde{X}, \tilde{Y}]_e = \frac{d}{dt}\bigg|_{t=0} \big(dR_{\exp(-tX)}\big)_{\exp(tX)}\big(\tilde{Y}_{\exp(tX)}\big) \tag{2.14}$$

Step 3: rewrite the inner expression through conjugation. The field $\tilde{Y}$ is left-invariant, so its value at $\exp(tX)$ is the left translate of Y,

$$\tilde{Y}_{\exp(tX)} = \big(dL_{\exp(tX)}\big)_e Y$$

Insert this into (2.14). The differential $\big(dR_{\exp(-tX)}\big)_{\exp(tX)}$ acts on the result, and since the maps $L_{\exp(tX)}$ and $R_{\exp(-tX)}$ commute as diffeomorphisms of G (left and right translations always commute), their composite is the conjugation map:

$$R_{\exp(-tX)} \circ L_{\exp(tX)} = \big(h \mapsto \exp(tX)\, h\, \exp(-tX)\big) = C_{\exp(tX)}$$

Both $L_{\exp(tX)}$ and $R_{\exp(-tX)}$ are diffeomorphisms, so the chain rule gives, for the composite evaluated on $Y \in T_eG$,

$$\big(dR_{\exp(-tX)}\big)_{\exp(tX)}\big(dL_{\exp(tX)}\big)_e Y = d\big(C_{\exp(tX)}\big)_e Y = \mathrm{Ad}(\exp tX)\, Y$$

the last equality by the definition of Ad in definition 2.5.1. Substituting into (2.14),

$$[\tilde{X}, \tilde{Y}]_e = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp tX)\, Y \tag{2.15}$$

Step 4: identify with ad. By definition 2.5.3 the right side of (2.15) is precisely $\mathrm{ad}(X)Y$, the velocity at $t = 0$ of the curve $t \mapsto \mathrm{Ad}(\exp tX)Y$ in $\mathfrak{g}$. Combining with (2.12) and (2.15),

$$[X, Y] = [\tilde{X}, \tilde{Y}]_e = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp tX)\, Y = \mathrm{ad}(X)Y$$

as we sought to show.

The theorem is the reconciliation the chapter was built toward. The bracket on $\mathfrak{g}$ was first defined abstractly, as whatever operation the vector-field bracket induces on T_eG, with the alternating and Jacobi axioms of definition 1.4.5 guaranteed but its meaning opaque. theorem 2.5.4 gives it a concrete face: $[X, Y]$ is the rate at which conjugation by $\exp(tX)$ moves Y, measured at $t = 0$. The bracket is thus the infinitesimal record of the failure of the two one-parameter subgroups $\exp(tX)$ and $\exp(sY)$ to commute, exactly the interpretation the matrix commutator carried in proposition 1.4.4, now established with no matrices in sight. Every structural fact about the bracket, the Jacobi identity included, is now a fact about the conjugation action of the group on itself.

The identification also explains the notation. The operator $\mathrm{ad}(X)$ acts on $\mathfrak{g}$ by $Y \mapsto [X, Y]$, so ad is the map sending X to left-bracketing by X. That ad is a Lie algebra homomorphism (proposition 2.4.2) then reads $\mathrm{ad}([X, Y]) = [\mathrm{ad}(X), \mathrm{ad}(Y)]$, which unwinds to $[[X, Y], Z] = [X, [Y, Z]] - [Y, [X, Z]]$ for all Z, a rearrangement of the Jacobi identity. The single homomorphism ad therefore encodes the entire bracket and its one nontrivial axiom at once.

Everything reduces to the commutator in the matrix case, closing the loop with Chapter 1. For a group $G \subseteq \mathrm{GL}_n(\mathbb{F})$ the intrinsic constructions must return the formulas computed by hand there, and the next result verifies this for Ad and ad simultaneously.

Proposition 2.5.5: Ad and ad for a Matrix Group

Let $G \subseteq \mathrm{GL}_n(\mathbb{F})$ be a matrix group with Lie algebra $\mathfrak{g} \subseteq M_n(\mathbb{F})$, identified with T_IG as in corollary 1.4.6. Then for all $g \in G$ and $X, Y \in \mathfrak{g}$,

$$\mathrm{Ad}(g)X = gXg^{-1}, \qquad \mathrm{ad}(X)Y = XY - YX = [X, Y]$$

The intrinsic adjoint action is conjugation of matrices, and the intrinsic bracket $\mathrm{ad}(X)Y$ is the commutator of proposition 1.4.4. The intrinsic bracket on $\mathfrak{gl}_n(\mathbb{F})$ is therefore the matrix commutator.

***Proof*:**
Step 1: the formula for Ad. Fix $g \in G$. The conjugation map $C_g(h) = ghg^{-1}$ is the restriction to G of the ambient linear map $M \mapsto gMg^{-1}$ on $M_n(\mathbb{F})$, which is its own differential at every point since it is linear. For $X \in \mathfrak{g} = T_IG$, represent X as the velocity of the curve $t \mapsto \exp(tX)$ through I (proposition 2.3.6 identifies the intrinsic exp with the matrix exponential). The conjugation-equivariance of the exponential (proposition 1.3.4) gives

$$C_g(\exp tX) = g \exp(tX) g^{-1} = \exp\big(t\, gXg^{-1}\big) \tag{2.16}$$

the last equality because $g \exp(tX)g^{-1} = \exp(g(tX)g^{-1})$ by the series computation $g\big(\sum_k \frac{(tX)^k}{k!}\big)g^{-1} =$

$\sum_k \frac{(g\,tX\,g^{-1})^k}{k!}$. Differentiating (2.16) at $t=0$,

$$\mathrm{Ad}(g)X = d(C_g)_I\,X = \frac{d}{dt}\bigg|_{t=0} \exp\bigl(t\,gXg^{-1}\bigr) = gXg^{-1}$$

using $\frac{d}{dt}\big|_0 \exp(tZ) = Z$ with $Z = gXg^{-1}$. The result gXg^{-1} lies in $\mathfrak{g}$, consistent with $\mathrm{Ad}(g)$ being an endomorphism of $\mathfrak{g}$.

Step 2: the formula for ad. Apply Step 1 with $g=\exp(sX)$, so $\mathrm{Ad}(\exp sX)Y = \exp(sX)\,Y\exp(-sX)$ for $Y\in\mathfrak{g}$. Differentiating in s at $s=0$, with $\frac{d}{ds}\big|_0 \exp(sX) = X$ and $\frac{d}{ds}\big|_0 \exp(-sX) = -X$, the product rule gives

$$\mathrm{ad}(X)Y = \frac{d}{ds}\bigg|_{s=0} \exp(sX)\,Y\exp(-sX) = XY - YX = [X,Y]$$

which is the commutator. By theorem 2.5.4 the left side is the intrinsic bracket, so the intrinsic bracket equals the matrix commutator, recovering proposition 1.4.4. For $G=\mathrm{GL}_n(\mathbb{F})$ this says the intrinsic Lie algebra $\mathfrak{gl}_n(\mathbb{F}) = M_n(\mathbb{F})$ carries the commutator bracket, completing the proof.

The proposition is the payoff of the whole chapter in one line: the bracket that section 2.2 manufactured from left-invariant vector fields, with no reference to matrices, becomes the ordinary commutator $XY-YX$ the moment G is realized inside some $\mathrm{GL}_n(\mathbb{F})$. Two constructions that looked unrelated, the vector-field bracket of theorem 2.2.3 and the commutator of proposition 1.4.4, are the same operation seen from two sides. The intrinsic definition is the more general one, applying to any Lie group, and proposition 2.5.5 certifies that it specializes correctly on the matrix groups where an independent computation exists.

The running example makes the abstract statement palpable. For SU(2) the adjoint representation is exactly the conjugation action that produced the double cover in Chapter 1, and ad is the cross product.

Example 2.5.6: Ad and ad for SU(2) and the Rotation Action

Take $G=\mathrm{SU}(2)$, with Lie algebra $\mathfrak{su}(2)$ the traceless skew-Hermitian 2×2 matrices of example 1.4.7, and its basis X_1,X_2,X_3 with $[X_j,X_k]=\sum_\ell \varepsilon_{jk\ell}X_\ell$. By proposition 2.5.5 the adjoint representation is conjugation,

$$\mathrm{Ad}(g)X = gXg^{-1}, \qquad g\in\mathrm{SU}(2),\ X\in\mathfrak{su}(2)$$

which is precisely the map $\varphi(g)\colon X\mapsto gXg^{-1}$ of theorem 1.5.5. That theorem showed $\varphi(g)$ preserves the Euclidean structure on $\mathfrak{su}(2)\cong\mathbb{R}^3$ given by the determinant form and lands in SO(3); in the present language this says

$$\mathrm{Ad}\colon \mathrm{SU}(2)\to\mathrm{SO}(3)\subset\mathrm{GL}(\mathfrak{su}(2))$$

is the adjoint representation, and the double cover $\varphi=\mathrm{Ad}$ is nothing but Ad regarded as a homomorphism onto the rotation group of the three-dimensional space $\mathfrak{su}(2)$. The kernel of Ad is the center $\{\pm I\}$, the two-to-one fibers of the cover.

On the algebra level, proposition 2.5.5 gives $\mathrm{ad}(X)Y=[X,Y]$, so on the basis

$$\mathrm{ad}(X_1)X_2 = [X_1,X_2] = X_3, \qquad \mathrm{ad}(X_1)X_3 = [X_1,X_3] = -X_2, \qquad \mathrm{ad}(X_1)X_1 = 0$$

In the ordered basis (X_1, X_2, X_3) the operator $\mathrm{ad}(X_1)$ has matrix

$$\mathrm{ad}(X_1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

the generator of rotation about the first axis, and $\mathrm{ad}(X_2)$, $\mathrm{ad}(X_3)$ are the analogous generators about the other two axes. Under the isomorphism $\mathfrak{su}(2) \cong (\mathbb{R}^3, \times)$ of example 1.4.7 the operator $\mathrm{ad}(X) = [X, -]$ becomes $v \mapsto v \times (-)$, cross product with the vector $v = \Phi(X)$, which is exactly the infinitesimal rotation about the axis v. The identity $\mathrm{Ad}(\exp tX) = \exp(t\,\mathrm{ad}(X))$ of (2.11) then states that conjugating by $\exp(tX)$ is the finite rotation whose generator is cross product with v, recovering the computation in Step 4 of theorem 1.5.5 that $\exp(\frac{\theta}{2}u)$ maps under Ad to the rotation by θ about u. The factor of two between the group parameter and the rotation angle, the arithmetic heart of the double cover, is precisely the passage through $\exp(t\,\mathrm{ad}(X))$ with $\mathrm{ad}(X)$ the generator normalized by the bracket relations (1.5).

The example closes the circle the book opened with. In Chapter 1 the conjugation action of SU(2) on $\mathfrak{su}(2)$ was constructed by hand to build the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, before the words "adjoint representation" were available; that action is Ad, and the double cover is Ad itself, the homomorphism onto the orthogonal group of the invariant form on $\mathfrak{su}(2)$. What Chapter 1 saw as a geometric accident, that a unit-quaternion group double-covers the rotation group, is the adjoint representation of SU(2) read through the isomorphism $\mathfrak{su}(2) \cong \mathbb{R}^3$.

The bracket now stands identified in three forms. It is the matrix commutator $XY - YX$ of proposition 1.4.4, the operation Chapter 1 extracted by differentiating a conjugation curve. It is the left-invariant-field bracket $[\tilde{X}, \tilde{Y}]_e$ of theorem 2.2.3, the intrinsic definition that requires no embedding. And it is the adjoint action $\mathrm{ad}(X)Y = \frac{d}{dt}\big|_0 \mathrm{Ad}(\exp tX)Y$ of theorem 2.5.4, the infinitesimal conjugation of the group on its own algebra. These are one operation, and the equivalence of the three descriptions is what makes the Lie algebra a faithful and computable shadow of the group. The bilinear form $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y)$ built from ad, invariant under the adjoint action of G, is the *Killing form* (defined and developed in exercise 2.5.1 (4) below); it measures the semisimplicity of $\mathfrak{g}$ and is the tool that drives the structure theory of Part II and the classification of Chapter 6 onward.

Exercise 2.5.1

1. Prove the identity $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}X}$ for all $X \in \mathfrak{g}$, where the right side is the exponential in $\mathrm{GL}(\mathfrak{g})$ of the endomorphism $\mathrm{ad}X \in \mathrm{End}(\mathfrak{g})$. Deduce that for a matrix group $G \subseteq \mathrm{GL}_n(\mathbb{F})$, conjugation $e^X Y e^{-X}$ is given by the convergent series $Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \cdots$.

2. Show that the kernel of $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ contains the center $Z(G)$, and that for a *connected* Lie group G the kernel is exactly $Z(G)$. Conclude that a connected G acts trivially by conjugation on its Lie algebra if and only if G is abelian.

3. Show that for each $X \in \mathfrak{g}$ the operator $\mathrm{ad}(X)\colon \mathfrak{g} \to \mathfrak{g}$ is a *derivation* of the bracket, that is

$$\mathrm{ad}(X)[Y, Z] = [\mathrm{ad}(X)Y, Z] + [Y, \mathrm{ad}(X)Z]$$

for all $Y, Z \in \mathfrak{g}$, and prove that this identity is equivalent to the Jacobi identity of definition 1.4.5.

4. Define the *Killing form* $\kappa(X,Y) = \mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y)$ on $\mathfrak{g}$. Show that κ is a symmetric bilinear form, and that it is Ad-*invariant*: $\kappa(\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y) = \kappa(X,Y)$ for all $g \in G$. Deduce the infinitesimal invariance $\kappa([X,Y],Z) + \kappa(Y,[X,Z]) = 0$, the associativity (or invariance) property of the Killing form.

5. For $\mathfrak{su}(2)$ with basis X_1, X_2, X_3 as in example 1.4.7, write out the three matrices $\mathrm{ad}(X_1), \mathrm{ad}(X_2), \mathrm{ad}(X_3)$ in the ordered basis (X_1, X_2, X_3), and compute the Killing form κ explicitly. Verify that κ is negative definite, and identify it as a negative multiple of the standard inner product on $\mathbb{R}^3$.

6. Give an example of a Lie group G for which $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ is not injective, and describe its kernel. Contrast with the case $G = \mathrm{SU}(2)$, where ad is injective. What feature of the Lie algebra distinguishes the two cases?

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